

Probability Fundamentals

Thomas Nield
O'Reilly Media

What to Expect

1. Introduction
2. Probability Math
3. Bayes Theorem
4. Normal Distribution

Section I

Why Learn Probability?

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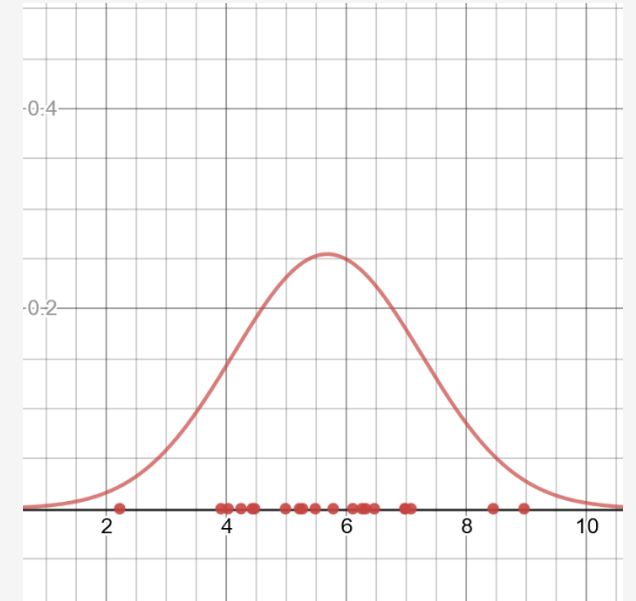
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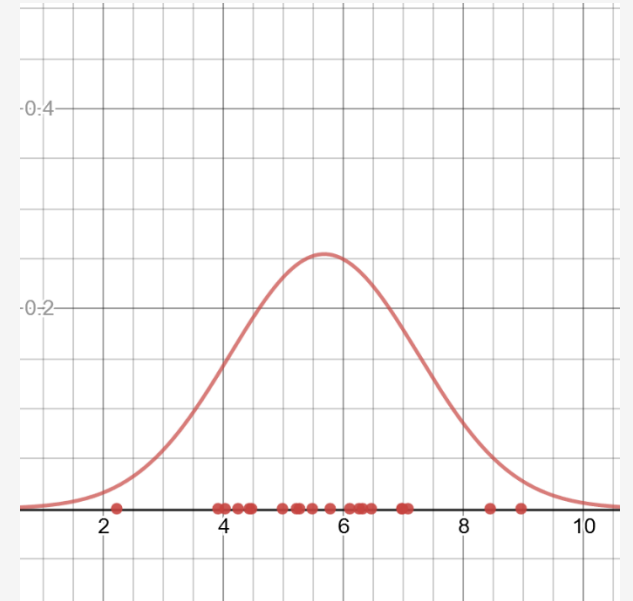
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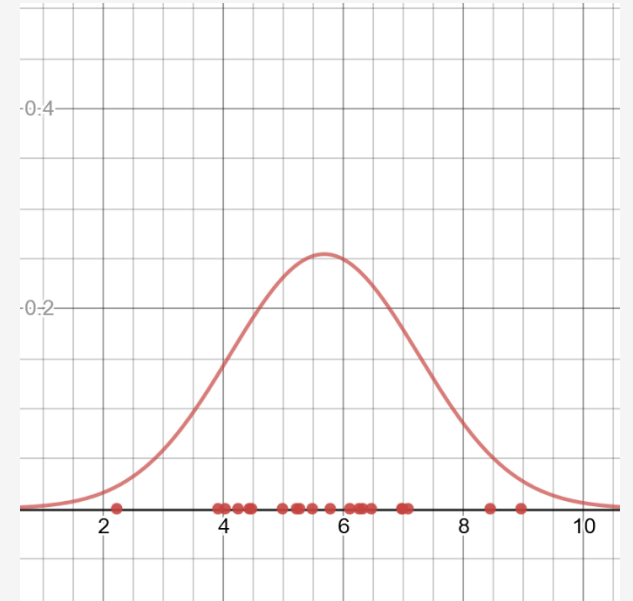


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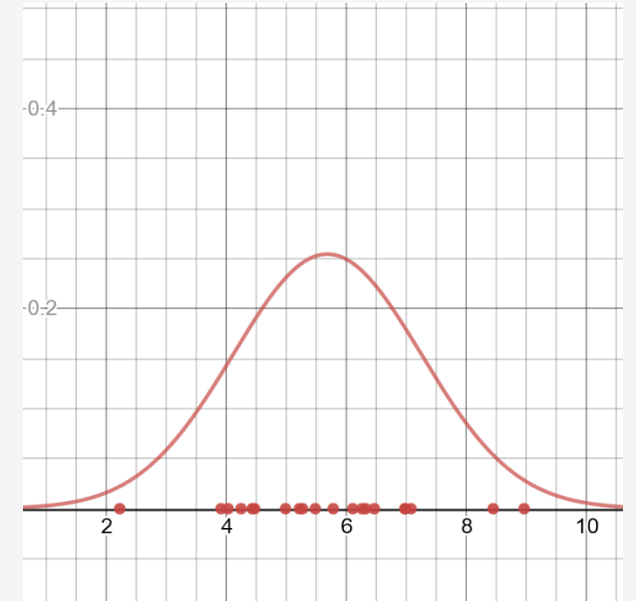
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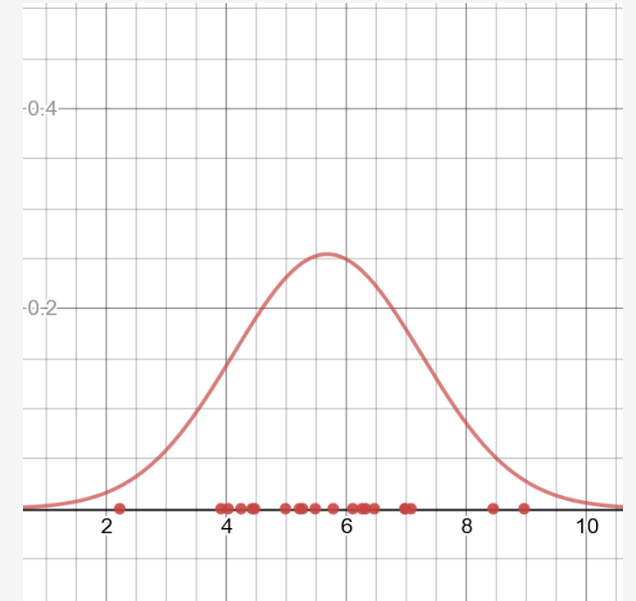
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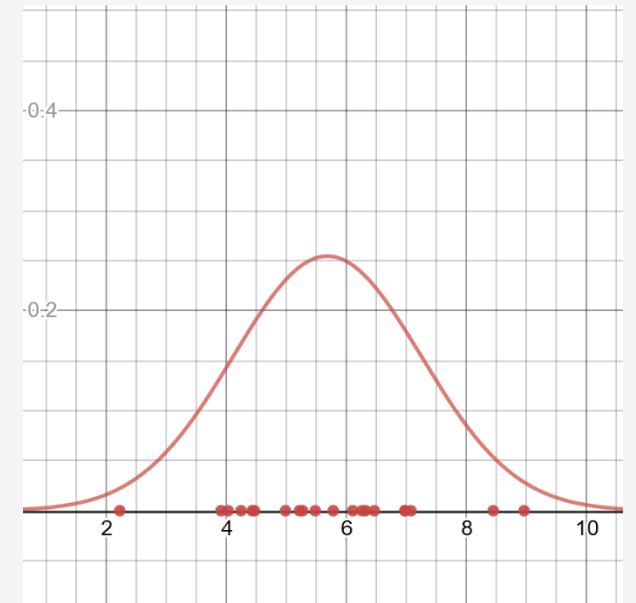
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We will put more emphasis on probability in this class, but we will work with data and statistics where it makes sense.



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What is Probability?



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Probability is how likely an event will happen, based on observations or belief.

- How likely is it I will get 7 heads in 10 fair coin flips?
- What are my chances in winning an election?
- What is the likelihood my flight will be late?
- How certain am I a product is defective?

Probability is often expressed in two ways:

- As a percentage: 60% chance my flight will be late
- As an odds: 3:2 odds my flight will be late



Section II

Probability Fundamentals

Probability Basics



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When we work with a single simple probability, it is known as a **marginal probability**.



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Probability based on data: If we sample 10 products from a factory line and find 4 items are defective, that would be a 40% defective rate.

Probability based on belief: An engineer realizes an inferior material was used and guesses the defective rate for the product will be 50%.

Probability based on data + belief: we can quantify the engineer's belief and the data, merge them together, and find a 44.44% probability is most likely.



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This would reduce to 1.5, which means we believe a product is 1.5 times more likely to be defective than not defective.



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To hyperbolize, if he pays \$200 for them winning but I must pay him \$1 if they lose, that means he *REALLY* believes the Dallas Cowboys are going to lose: 200x more likely ($\frac{200}{1} = 200$) and feels no need to hedge his bet.



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Putting money on something (horse races, sports teams, stock markets) forces you to quantify how strongly you believe in an event, and can be effective measure to turn subjective beliefs into something quantitative.



Turning Odds into Probabilities

If you need to turn an odds $O(X)$ into a probability, you can do so with this formula:

$$P(X) = \frac{O(X)}{1 + O(X)}$$

So if you believe something is 3x likely to happen versus not happen (3:1), the probability for that event is 75%:

```
def prob_to_odds(p):  
    return p / (1.0 - p)  
  
def odds_to_prob(o):  
    return o / (1.0 + o)  
  
p_x = .75  
o_x = prob_to_odds(p_x)  
  
# prints ODDS: 3.0  
print("ODDS: {}".format(o_x))
```



Joint Probabilities

$$P(A \cap B)$$

$$P(A \text{ and } B)$$

$$P(A, B)$$



and



```
p_heads = 1.0 / 2.0  
p_six = 1.0 / 6.0  
  
p_heads_and_six = p_heads * p_six  
  
print(p_heads_and_six) # 0.08333333333333333
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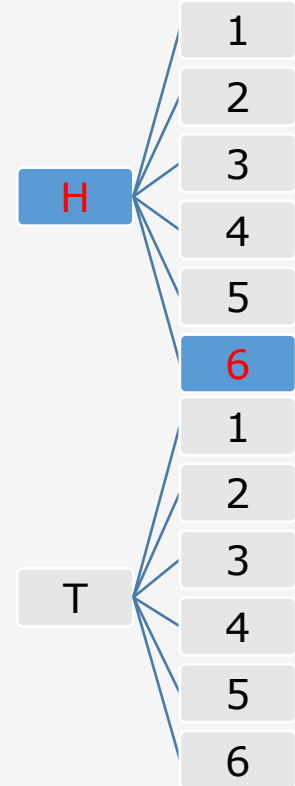
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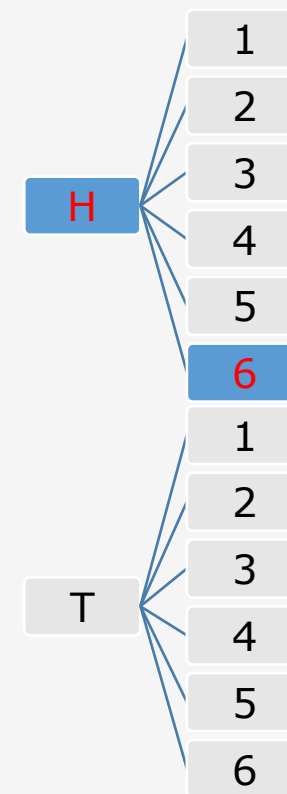
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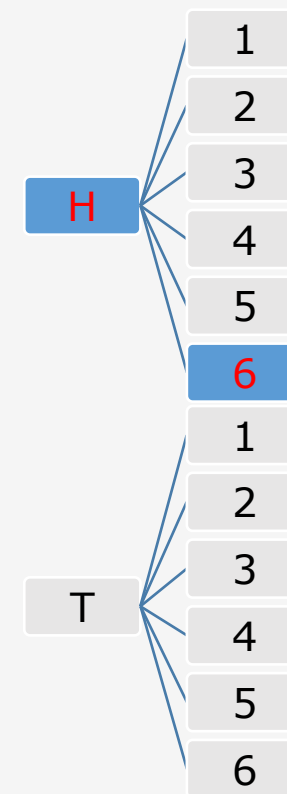
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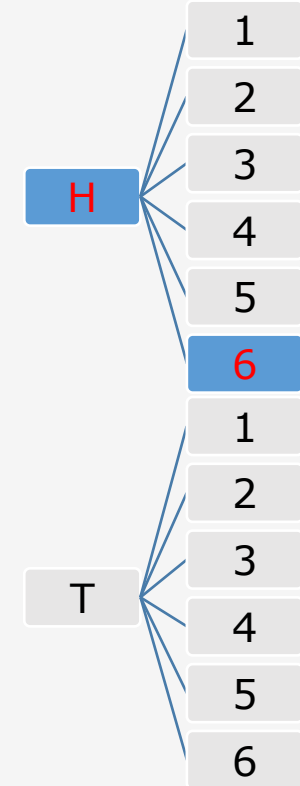
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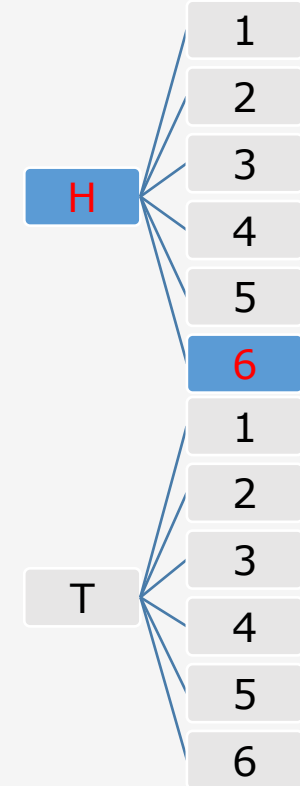
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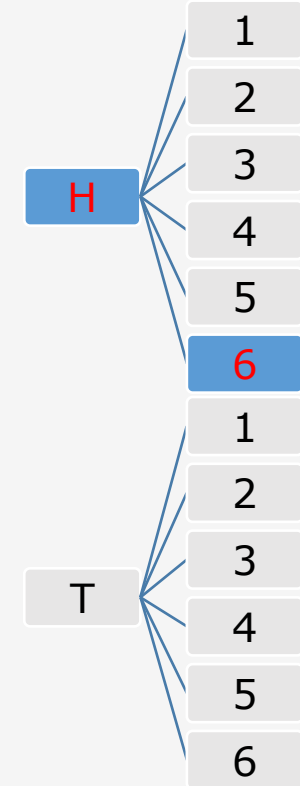
The multiplication allows us to avoid generating and counting all these combinations and is known as the **product rule**.

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Joint Probabilities



$$\left(\frac{1}{6}\right)^{10} = \frac{1}{60466176} = 0.0000000165382$$

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# Probability of rolling a "six"
p = 1.0 / 6.0

# Probability of rolling a "six" ten times in a row
p_10_sixes = p ** 10

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You can use the product rule to combine as many probabilities as you want.



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Of course, since this multiplication is repetitive, we can use an exponent:

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Exercise #1

There is a 30% chance of rain today, and a 40% chance your umbrella order will arrive on time. You are eager to walk in the rain today and cannot do so without either!

What is the probability it will rain **AND** your umbrella will arrive?



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There is a 30% chance of rain today, and a 40% chance your umbrella order will arrive on time. You are eager to walk in the rain today and cannot do so without either!

What is the probability it will rain **AND** your umbrella will arrive?

$$P(A \text{ and } B) = P(A) * P(B)$$

$$P(\text{Rain and umbrella arrives}) = .30 * .40 = .12$$

12% chance it will rain and your umbrella will arrive.



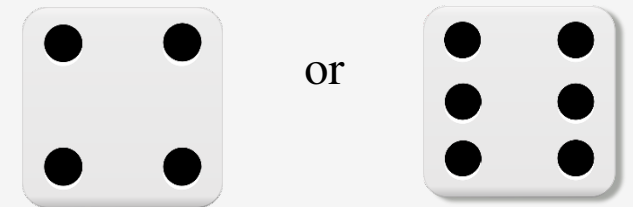
Union Probability – Mutually Exclusive

Things get a little more nuanced when we work with **union probabilities**, or the probability that at least one of multiple events will occur.

When two events are **mutually exclusive**, meaning that only one of the events can occur but not both, then it is as easy as adding their probabilities together.

For example, what is the probability of getting a “4” or “6” on a die roll? Since we cannot get “4” and a “6” simultaneously, we just add these probabilities together.

$$P(A \cup B)$$
$$P(A \text{ or } B)$$



Union Probability – Non-Mutually Exclusive

$$P(A \cup B)$$

$$P(A \text{ or } B)$$

$$\frac{7}{12} = .5833$$

$$\frac{1}{2} + \frac{1}{6} = \frac{2}{3} = .6666$$



or



Union Probability – Non-Mutually Exclusive

When two events are **non-mutually exclusive**, meaning that two events can occur simultaneously, their union probability gets tricky.

$$P(A \cup B)$$
$$P(A \text{ or } B)$$

$$\frac{7}{12} = .5833$$

$$\frac{1}{2} + \frac{1}{6} = \frac{2}{3} = .6666$$



or



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H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6

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The outcomes in red are the ones that meet our condition, and counting them would lead us to a correct answer:

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The outcomes in red are the ones that meet our condition, and counting them would lead us to a correct answer:

$$\frac{7}{12} = .5833$$

But if we added those two probabilities together, we would get a wrong answer! So why does this not work?

$$P(A \cup B)$$

$$P(A \text{ or } B)$$



or



Union Probability – Non-Mutually Exclusive

$$P(A \cup B)$$

$$P(A \text{ or } B)$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$P(\text{heads or six}) = \frac{1}{2} + \frac{1}{6} - \left(\frac{1}{2} * \frac{1}{6} \right) = \frac{7}{12}$$



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Union Probability – Non-Mutually Exclusive

H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6

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Union Probability – Non-Mutually Exclusive

H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6

The problem with adding non-mutually exclusive events is it causes double-counting of joint probability events as highlighted above.

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The problem with adding non-mutually exclusive events is it causes double-counting of joint probability events as highlighted above.

It's subtle but notice above that if we add the probability of "heads" (1/2) with the probability of getting a "six" (1/6), we have counted that "six" probability of 1/6 twice!

$P(A \cup B)$

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We can remedy this by subtracting the joint probability for non-mutually exclusive events:

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Union Probability – Non-Mutually Exclusive

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This is known as the **sum rule**, where we can find the OR probability between two or more events by summing them and then subtracting their joint probability.

$P(A \cup B)$

$P(A \text{ or } B)$



or



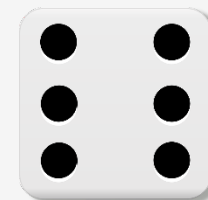
The Sum Rule

$$P(A \cup B)$$

$$P(A \text{ or } B)$$



or



The Sum Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A) * P(B)$$

$P(A \cup B)$

$P(A \text{ or } B)$



or



The Sum Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A) * P(B)$$

Notice that the sum rule applies to both mutually exclusive and non-mutually exclusive probabilities.

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$$P(A \text{ or } B)$$



or



The Sum Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A) * P(B)$$

Notice that the sum rule applies to both mutually exclusive and non-mutually exclusive probabilities.

When events A and B are mutually exclusive, the joint probability is 0 since they both cannot occur simultaneously.

$$P(A \cup B)$$
$$P(A \text{ or } B)$$



or



The Sum Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A) * P(B)$$

Notice that the sum rule applies to both mutually exclusive and non-mutually exclusive probabilities.

When events A and B are mutually exclusive, the joint probability is 0 since they both cannot occur simultaneously.

But when events A and B are not mutually exclusive, the joint probability plays a role in subtracting the double-counting caused by both events occurring.

$$P(A \cup B)$$
$$P(A \text{ or } B)$$



or



Exercise #2

There is a 30% chance of rain today, and a 40% chance your umbrella order will arrive on time.

You will only be able to run errands if it does not rain or your umbrella arrives.

What is the probability it will not rain **OR** your umbrella arrives?



Exercise #2

There is a 30% chance of rain today, and a 40% chance your umbrella order will arrive on time.

You will only be able to run errands if it does not rain or your umbrella arrives.

What is the probability it will not rain **OR** your umbrella arrives?

$$P(\text{rain}) = .30$$

$$P(\text{no rain}) = 1.0 - P(\text{rain}) = 1.0 - .30 = .70$$

$$P(\text{umbrella arrives}) = .40$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$P(\text{no rain or umbrella arrives}) = .70 + .40 - (.70 * .40) = .82$$



Katacoda Interactive Scenario – Probability Logic



Probability Logic

Step 1 of 5 ▶

```
import random

heads = 0

for i in range(0,100):
    if random.uniform(0,1) <=
.6:
    heads += 1

print("# HEADS:
{}/100".format(heads))
```

python3 probability_math.py
✓

probability_math.py

```
1 def odds_to_prob(odds):
2     return odds / (1.0 + odds)
3
4 defective_odds = 1.5
5 defective_probability = odds_to_prob(defective_odds)
6 print(defective_probability)
7
```

Terminal

```
Introduction
$ cd /home/scrapbook/tutorial
$ python3 probability_math.py
0.6
$
```

<https://learning.oreilly.com/scenarios/probability-from-scratch/9781492080541/>

Conditional Probability



$$P(A \mid B)$$

$$P(A \text{ given } B)$$

Conditional Probability

Another nuanced idea in probability is **conditional probability**, or the probability of A given B has occurred.



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

Another nuanced idea in probability is **conditional probability**, or the probability of A given B has occurred.

If B has no impact on whether A occurs, then $P(A) = P(A \text{ given } B)$.



$$P(A | B)$$

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Conditional Probability

Another nuanced idea in probability is **conditional probability**, or the probability of A given B has occurred.

If B has no impact on whether A occurs, then $P(A) = P(A \text{ given } B)$.

But if B does increase or decrease the probability of A occurring, then the $P(A)$ is going to be different than $P(A \text{ given } B)$.



$P(A | B)$

$P(A \text{ given } B)$

Conditional Probability

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The probability of someone being colorblind is 4.25%, but the probability of a male being colorblind is 8%.



$$P(A | B)$$

$$P(A \text{ given } B)$$

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The probability of someone being colorblind is 4.25%, but the probability of a male being colorblind is 8%.

$$P(\text{colorblind}) = .0425$$



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$$P(\text{colorblind}) = .0425$$

$$P(\text{colorblind given male}) = .08$$



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

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The probability of someone being colorblind is 4.25%, but the probability of a male being colorblind is 8%.

$$P(\text{colorblind}) = .0425$$

$$P(\text{colorblind given male}) = .08$$

Does this mean any colorblind person is 8% likely to be male? Or that any male is 8% likely to be colorblind?



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability



$$P(\text{male}) = .5$$

$$P(\text{colorblind}) = .0425$$

$$P(\text{colorblind given male}) = .08$$

$$P(\text{male and colorblind}) = P(\text{male}) \quad * \quad ?$$

$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

Direction of the conditional probability matters! The *probability of a male given they are colorblind* is not the same as the *probability of being colorblind given they are male*!



$$P(\text{male}) = .5$$

$$P(\text{colorblind}) = .0425$$

$$P(\text{colorblind given male}) = .08$$

$$P(\text{male and colorblind}) = P(\text{male}) \quad * \quad ?$$

$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

Direction of the conditional probability matters! The *probability of a male given they are colorblind* is not the same as the *probability of being colorblind given they are male*!

We will learn how to flip this probability with Bayes Theorem shortly, but first let's see why conditional probability plays a critical role in **AND** and **OR** probabilities.

$$P(\text{male}) = .5$$

$$P(\text{colorblind}) = .0425$$

$$P(\text{colorblind given male}) = .08$$

$$P(\text{male and colorblind}) = P(\text{male}) \quad * \quad ?$$



$$P(A | B)$$

$$P(A \text{ given } B)$$

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Direction of the conditional probability matters! The *probability of a male given they are colorblind* is not the same as the *probability of being colorblind given they are male*!

We will learn how to flip this probability with Bayes Theorem shortly, but first let's see why conditional probability plays a critical role in **AND** and **OR** probabilities.

Let's say we draw a random person from the population, and the probability of them being male $P(\text{male})$ is 50%.

$$P(\text{male}) = .5$$

$$P(\text{colorblind}) = .0425$$

$$P(\text{colorblind given male}) = .08$$

$$P(\text{male and colorblind}) = P(\text{male}) \quad * \quad ?$$



$$P(A | B)$$

$$P(A \text{ given } B)$$

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Direction of the conditional probability matters! The *probability of a male given they are colorblind* is not the same as the *probability of being colorblind given they are male*!

We will learn how to flip this probability with Bayes Theorem shortly, but first let's see why conditional probability plays a critical role in **AND** and **OR** probabilities.

Let's say we draw a random person from the population, and the probability of them being male $P(\text{male})$ is 50%.

$P(\text{male}) = .5$
If we want to calculate the probability they are male and colorblind, do we multiply $P(\text{male})$ with $P(\text{colorblind})$ or

$P(\text{male and colorblind}) = P(\text{male}) * ?$



$P(A | B)$

$P(A \text{ given } B)$

Conditional Probability



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

With joint probability we use the conditional probability when it is available! If we want to calculate the probability of someone being male and colorblind, use the conditional probability that accounts for them being male!



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

$$P(\text{male}) = .5$$

$$~~P(\text{colorblind}) = .0425~~$$

$$P(\text{colorblind given male}) = .08$$

$$P(\text{male and colorblind}) = P(\text{male}) * P(\text{colorblind given male})$$

$$P(\text{male and colorblind}) = .5 * .08$$

$$P(\text{male and colorblind}) = .04$$

With joint probability we use the conditional probability when it is available! If we want to calculate the probability of someone being male and colorblind, use the conditional probability that accounts for them being male!

If we do not use the conditional probability, then the probability of them being male and colorblind will be no different than being female and colorblind!



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

$$P(A \text{ or } B) = P(A) + P(B) - P(A) * P(B \text{ given } A)$$



$P(A | B)$

$P(A \text{ given } B)$

Conditional Probability

We can update our joint probability and union probability formulas to account for conditional probability:

$$P(A \text{ or } B) = P(A) + P(B) - P(A) * P(B \text{ given } A)$$



$$P(A | B)$$

$$P(A \text{ given } B)$$

Conditional Probability

We can update our joint probability and union probability formulas to account for conditional probability:

$$P(A \text{ and } B) = P(A)P(B \text{ given } A) = P(A) + P(B) - P(A) * P(B \text{ given } A)$$



$P(A | B)$

$P(A \text{ given } B)$

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Remember that if A and B are independent and have no impact on each other, then these formulas still work because $P(B) = P(B \text{ given } A)$.



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It can be hard to determine if two events are conditional and related, and therefore it is common to assume in statistics they are independent.



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It can be hard to determine if two events are conditional and related, and therefore it is common to assume in statistics they are independent.

We will see this assumption in machine learning applications like Naïve Bayes.



$P(A | B)$

$P(A \text{ given } B)$

Exercise

There is a 30% chance of rain today, and a 40% chance your umbrella order will arrive on time.

However, you found out if it rains there is only 20% chance your umbrella will arrive on time.

What is the probability it will rain and your umbrella will arrive on time?



Exercise

There is a 30% chance of rain today, and a 40% chance your umbrella order will arrive on time.

However, you found out if it rains there is only 20% chance your umbrella will arrive on time.

What is the probability it will rain and your umbrella will arrive on time?

$$P(A \text{ and } B) = P(A) * P(B \text{ given } A)$$

$$P(\text{rain and umbrella arrives}) = .30 * .20 = .06$$

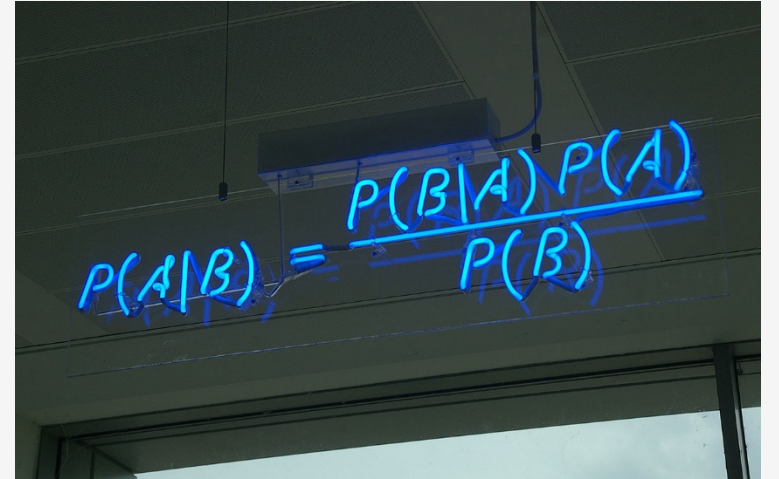
6% chance it will rain and your umbrella will arrive.



Bayes Theorem

It is worth bringing up Bayes Theorem it allows us to flip a conditional probability, turning $P(A \text{ given } B)$ into $P(B \text{ given } A)$.

$$P(A \text{ given } B) = \frac{P(B \text{ given } A) * P(A)}{P(B)}$$



A photograph of a whiteboard with the Bayes Theorem formula written in blue marker. The formula is $P(A|B) = \frac{P(B|A) P(A)}{P(B)}$. The whiteboard is mounted on a wall, and the lighting is somewhat dim, with the blue marker providing a strong contrast.

Bayes Theorem

Let's look at the color blindness problem again. We have these data points:

$$P(\text{color blind}) = .0425$$

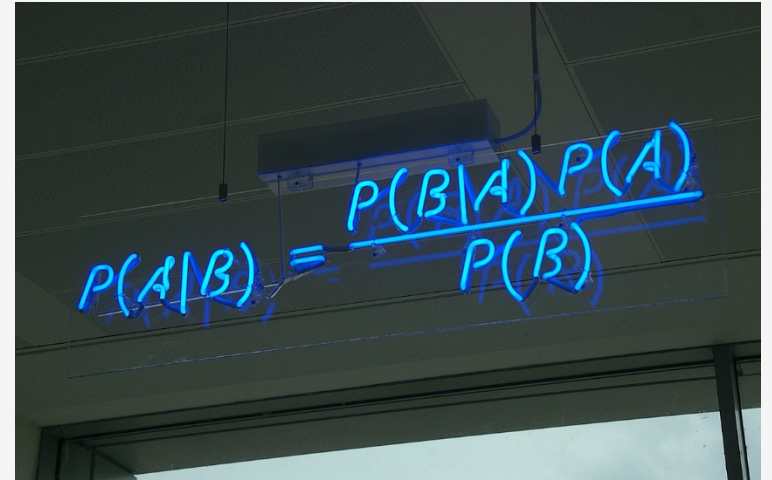
$$P(\text{male}) = .50$$

$$P(\text{color blind given male}) = .08$$

What is the probability of a *person being male given they are color blind*?

$$P(A \text{ given } B) = \frac{P(B \text{ given } A) * P(A)}{P(B)}$$

$$P(\text{male given color blind}) = \frac{.08 * .5}{.0425} = .9411$$

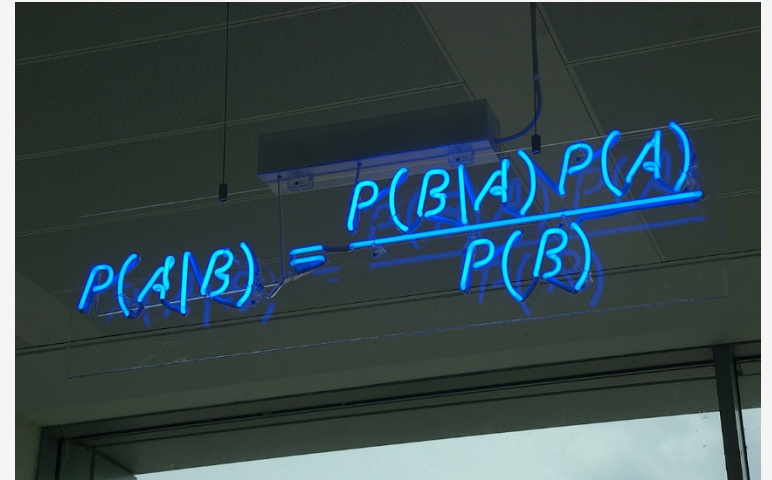


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Bayes Theorem

The probability of a person *being male given they are color blind* is 94.11%, which is much different than the probability of *being color blind given they are a male*, which is 8%.

Bayes Theorem is the core of conditional probability, as it allows us to flip inferences based on conditions.


$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Section IV

Discovering the Binomial and Beta Distribution

Discussion: Measuring Uncertainty



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Discussion: Measuring Uncertainty

You and an engineer colleague are testing a system to figure out its rate of failure, and you want at least 90% success rate.



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Discussion: Measuring Uncertainty

You and an engineer colleague are testing a system to figure out its rate of failure, and you want at least 90% success rate.

Each test is expensive and time-consuming, but you were able to complete 10 tests.



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Discussion: Measuring Uncertainty

You and an engineer colleague are testing a system to figure out its rate of failure, and you want at least 90% success rate.

Each test is expensive and time-consuming, but you were able to complete 10 tests.

8/10 of the tests succeeded but you are dissatisfied that just 10 tests already yielded two failures, and thus conclude an 80% failure rate.



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Discussion: Measuring Uncertainty

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✓✓✓✓✓✗✓✗✓✓

TEST OUTCOME:

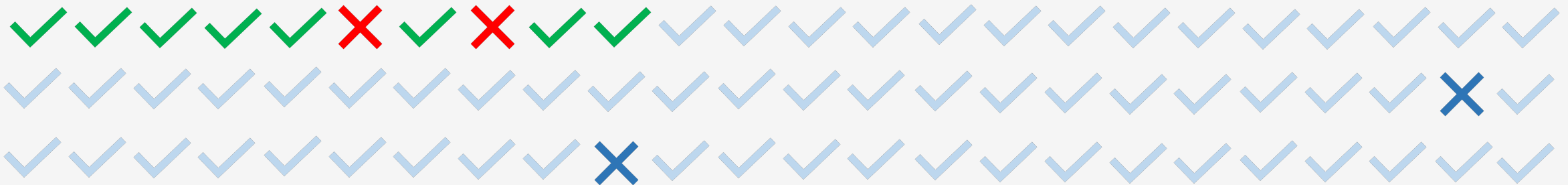


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Discussion: Measuring Uncertainty



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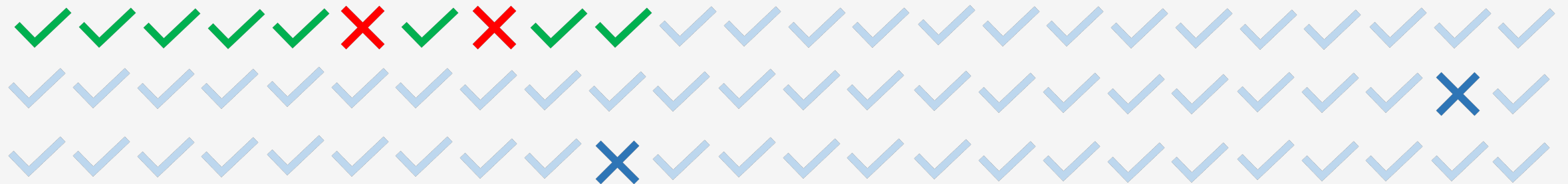


Discussion: Measuring Uncertainty

You propose going back to the drawing board, but the engineer wants to run more tests arguing this could be a fluke. The only way we will know for sure is to run more tests.



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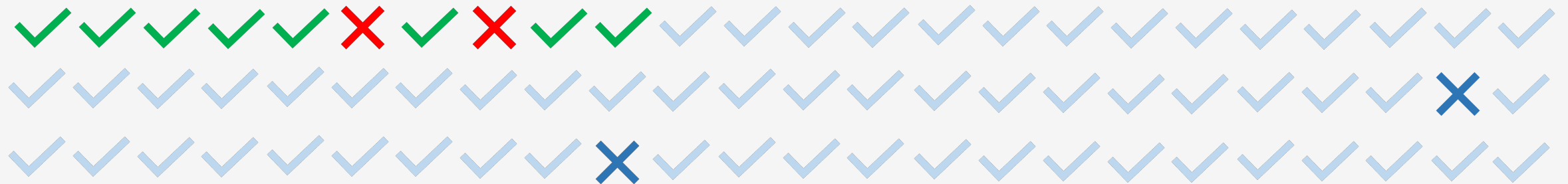
Discussion: Measuring Uncertainty

You propose going back to the drawing board, but the engineer wants to run more tests arguing this could be a fluke. The only way we will know for sure is to run more tests.

He argues "what if more tests might yield 90% or greater success? Consider the possible scenario below:"



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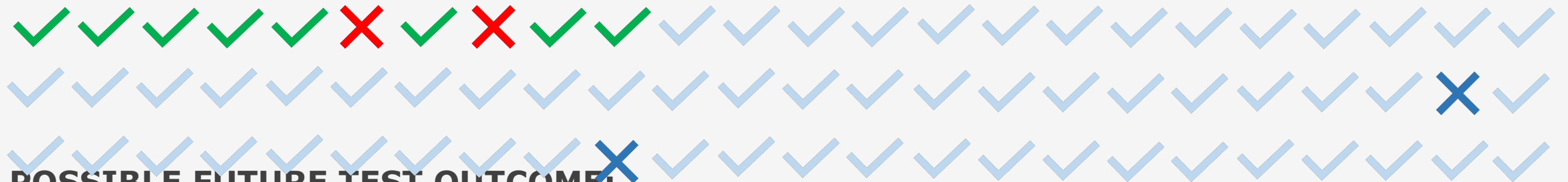
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POSSIBLE FUTURE TEST OUTCOME:

Discussion: Measuring Uncertainty



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Discussion: Measuring Uncertainty

You *really* do not want to do more tests, but you consider the engineer's argument. After all, if you flip a coin 10 times and get 8 heads, it does not mean the coin is fixed at 80%.



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DISCUSSION



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Discussion: Measuring Uncertainty

You *really* do not want to do more tests, but you consider the engineer's argument. After all, if you flip a coin 10 times and get 8 heads, it does not mean the coin is fixed at 80%.

DISCUSSION

Is the success rate really 80%?



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Why or why not?



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Discussion: Measuring Uncertainty



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Getting more data is always helpful, but sometimes expense and availability prevents us from doing so.



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In situations where data is limited, a Bayesian approach can be advantageous over a Frequentist one.



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Discussion: Measuring Uncertainty

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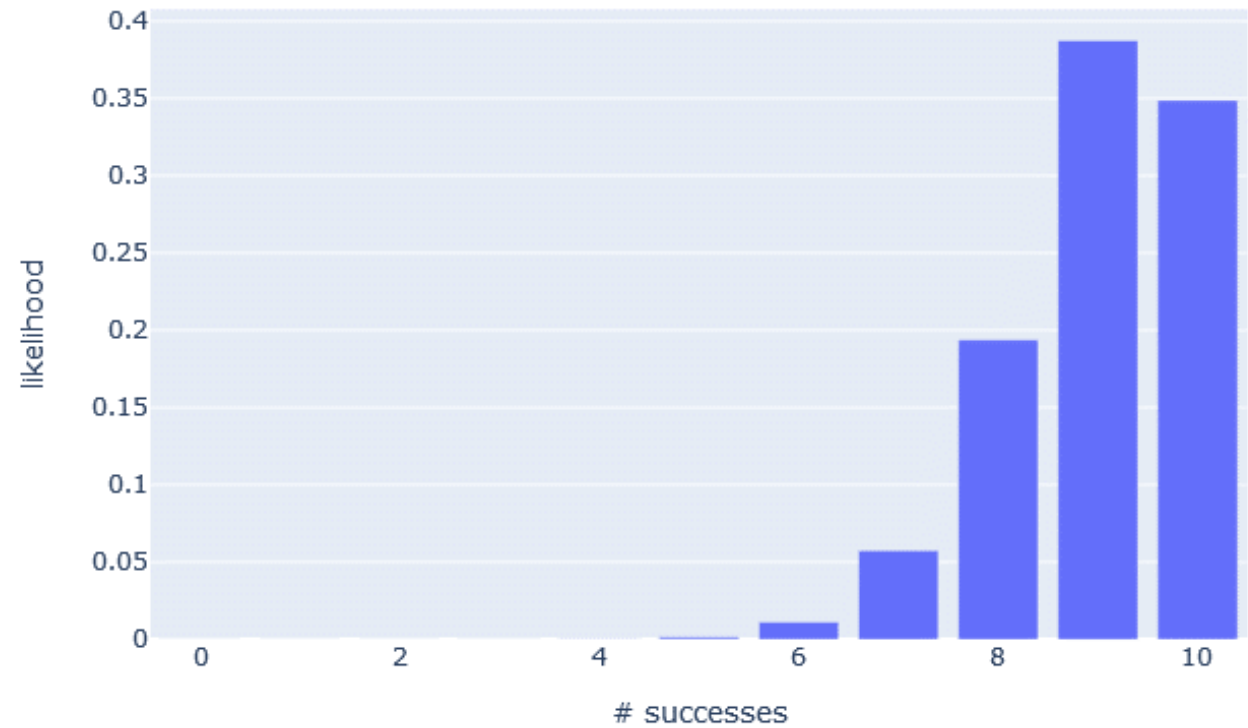
In situations where data is limited, a Bayesian approach can be advantageous over a Frequentist one.

Think about the probabilities of probabilities... if an event inherently has a 90% probability of occurring, what is the likelihood I would get 8/10 successes?



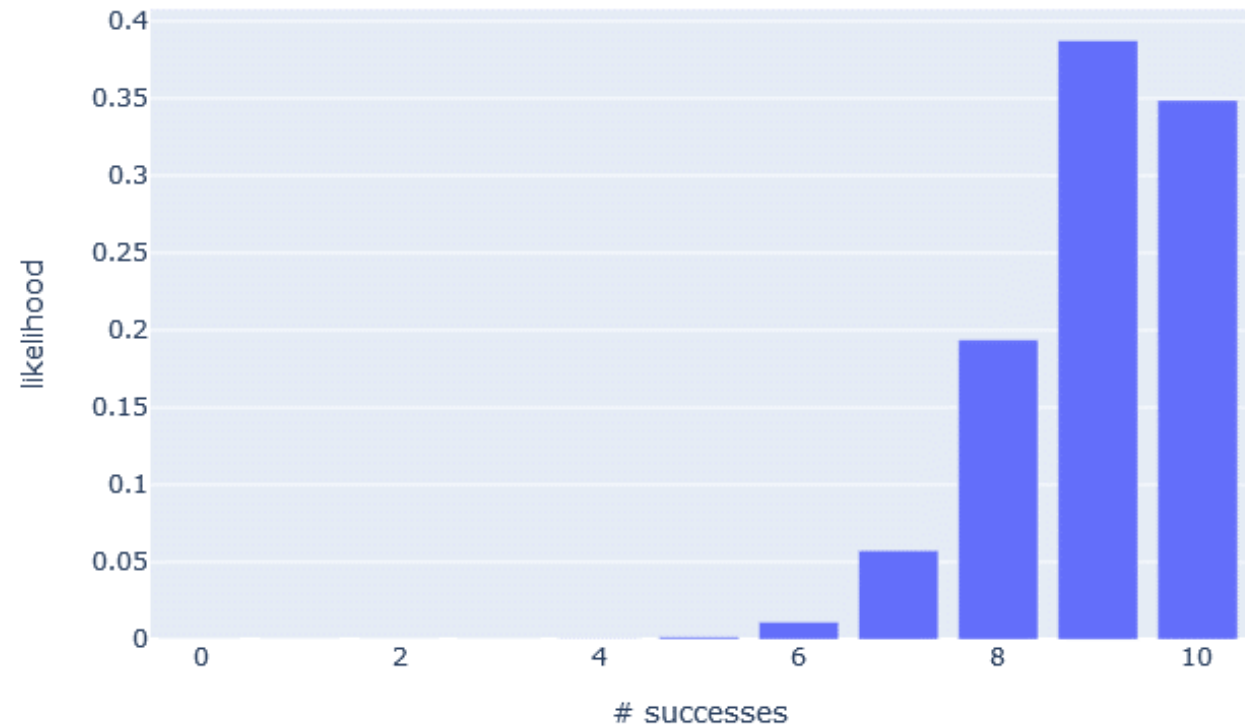
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The Binomial Distribution



The Binomial Distribution

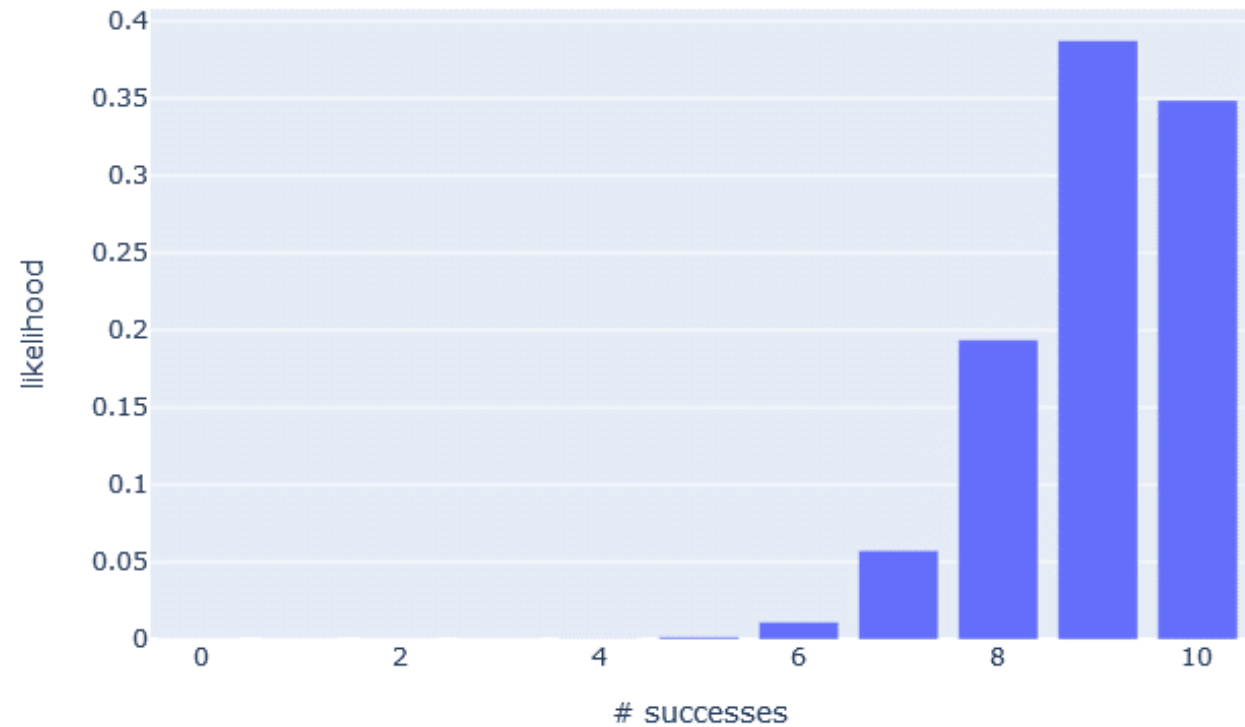
Let's first consider the **binomial distribution** to the right, which shows the y *likelihood* of x *successes* when the probability of success is 90% and there are 10 trials.



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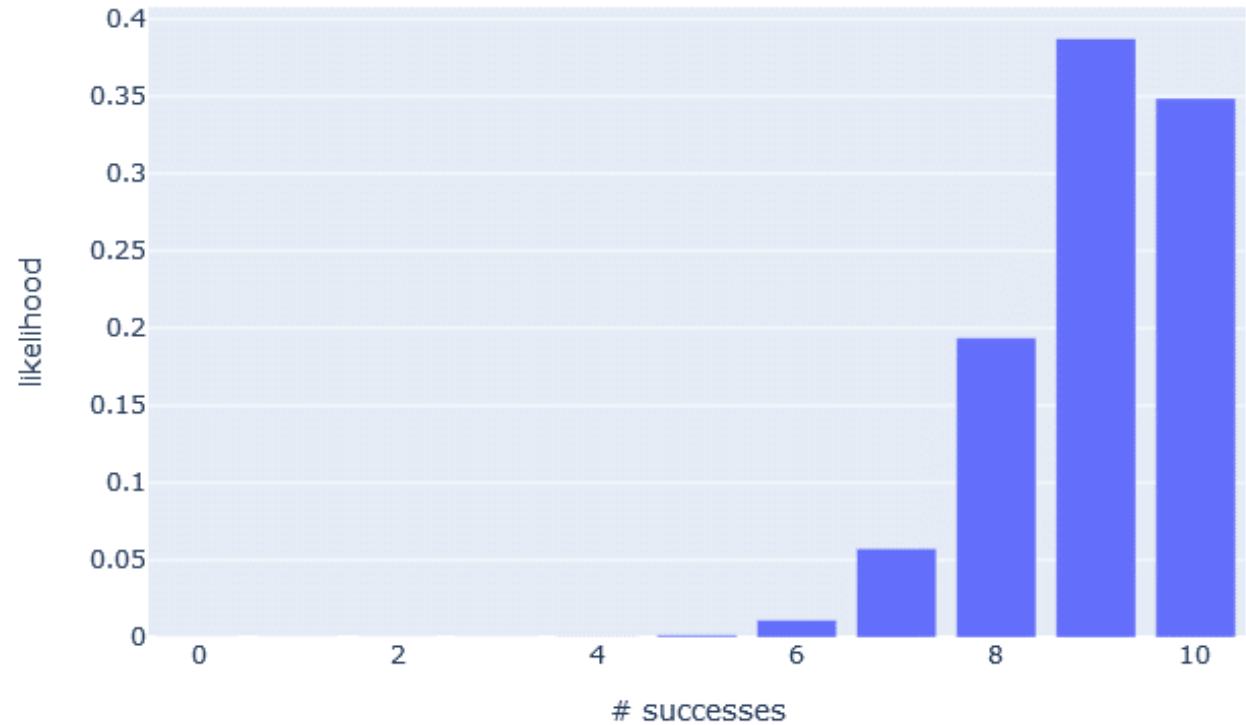


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If the probability of success is 90%, there is a 19.37% probability I will get 8 successes out of 10 trials.



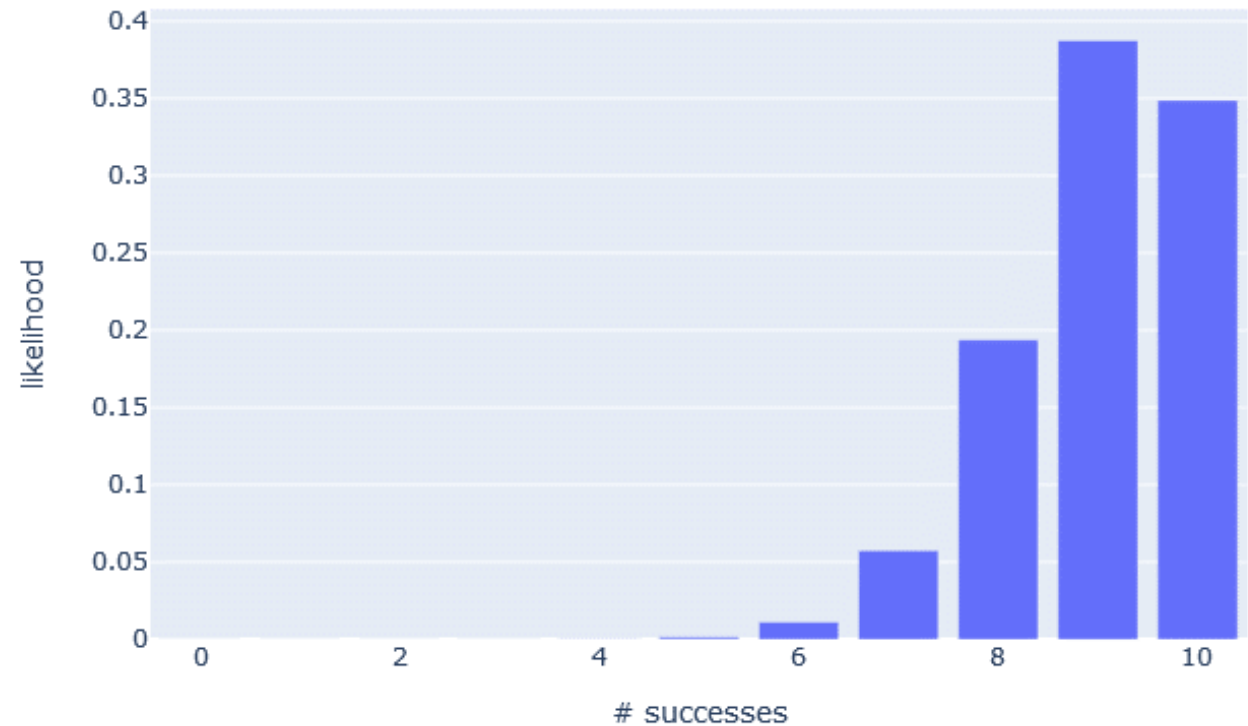
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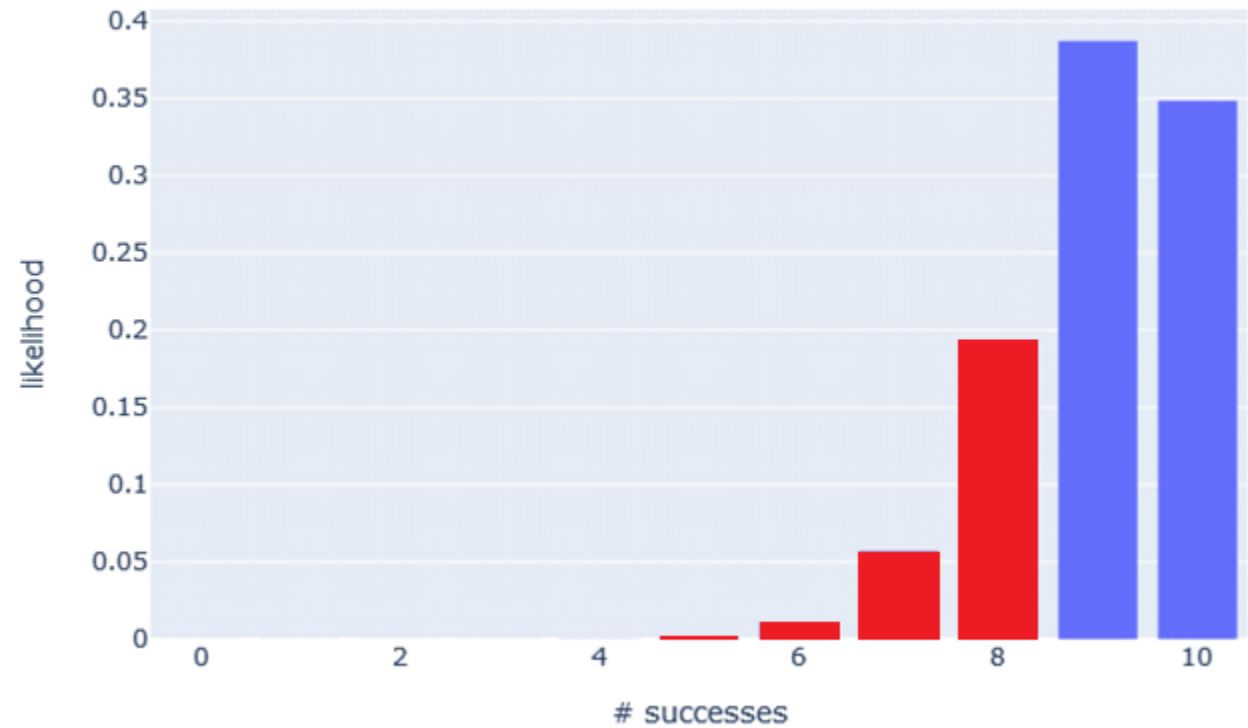
Each bar represents the probability of getting x successes.

If the probability of success is 90%, there is a 19.37% probability I will get 8 successes out of 10 trials.

Notice that all the outcomes add up to 1.0, or 100%, so this is a **probability distribution**.

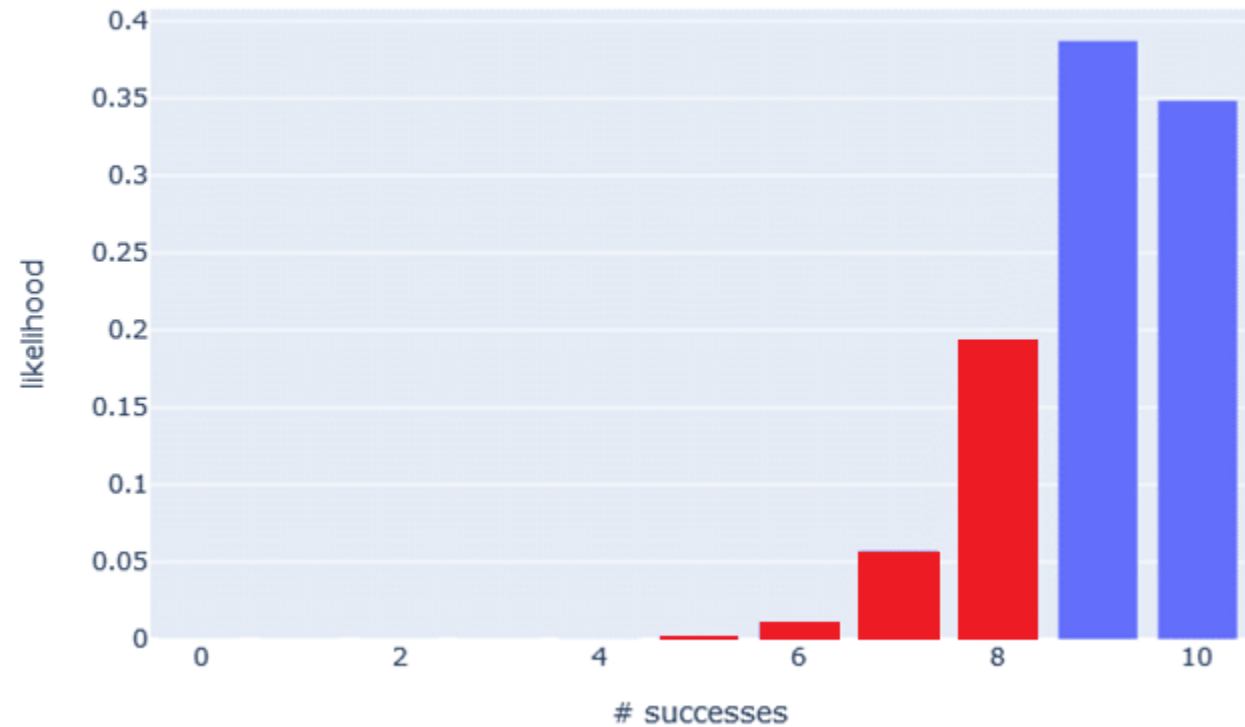


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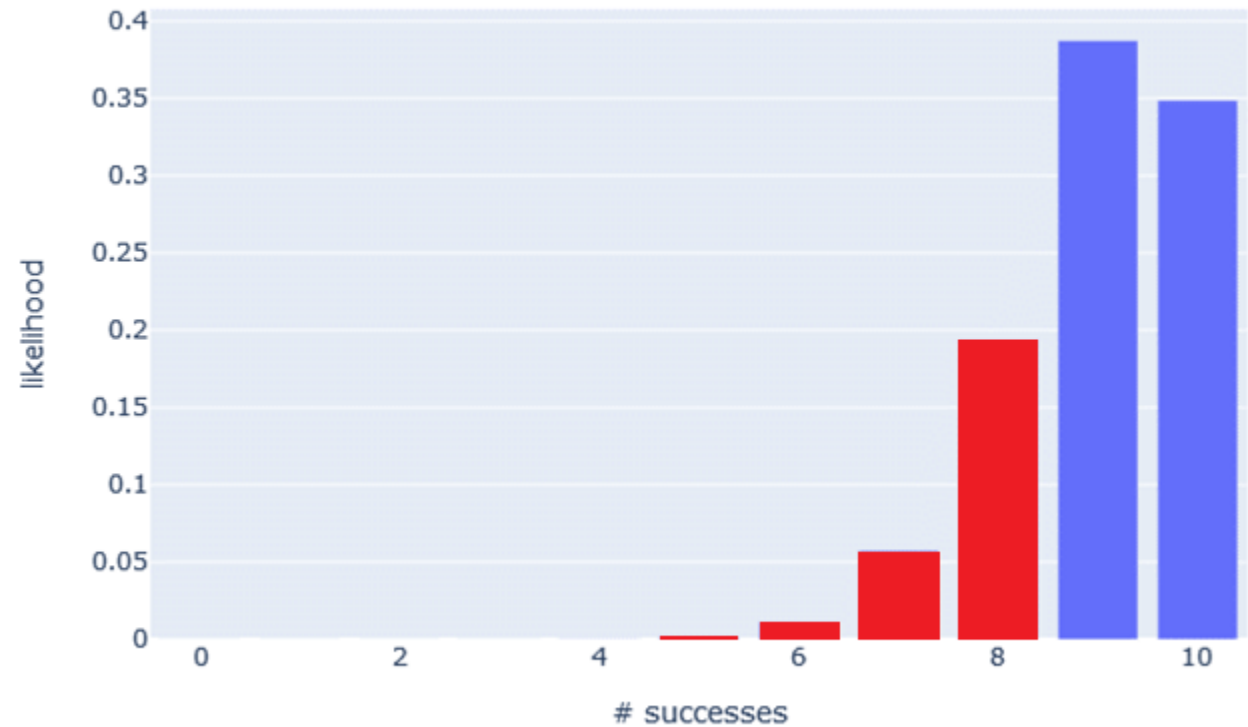
How would I calculate the probability of 8 or less successes?



The Binomial Distribution

How would I calculate the probability of 8 or less successes?

Sum the probabilities (the bars in red) and you will find the probability of 8 or less successes to be 26.39%

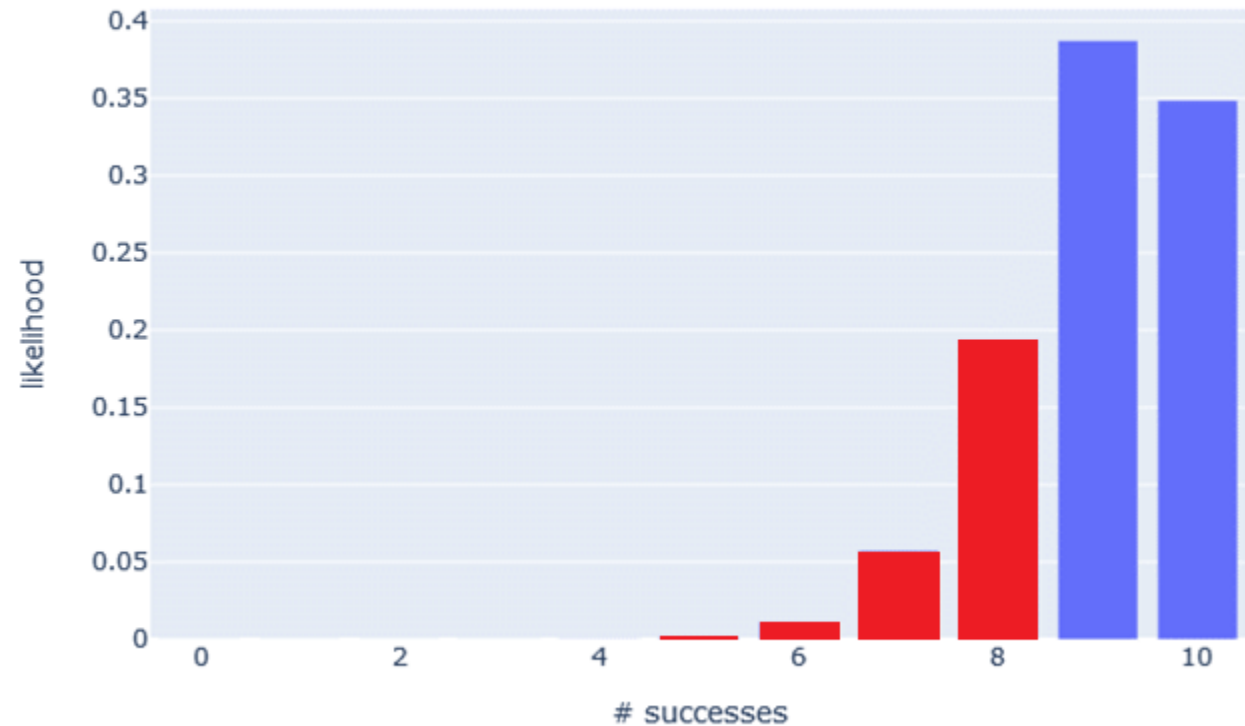


The Binomial Distribution

How would I calculate the probability of 8 or less successes?

Sum the probabilities (the bars in red) and you will find the probability of 8 or less successes to be 26.39%

If we are interested in having 90% or greater success, then the probability of seeing less than 90% success is 26.39% **EVEN IF** each outcome has a 90% success rate.



Binomial Distribution in Python

```
from scipy.stats import binom
```

```
n = 10
```

```
p = 0.9
```

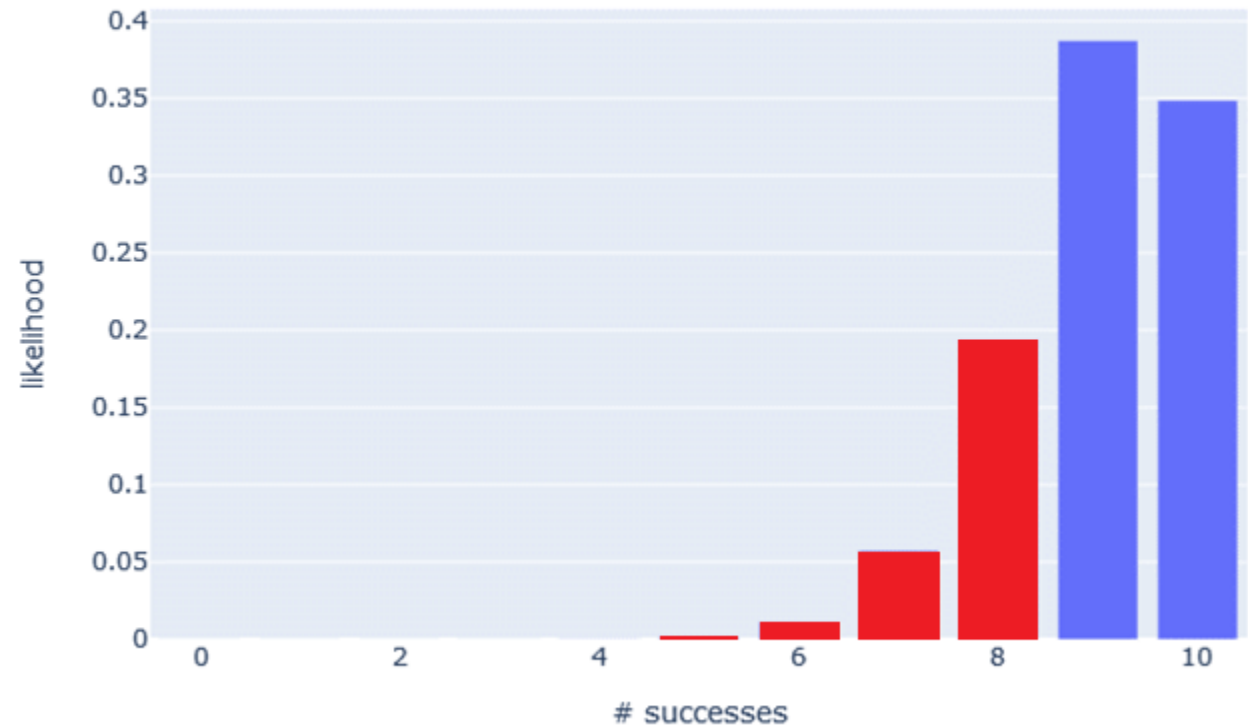
```
for x in range(n + 1):
```

```
    probability = binom.pmf(x, n, p)
```

```
    print("{0} - {1}".format(x, probability))
```

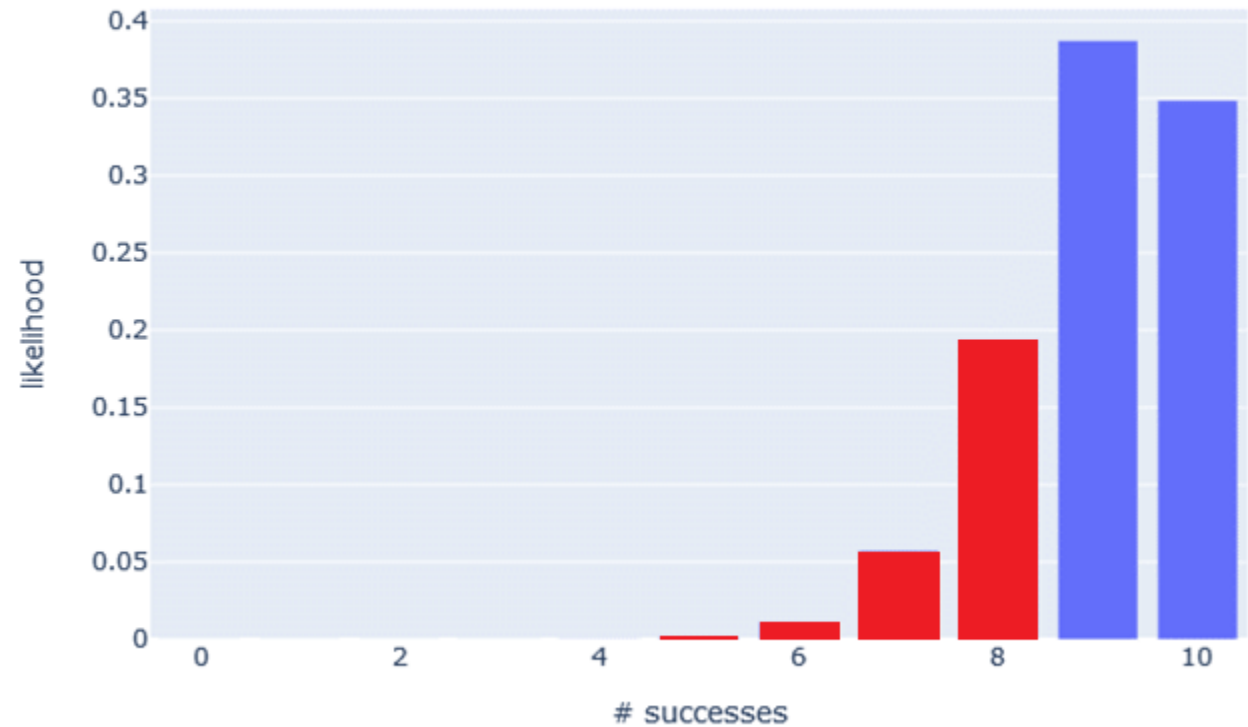
```
0 - 9.9999999999999996e-11
1 - 8.9999999999999996e-09
2 - 3.6449999999999996e-07
3 - 8.7480000000000003e-06
4 - 0.00013778099999999999
5 - 0.00148803479999999988
6 - 0.011160260999999996
7 - 0.057395628000000001
8 - 0.19371024449999993
9 - 0.387420489000000037
10 - 0.348678440100000004
```

The Binomial Distribution



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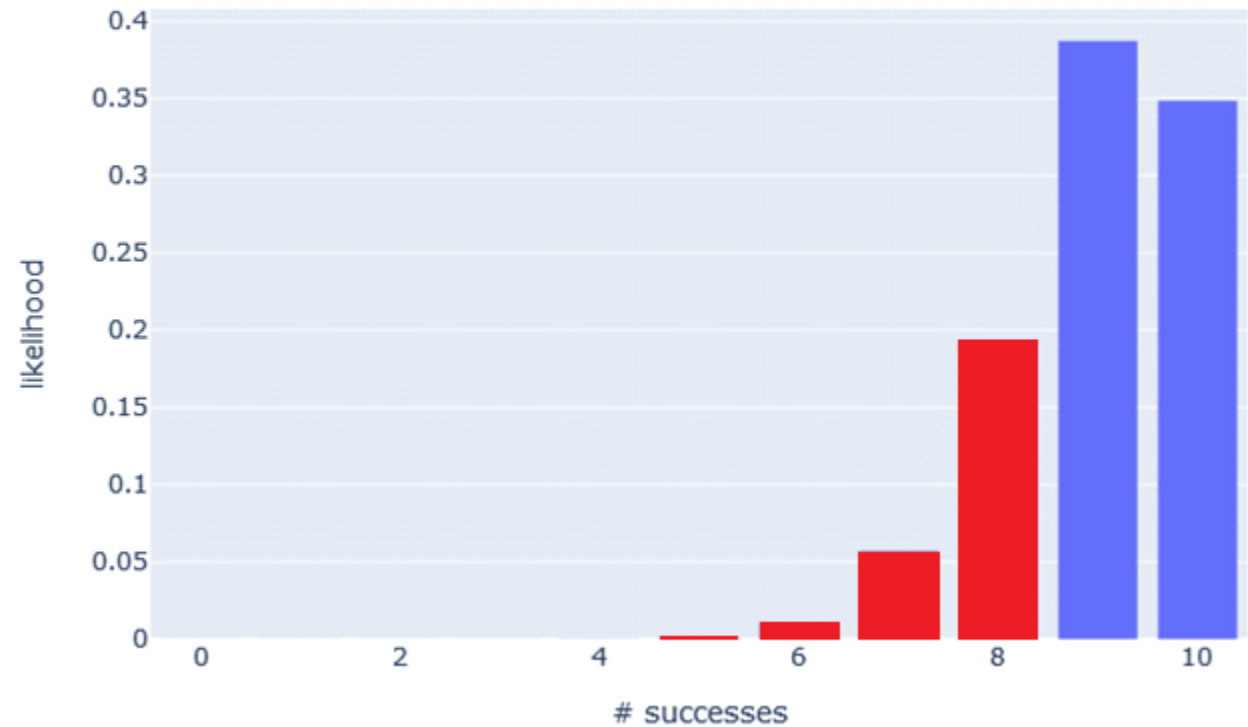
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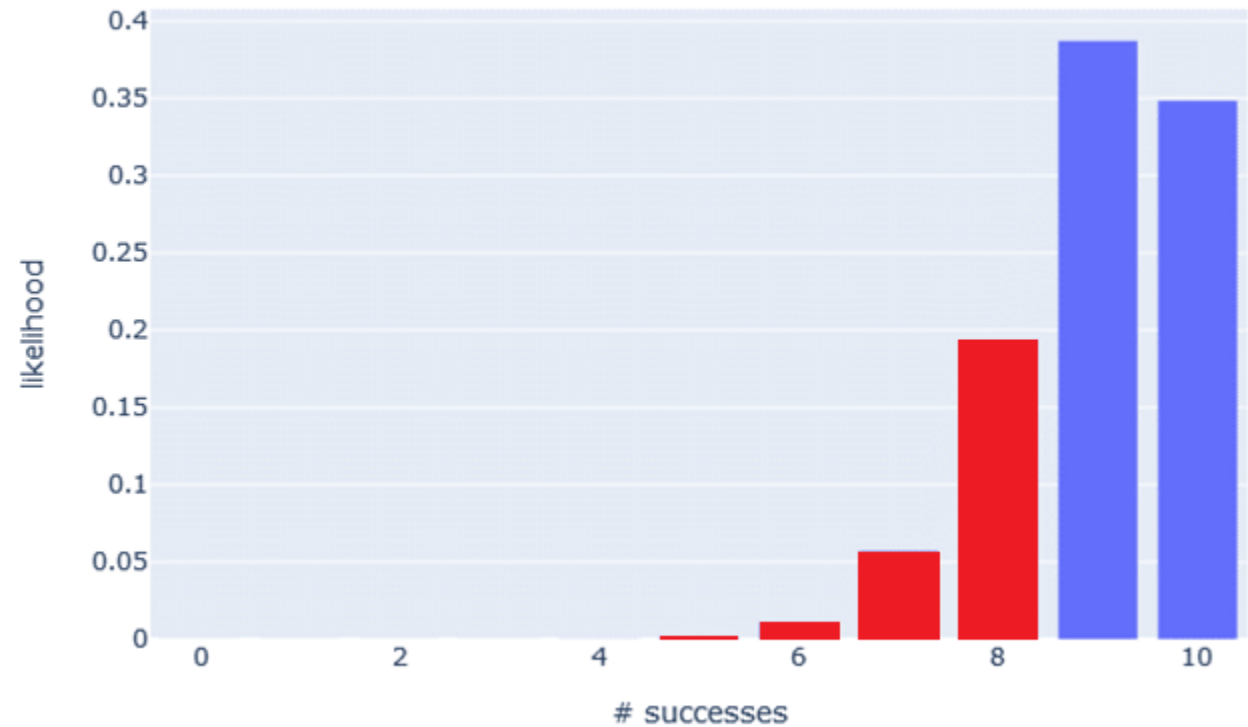
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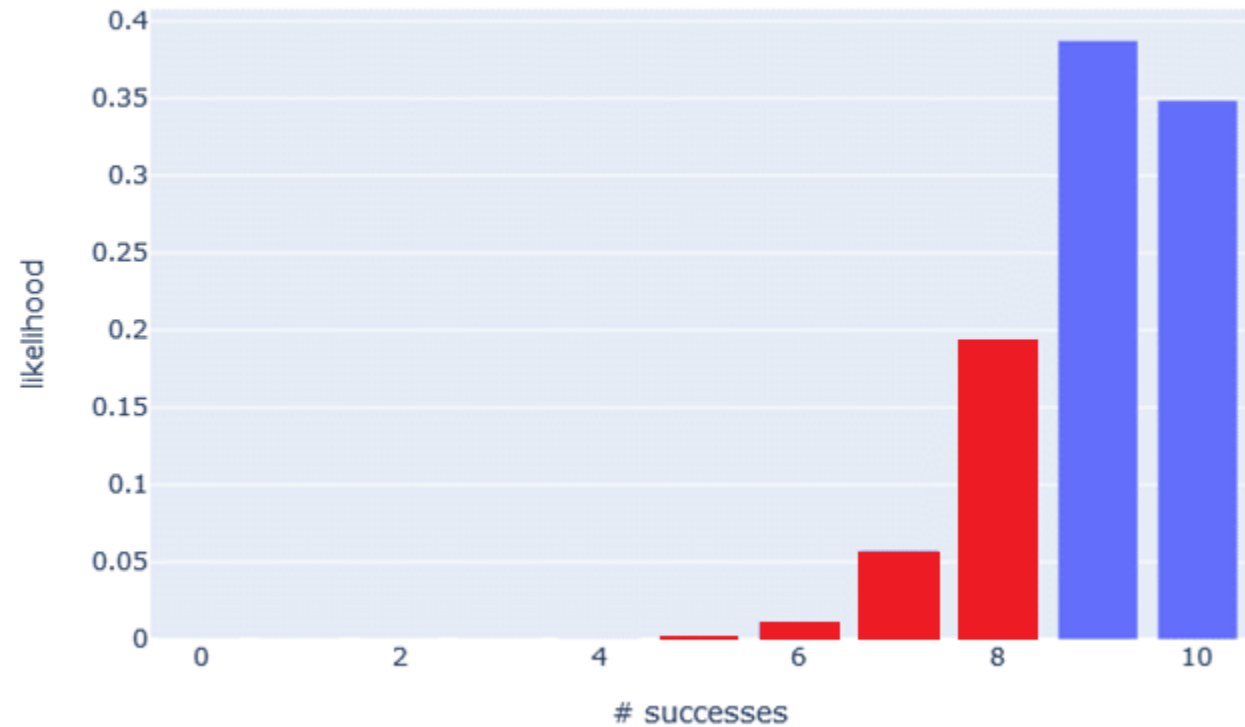


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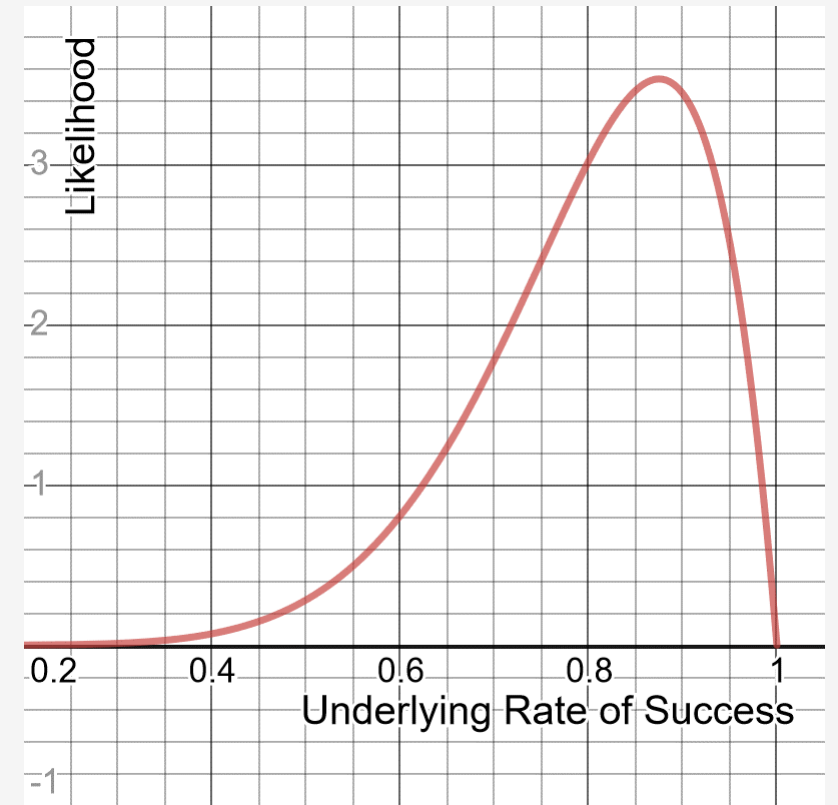
So there is a 26.39% chance the engineer is correct, **ASSUMING** each success has a 90% success rate.

This is informative showing it's 26.39% possible to see 8 or less successes with an underlying 90% probability.

But notice that our model **ASSUMED** the probability of success is 90%. What if we flipped the question and explored other probabilities besides 90%, and how likely each would produce 8/10 successes?



The Beta Distribution

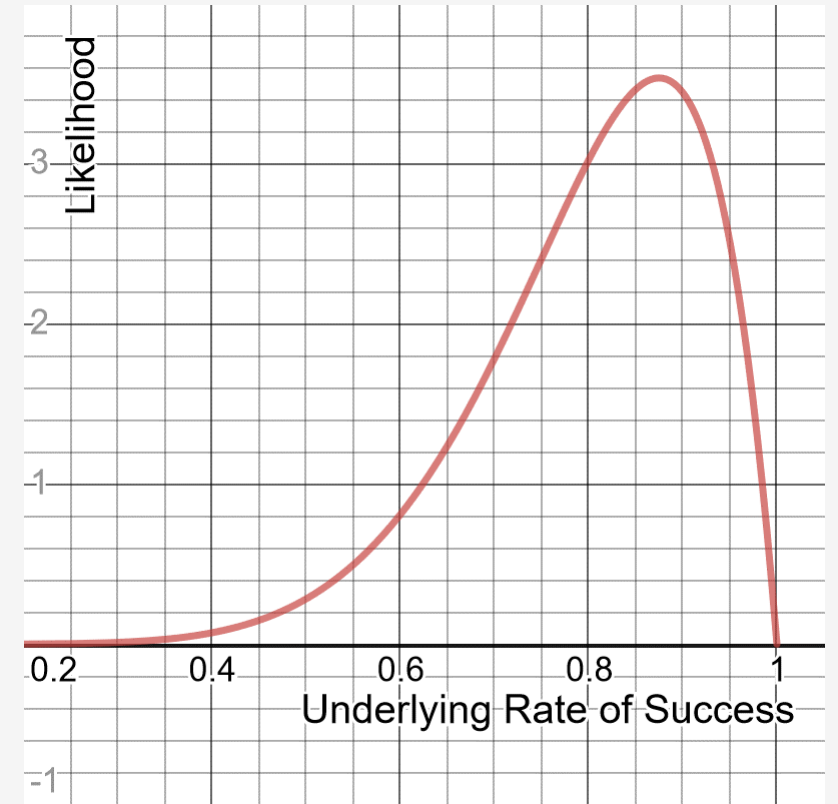


<https://www.desmos.com/calculator/zyranflxo4>



The Beta Distribution

Here's another way to look at this: consider all underlying rates of success and how likely each is to produce 8/10 successes.

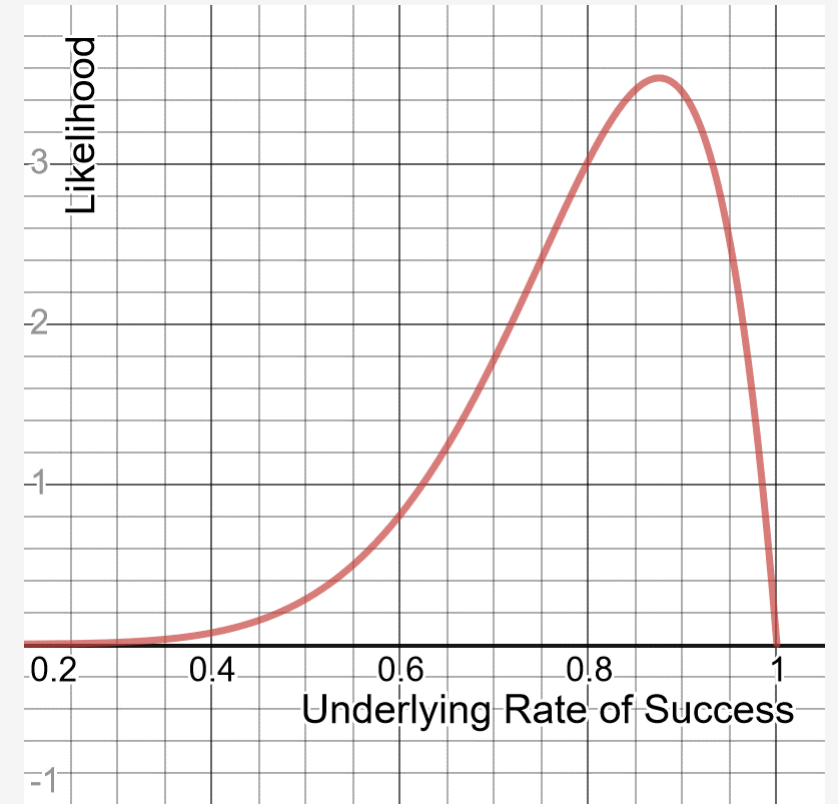


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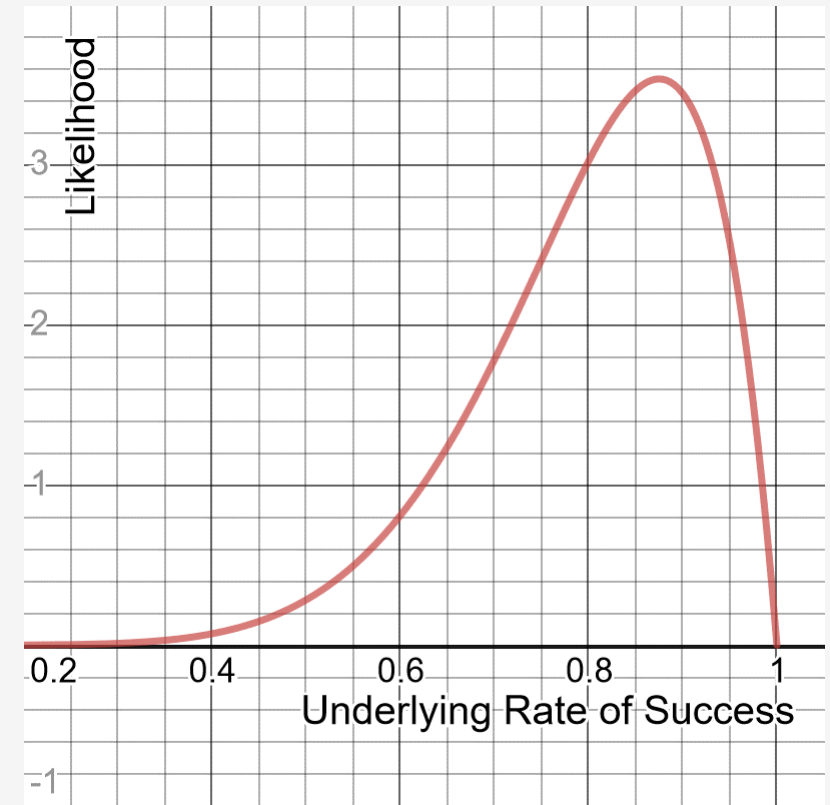
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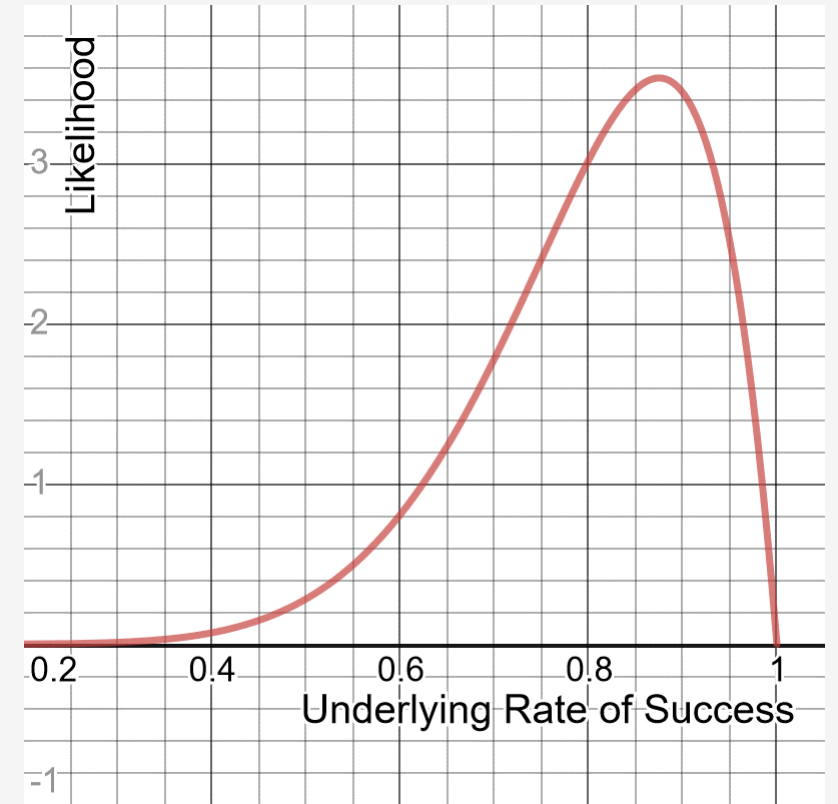
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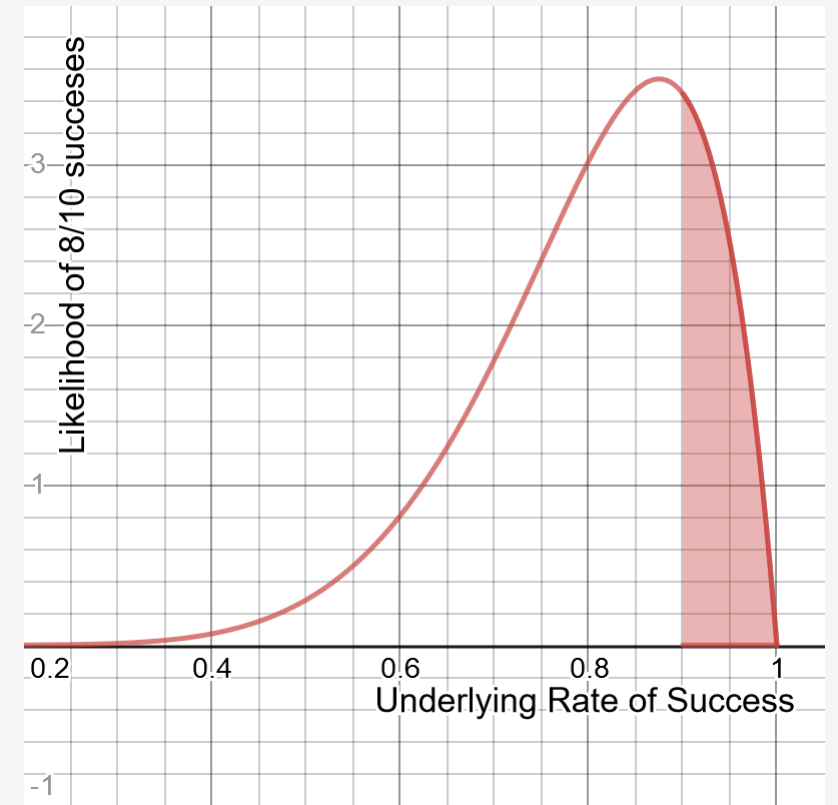
To the right shows the range of likelihoods for each actual rate of success, when 10 trials yields 8 successes.

✓✓✓✓✓✗✓✗✓✓ The beta distribution accepts two arguments, the **alpha** which is the number of successes, and the **beta** which is the number of failures.



<https://www.desmos.com/calculator/zyranflxo4>

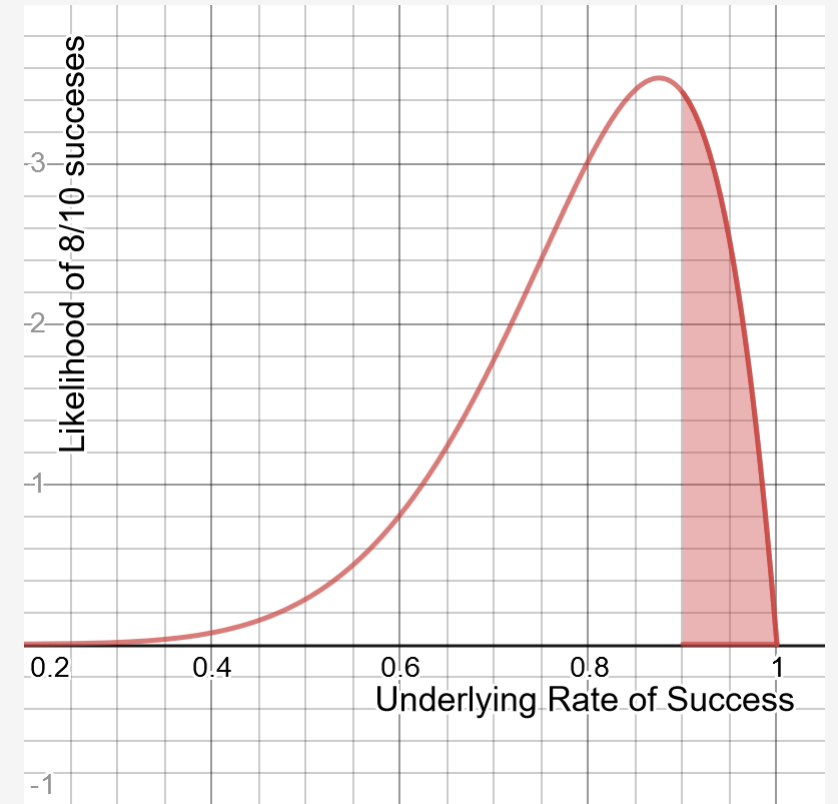
The Beta Distribution



<https://www.desmos.com/calculator/cakj42wlie>

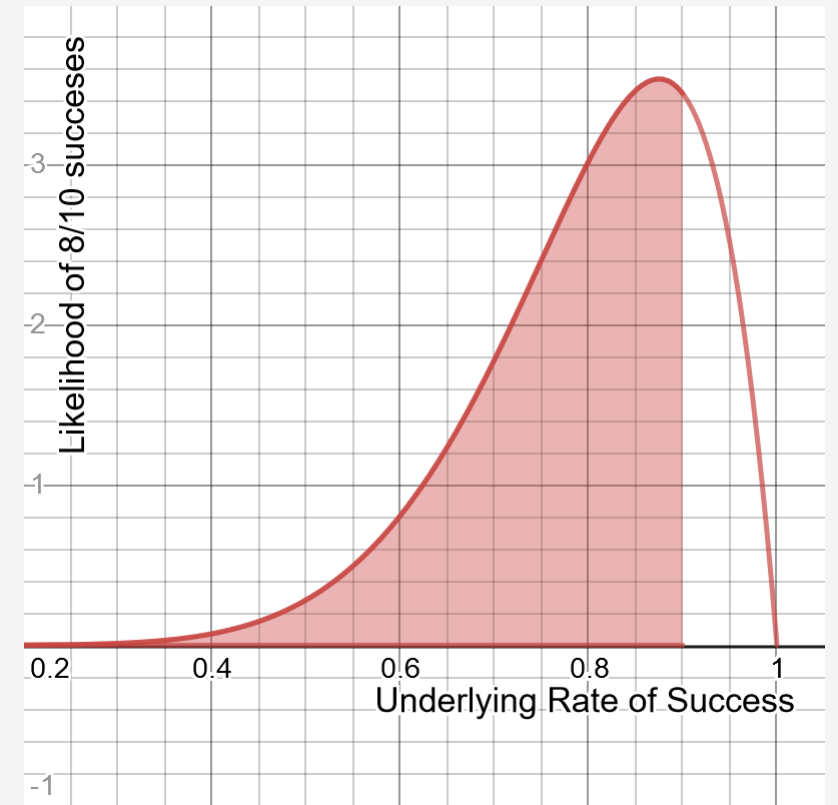
The Beta Distribution

Given 8/10 successes, the probability that the underlying rate of success is 90% or higher is 22.5%.



<https://www.desmos.com/calculator/cakj42wlie>

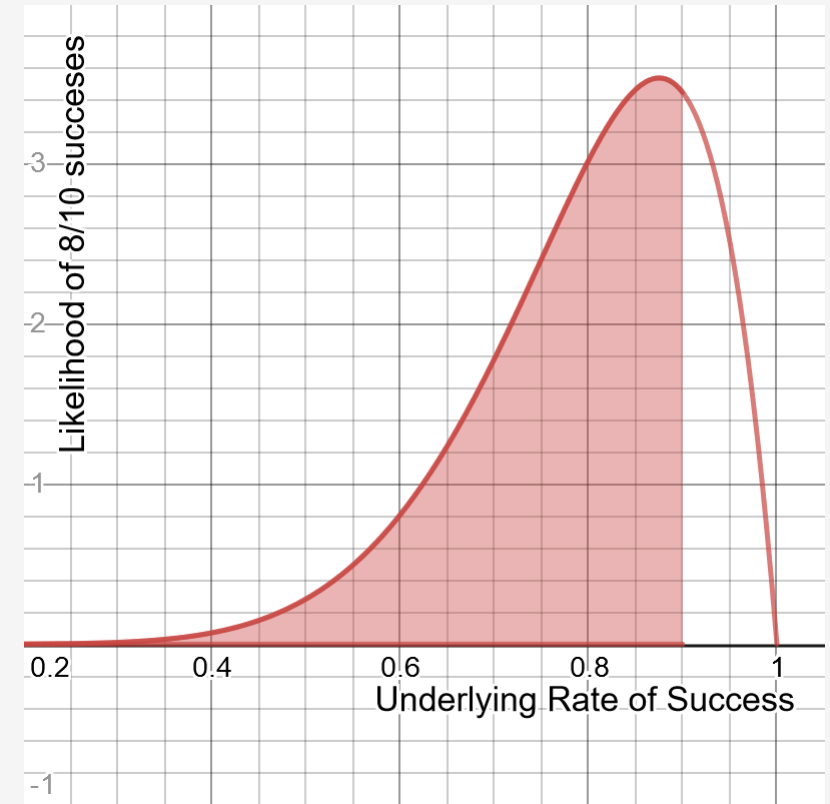
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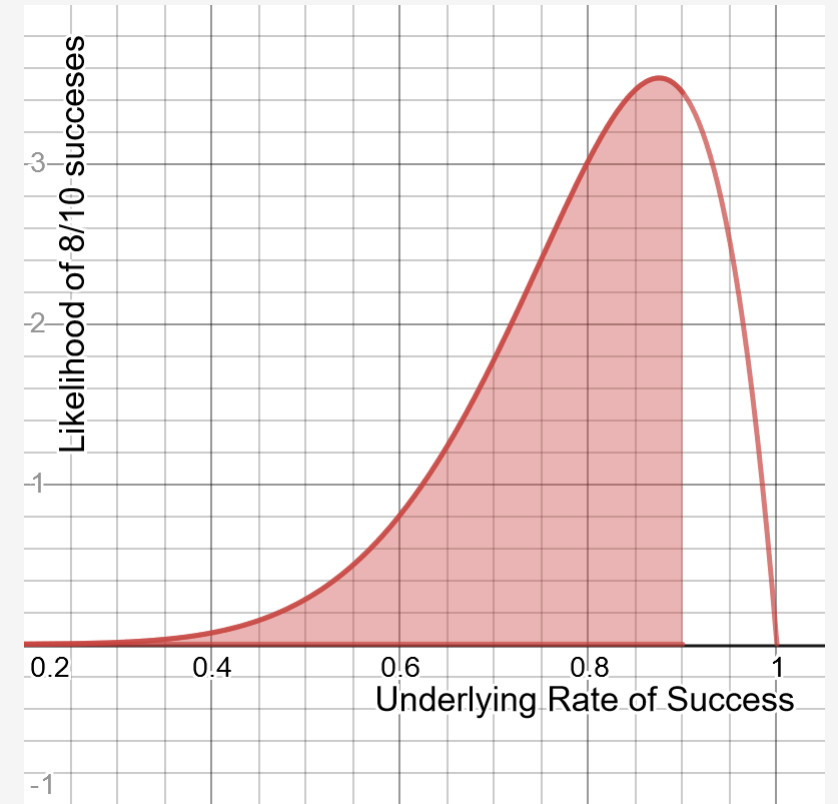


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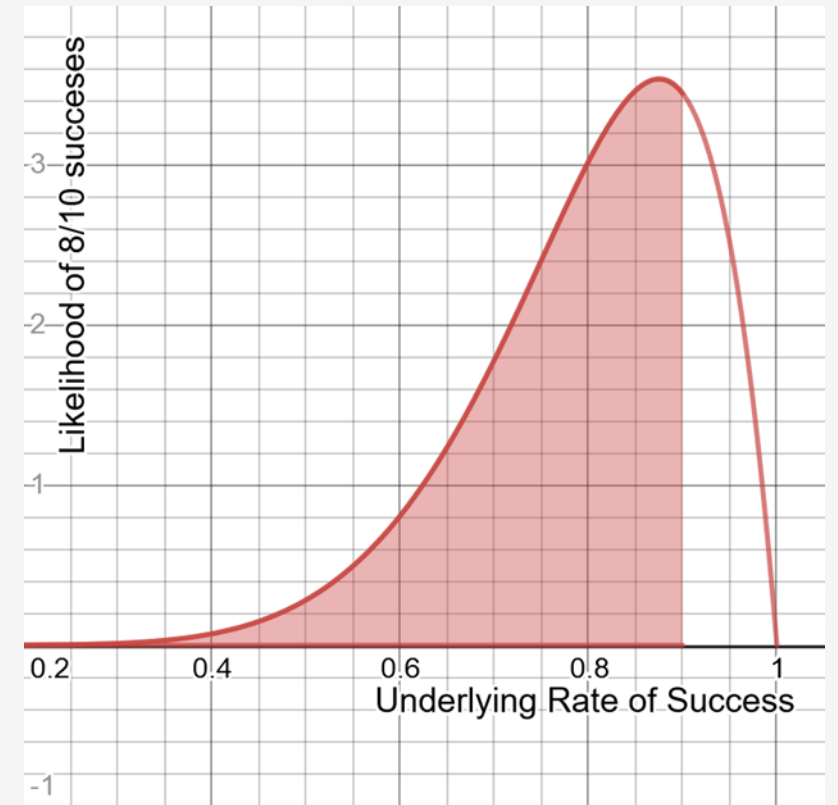
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But the probability of the underlying rate being less than 90% is 77.5%.



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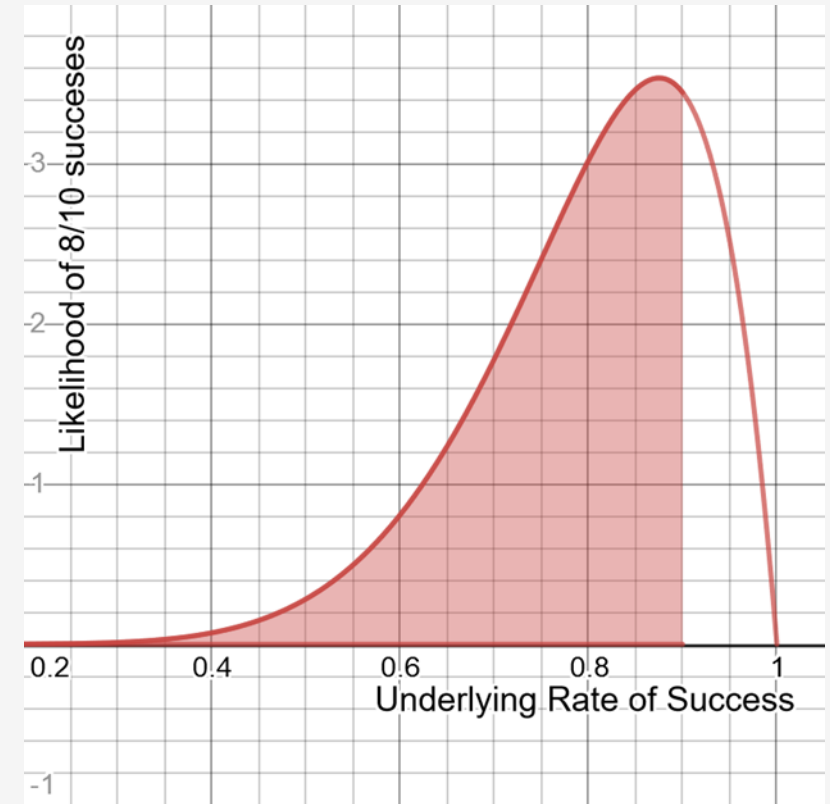
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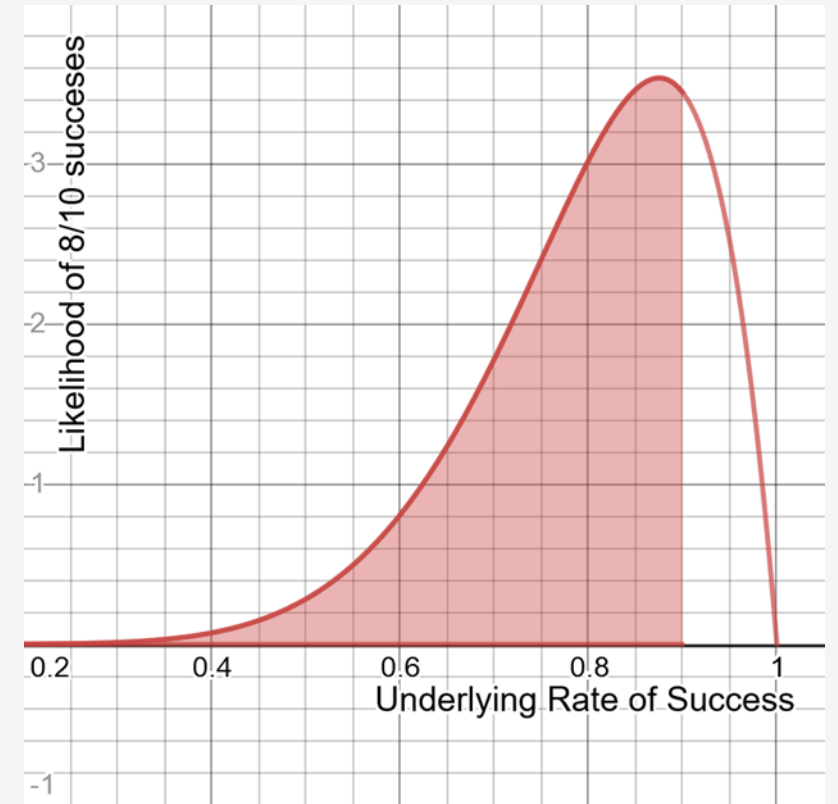


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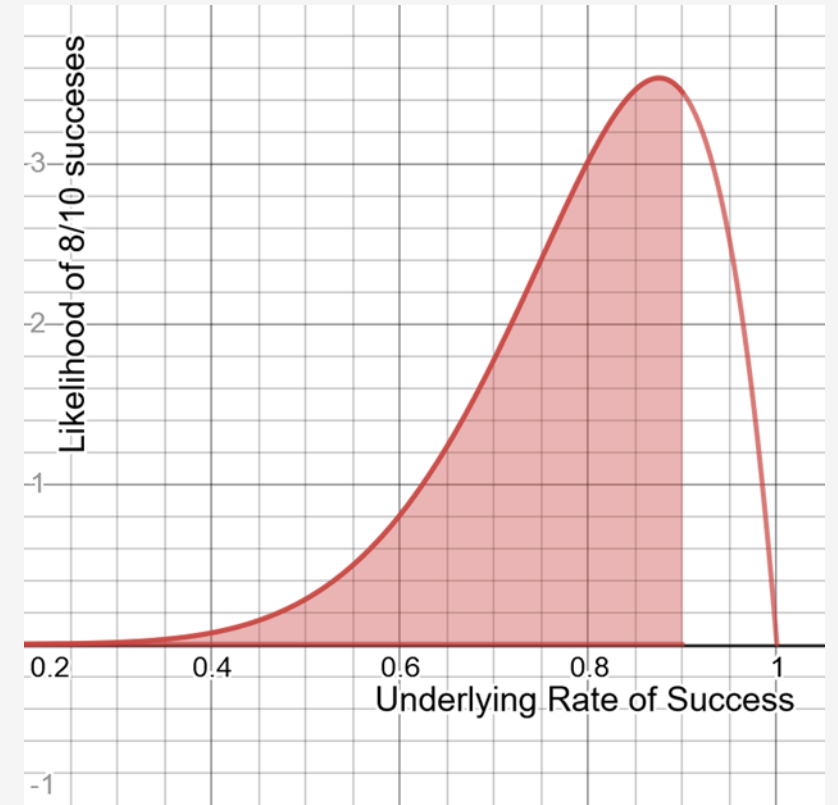
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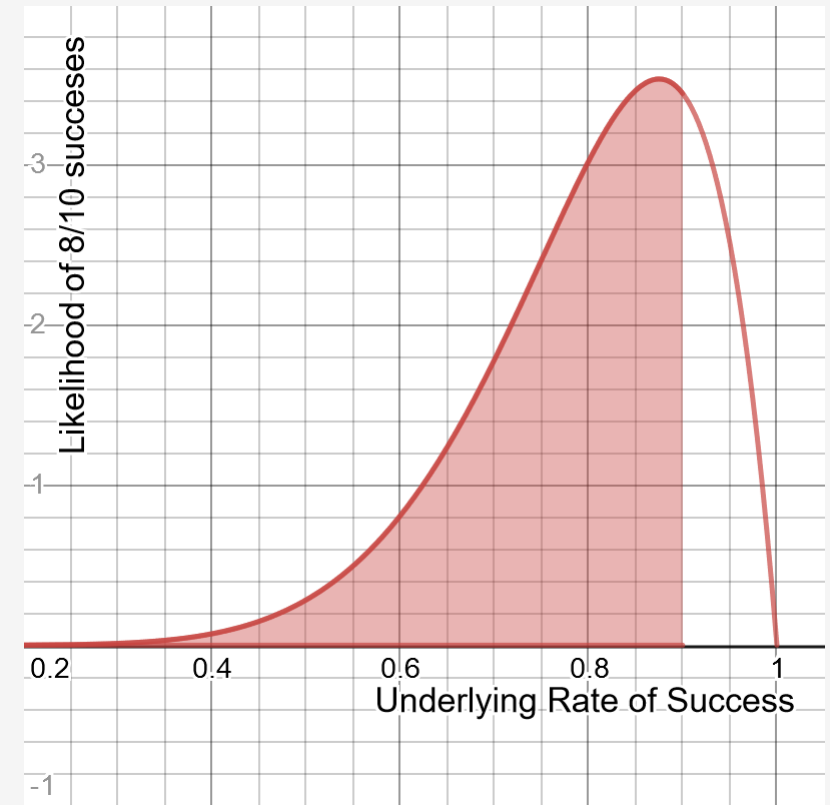
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Given these numbers, is it worth doing more tests?



<https://www.desmos.com/calculator/cakj42wlie>

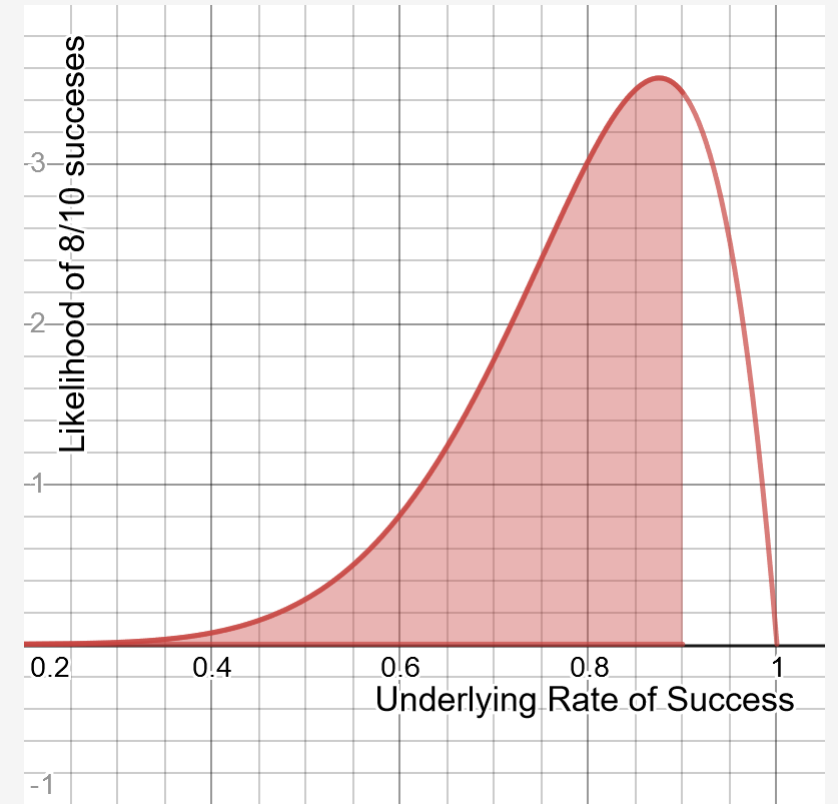
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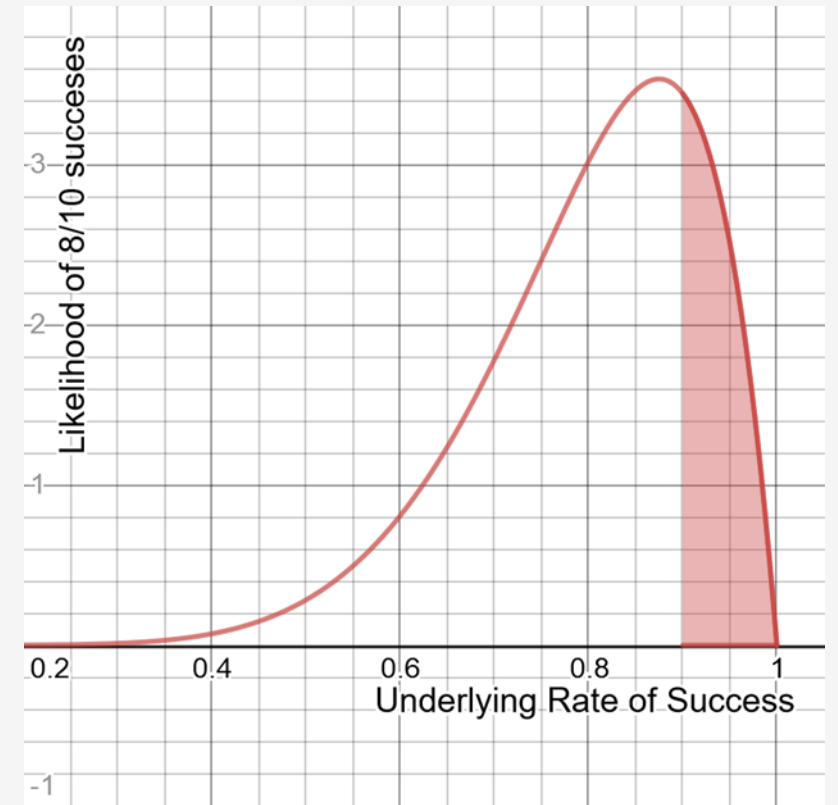
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What we essentially have found is there is a 77.5% chance the underlying success rate is below the required threshold of 90%.



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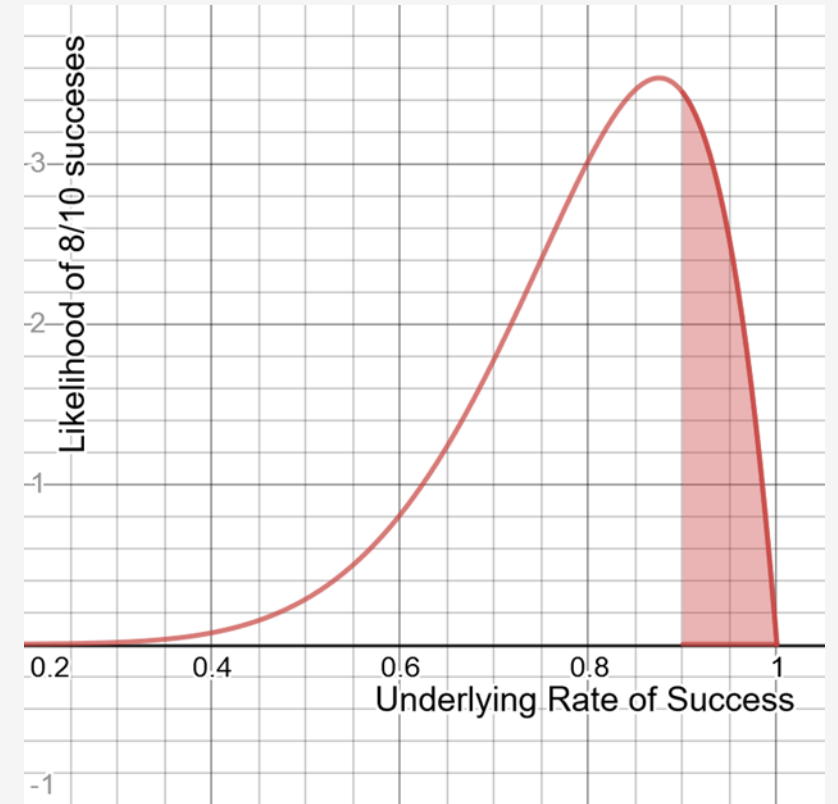
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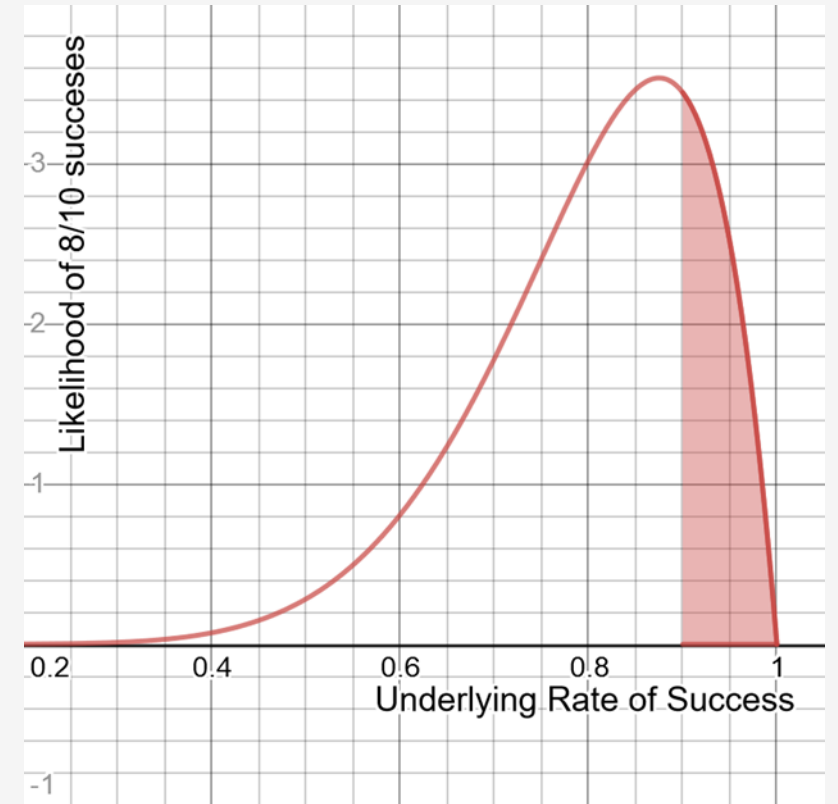


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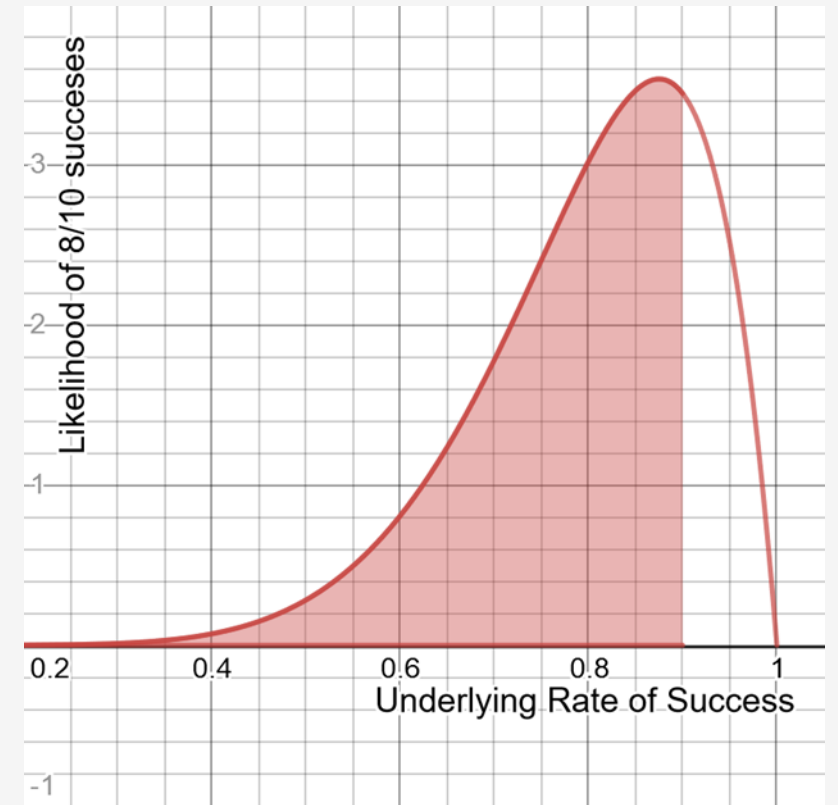
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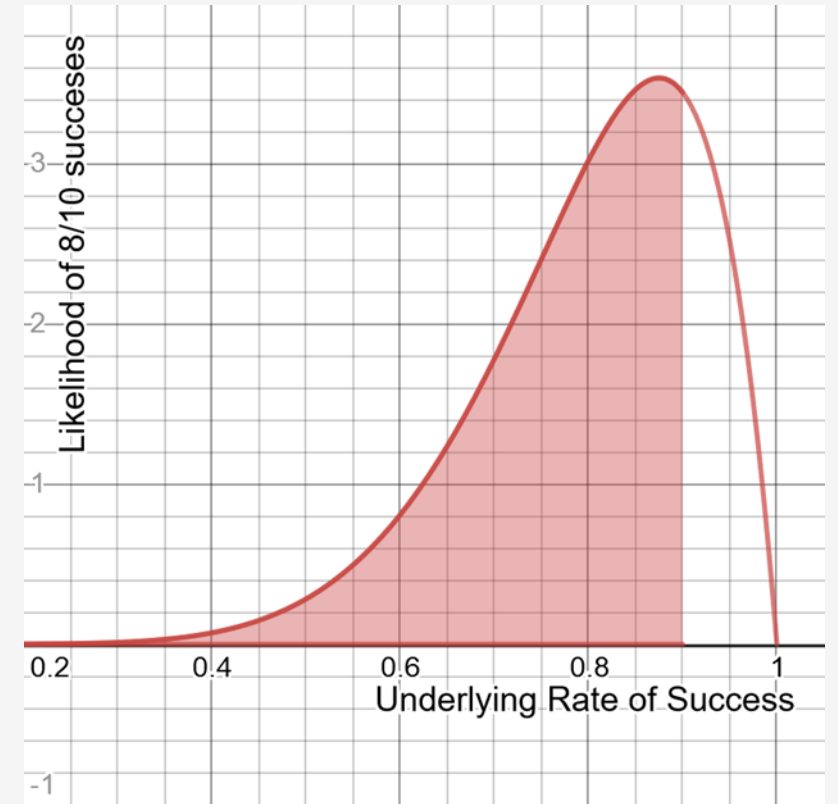
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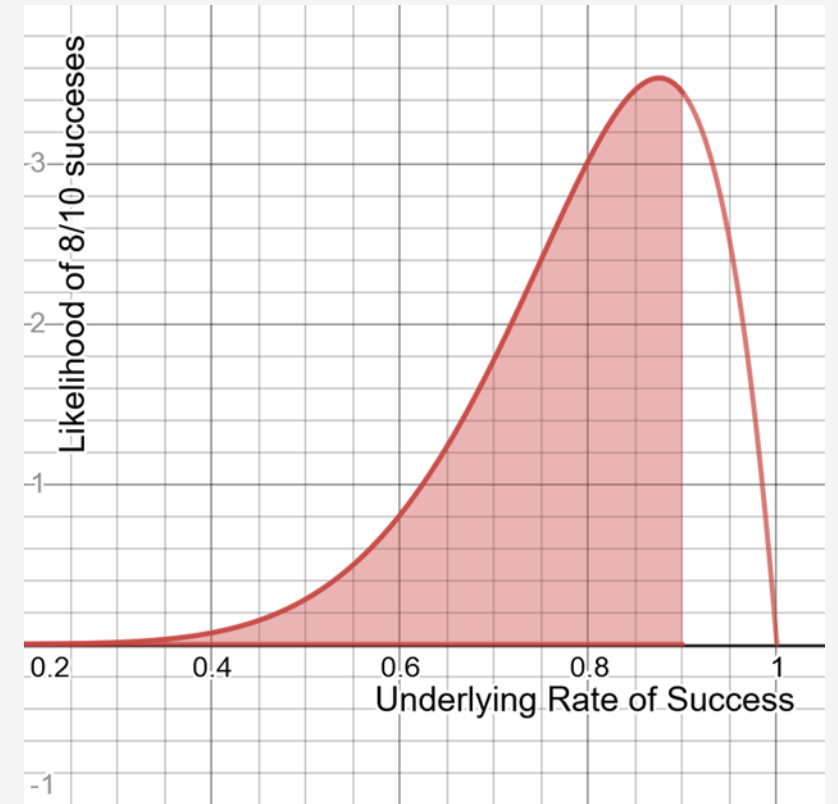


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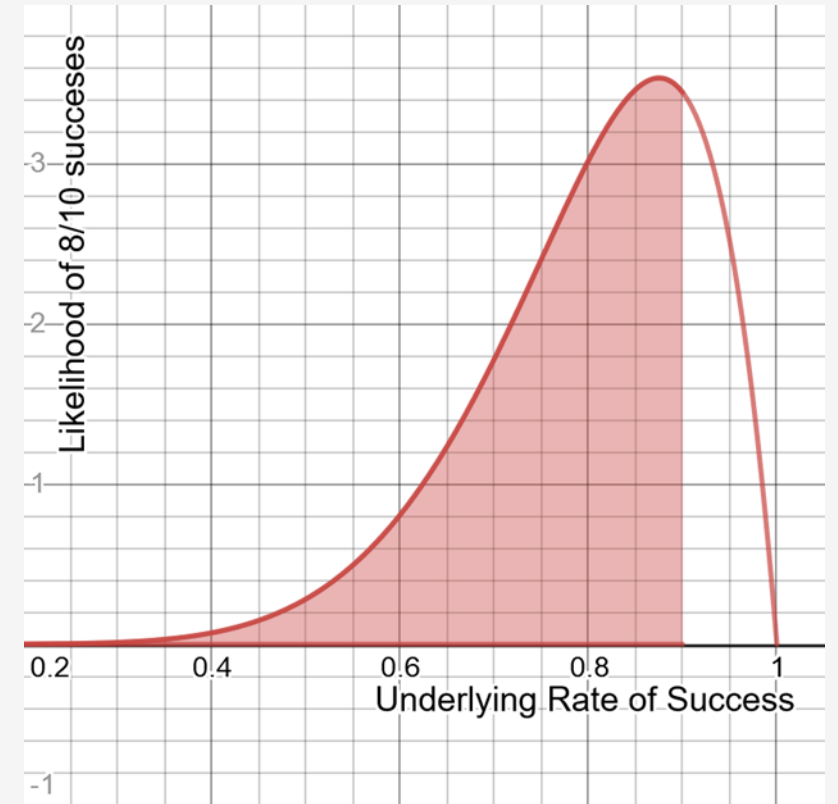
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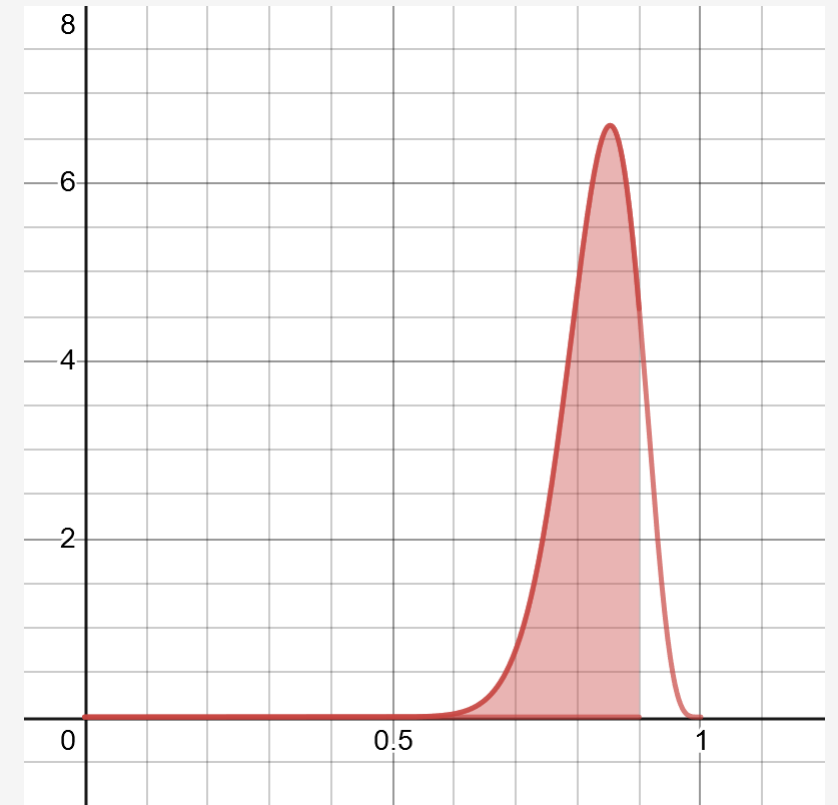
But if we want to gamble on that other 22.5% and order a few more tests to confirm, we can make that call too.

But it may be a risk! Judging just by the numbers, we may just want to abandon testing because there's a 77.5% chance the test truly failed to meet the 90% success rate.



<https://www.desmos.com/calculator/cakj42wlie>

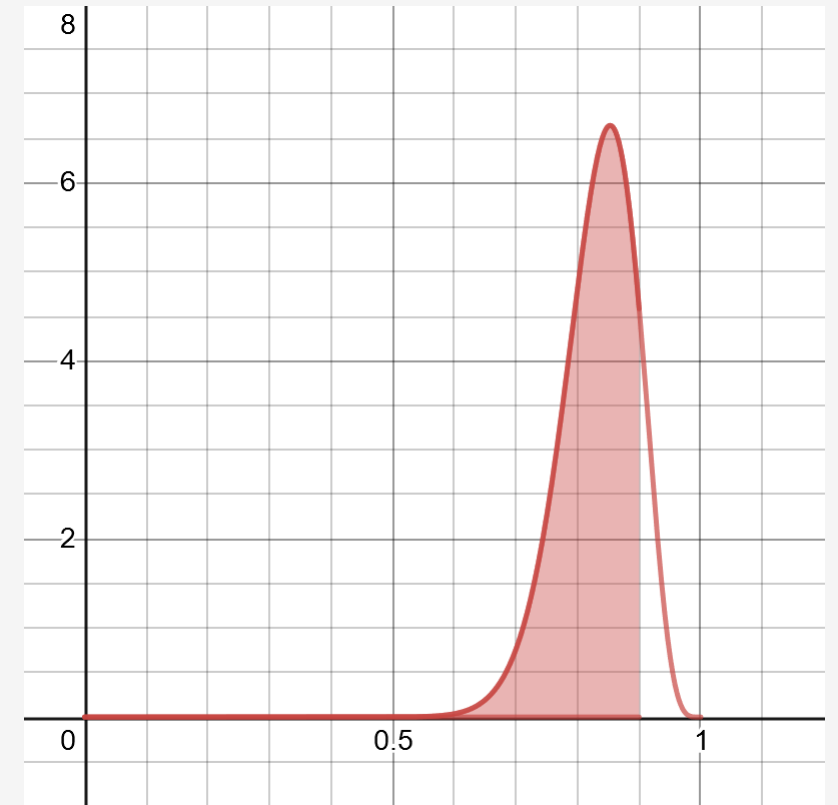
Using Beta Distribution to Update Predictions



<https://www.desmos.com/calculator/xnpjwqclnc>

Using Beta Distribution to Update Predictions

Let's say our CFO granted funding for 26 more tests and wanted to take that risk.

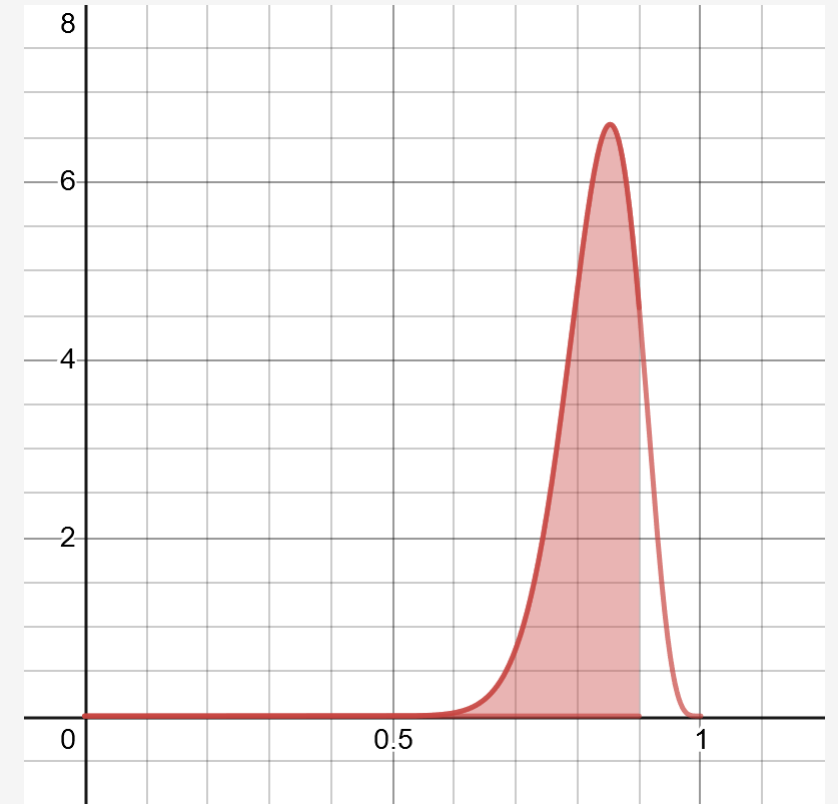


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Using Beta Distribution to Update Predictions

Let's say our CFO granted funding for 26 more tests and wanted to take that risk.

Accumulating on top of the 8 successes and 2 failures, we now have 30 successes and 6 failures.



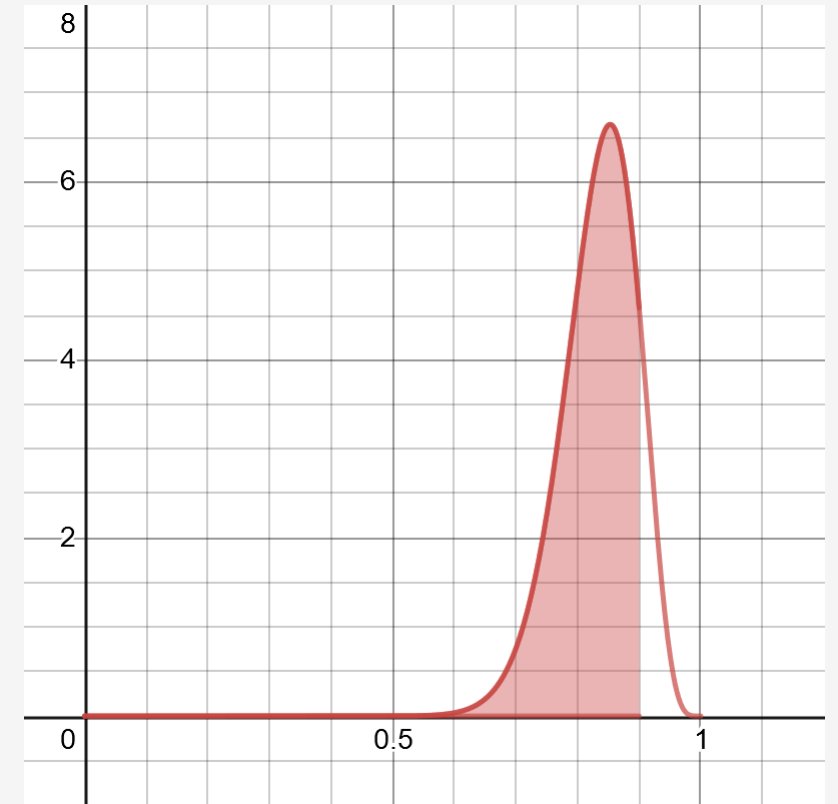
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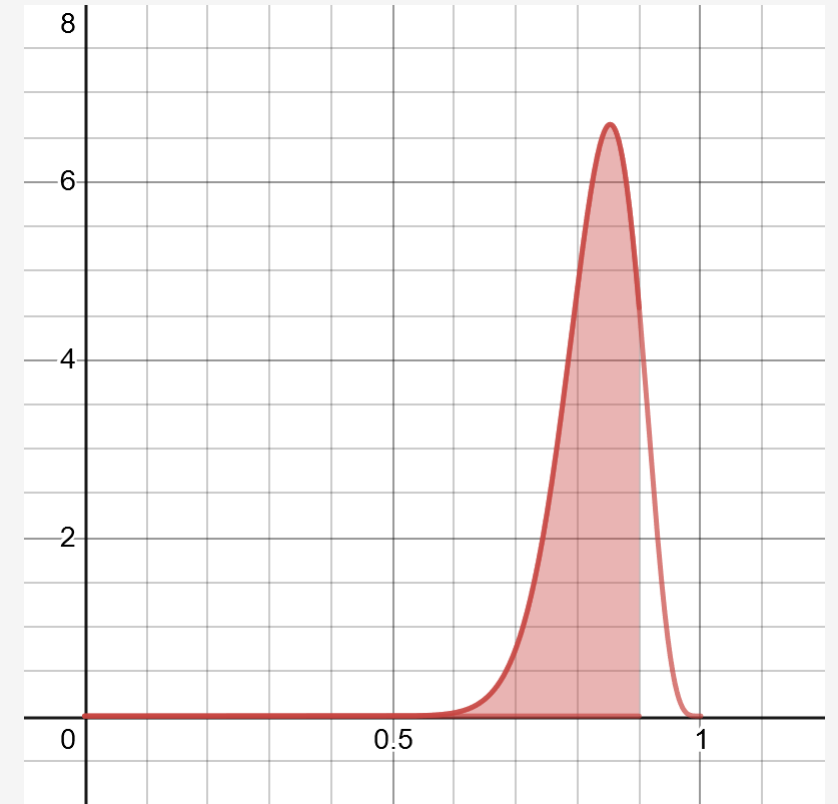
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The probability of the underlying success rate being less than 90% is now 86.8%, leaving only 13.2% possibility that our system meets requirements.



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Using Beta Distribution to Update Predictions

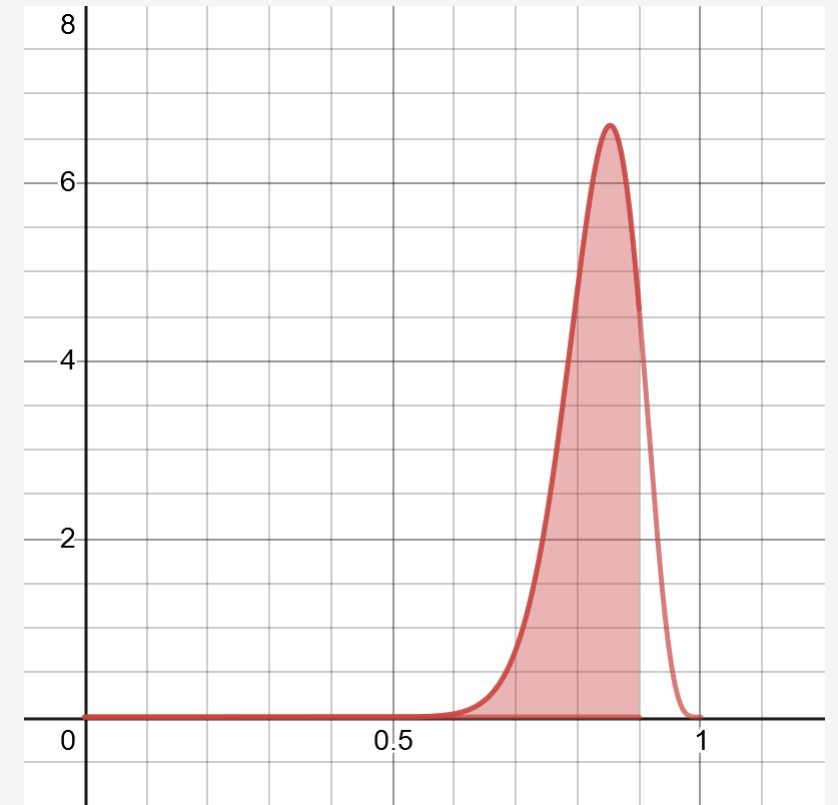
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The probability of the underlying success rate being less than 90% is now 86.8%, leaving only 13.2% possibility that our system meets requirements.

At this point, it may be time to abandon testing and go back to the drawing board, unless you want to continue gambling on that 13.2%!



<https://www.desmos.com/calculator/xnpjwqclnc>

Beta Distribution in Python

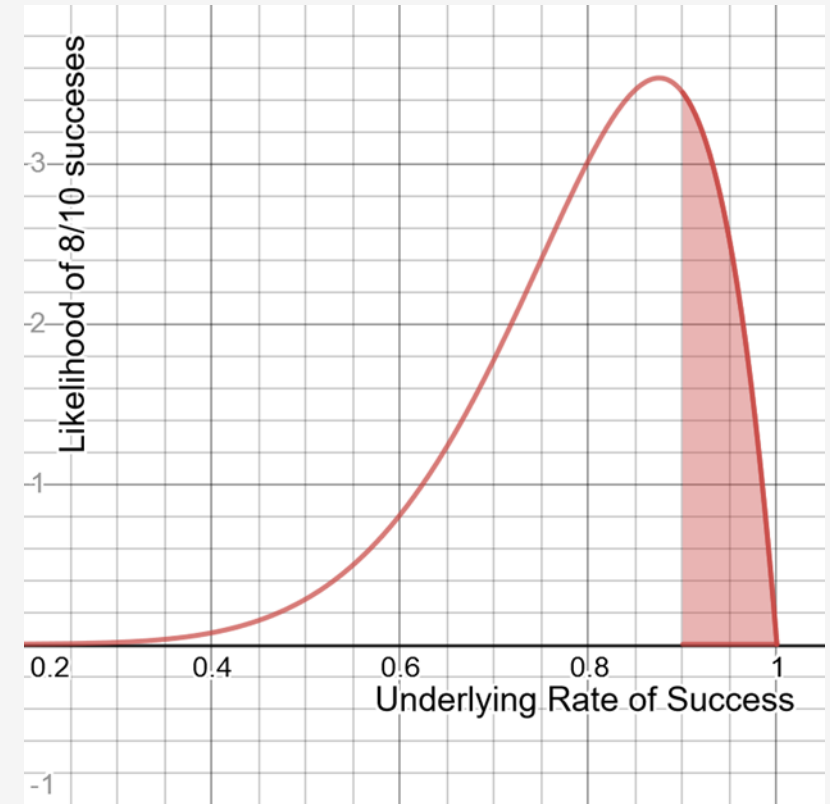
```
from scipy.stats import beta
```

```
a = 8
```

```
b = 2
```

```
p = beta.cdf(1.0, a, b) - beta.cdf(.90, a, b)
```

```
print(p)
```



Katacoda Interactive Scenario – Binomial and Beta Distribution



Discovering Binomial and Beta Distribution

Step 1 of 4 ▶

```
# Generates the coefficient needed for the binomial distribution
def binomial_coefficient(n: int, k: int):
    return factorial(n) / (factorial(k) * factorial(n - k))
```

Before moving on, feel free to play with these functions in the editor. Use the

scratch.pybinomial_beta.py

```
5 def factorial(n: int):
6     f = 1
7     for i in range(n):
8         f *= (i + 1)
9     return f
10
11 # Generates the coefficient needed for the binomial distribution
12 def binomial_coefficient(n: int, k: int):
13     return factorial(n) / (factorial(k) * factorial(n - k))
14
15
```

Terminal +

```
Introduction
$ cd /home/scrapbook/tutorial
$ python3 scratch.py
python3: can't open file 'scratch.py': [Errno 2] No such file or directory
$
```

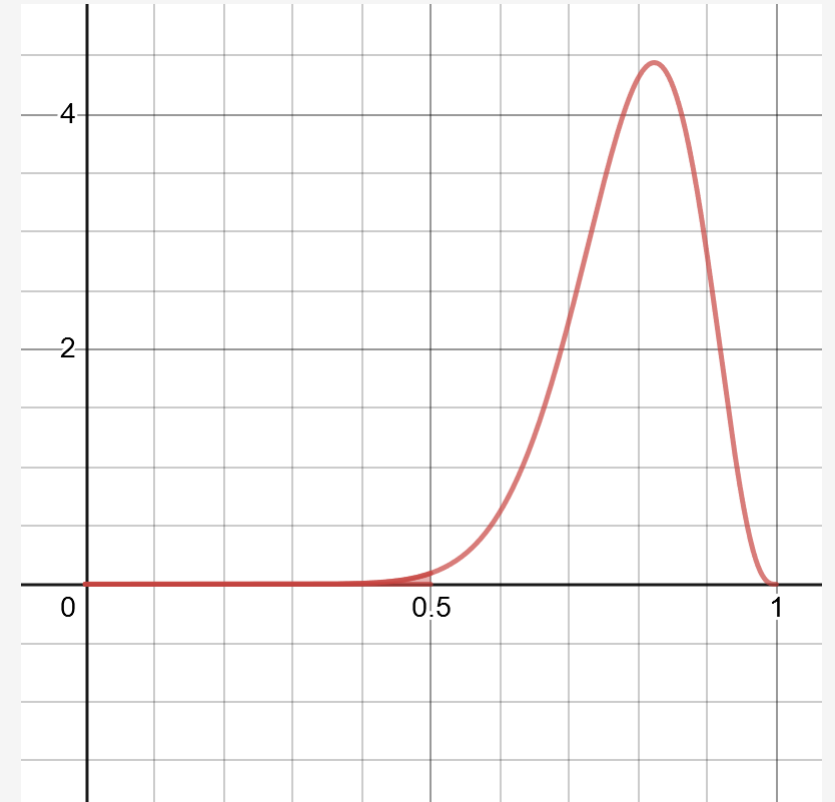
Powered by  Katacoda

<https://learning.oreilly.com/scenarios/probability-from-scratch/9781492080565/>

Exercise

You flipped a coin 19 times and got heads 15 times and tails 4 times.

Do you think this coin has any good probability of being fair?



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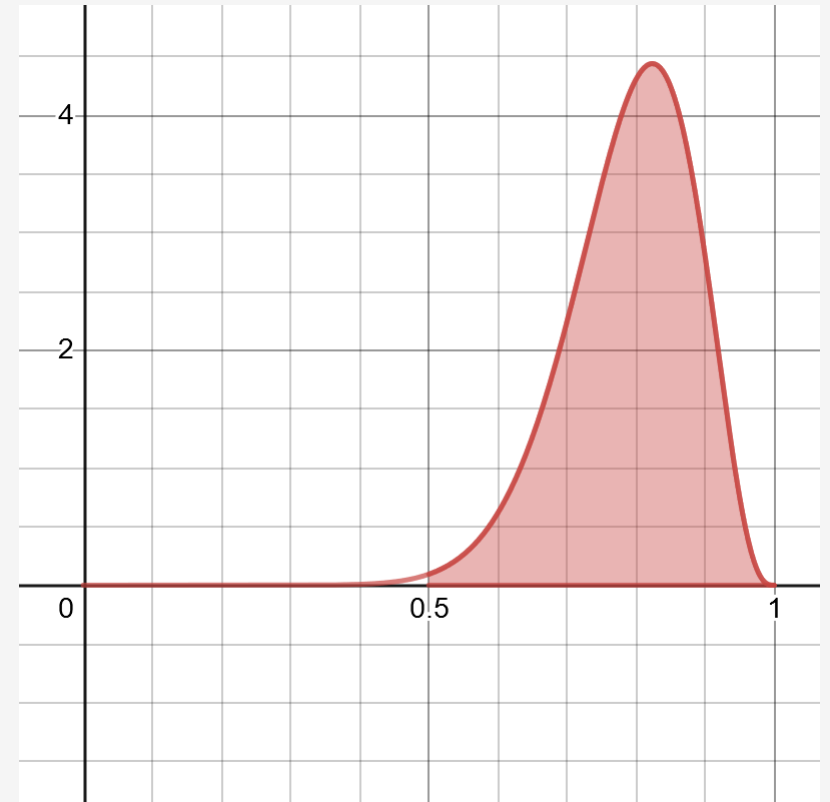
```
a = 15
```

```
b = 4
```

```
x = beta.cdf(1.0, a, b) - beta.cdf(.50, a, b)
```

```
print(x)
```

s above 50%,



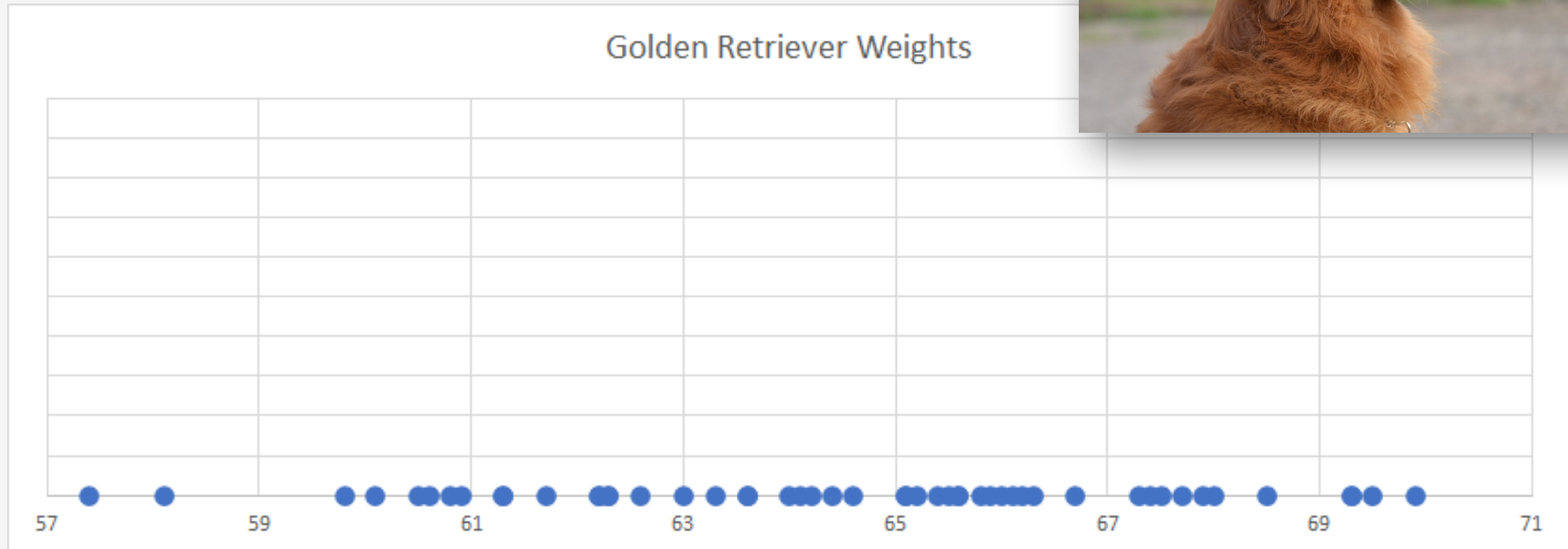
Section V

Discovering the Normal Distribution

Discovering the Normal Distribution

Below is a number line showing recorded weights (in pounds) for 50 adult golden retrievers.

What do you notice about this data?

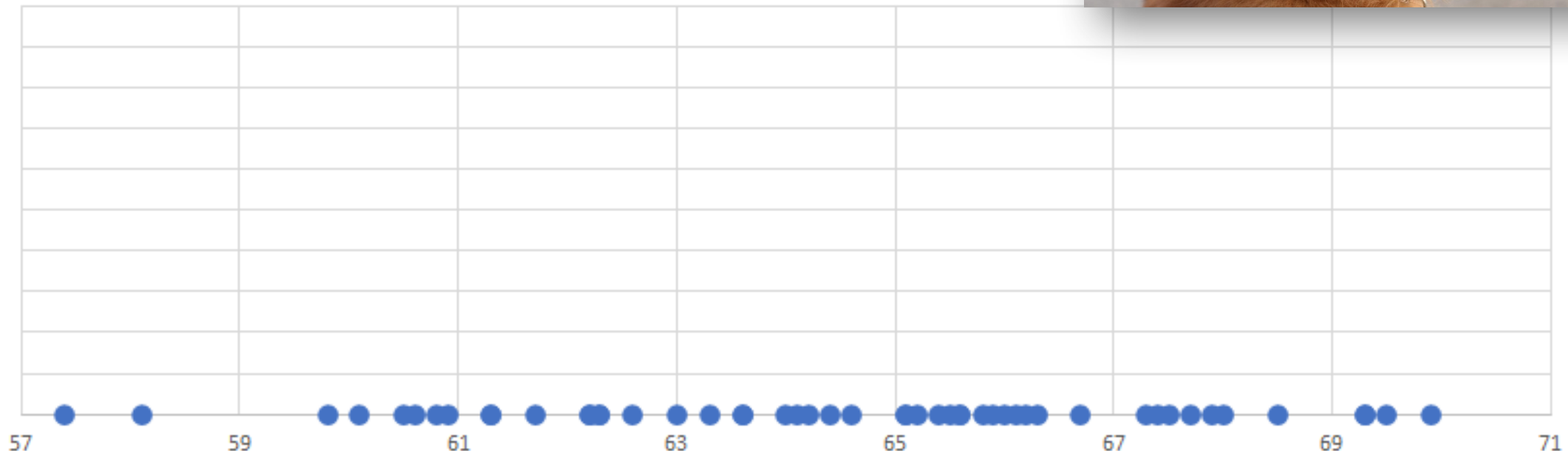


Discovering the Normal Distribution

Notice that towards the “center” there are more points clustered together, while the “tails” on either end have less points.

We can infer that we are more likely to see more golden retrievers in the 61-67 lb range as opposed to lower or higher weights like 57 and 71.

Golden Retriever Weights

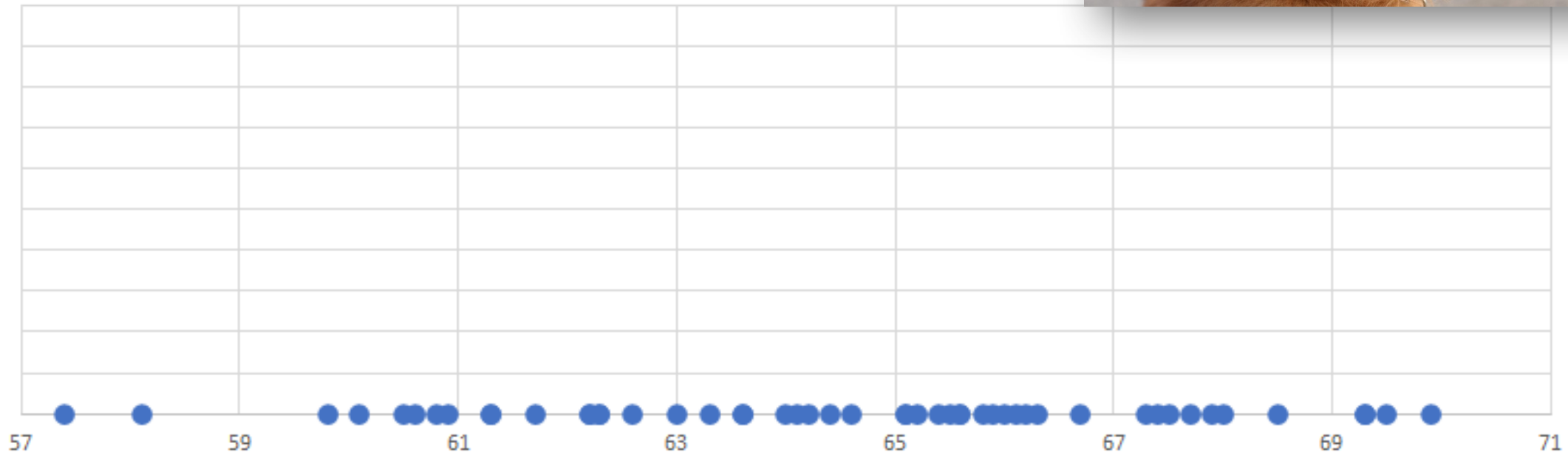


Discovering the Normal Distribution

Is there a better way we can visualize this data? And get an idea where we would see more golden retrievers in terms of their weight?



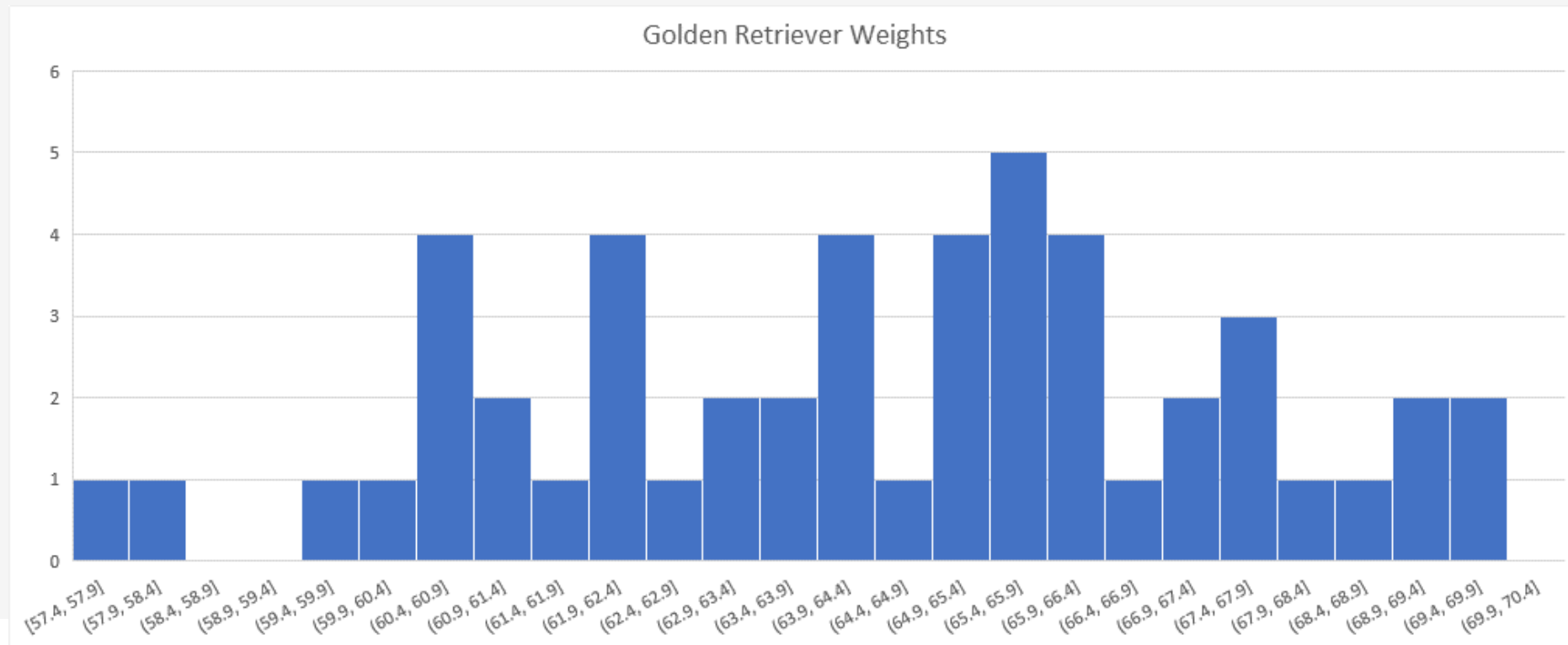
Golden Retriever Weights



Discovering the Normal Distribution

We can try to **bin** up points on equally-sized ranges, and then count the number of instances as a bar chart to create a **histogram**.

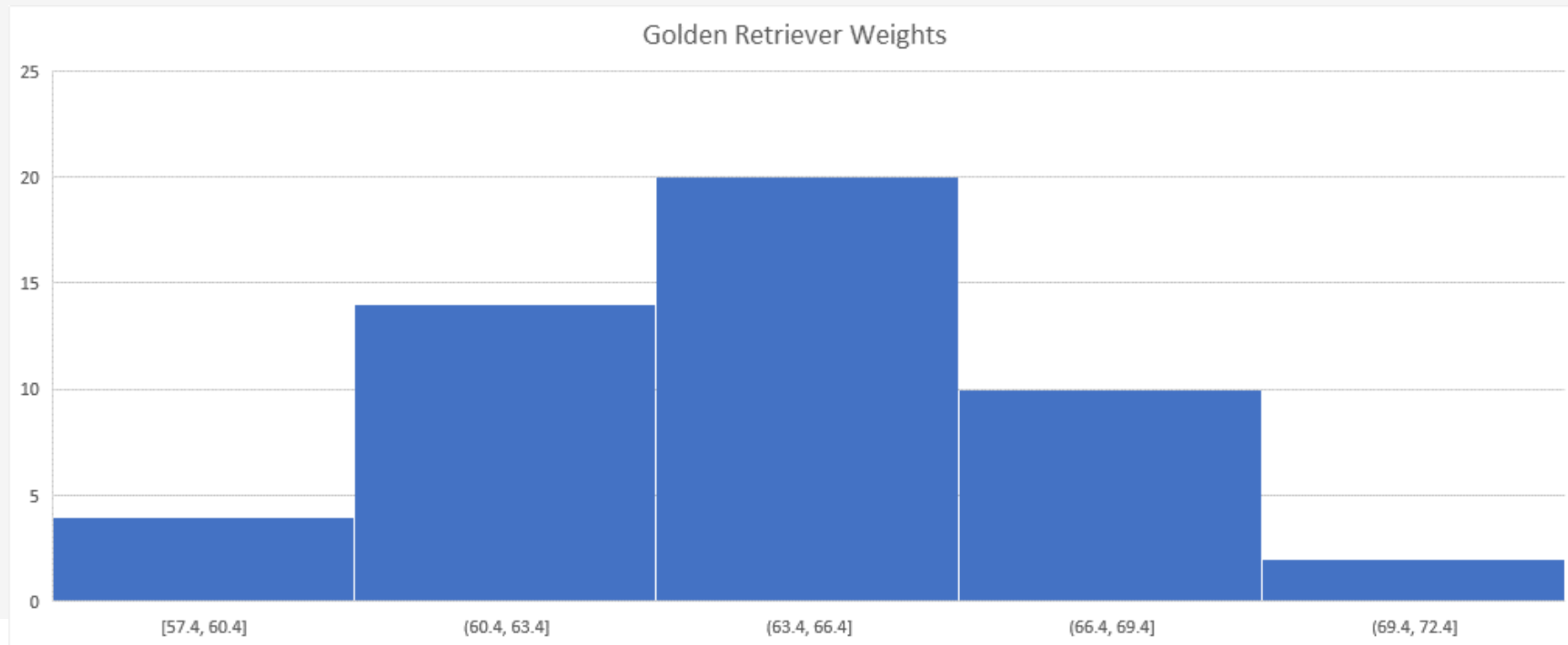
Unfortunately, we don't have an infinite amount of data so making the bin ranges too small will not reveal much about the underlying distribution.



Discovering the Normal Distribution

But if we make the bin sizes just right, we might be able to see a shape resembling a probability distribution.

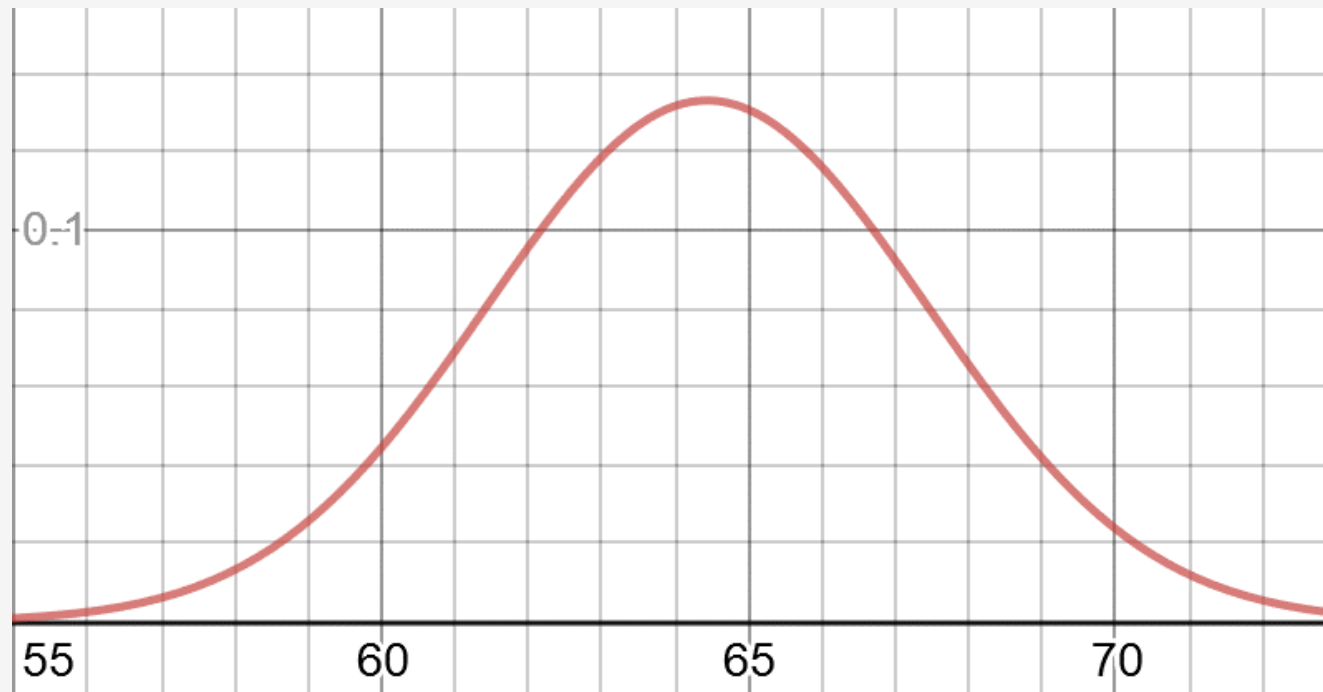
There are many types of distributions, but in this case we can see a nice bell curve known as the **normal distribution**, also called the **Gaussian Distribution**.



Discovering the Normal Distribution

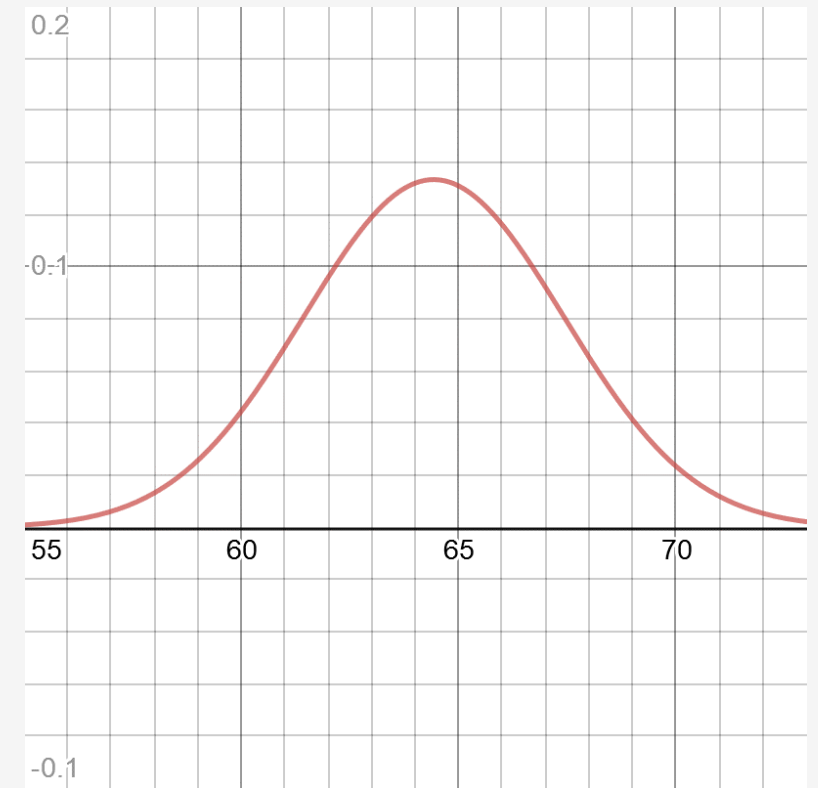
We can fit this curve onto a histogram to make inferences about the entire population, as it's unlikely I will be able to weigh every golden retriever in existence.

There are ways to measure uncertainty whether a bell curve resembles the population's bell curve, but we will steer clear of statistics in this course.



<https://www.desmos.com/calculator/yftpag0tse>

Discovering the Normal Distribution



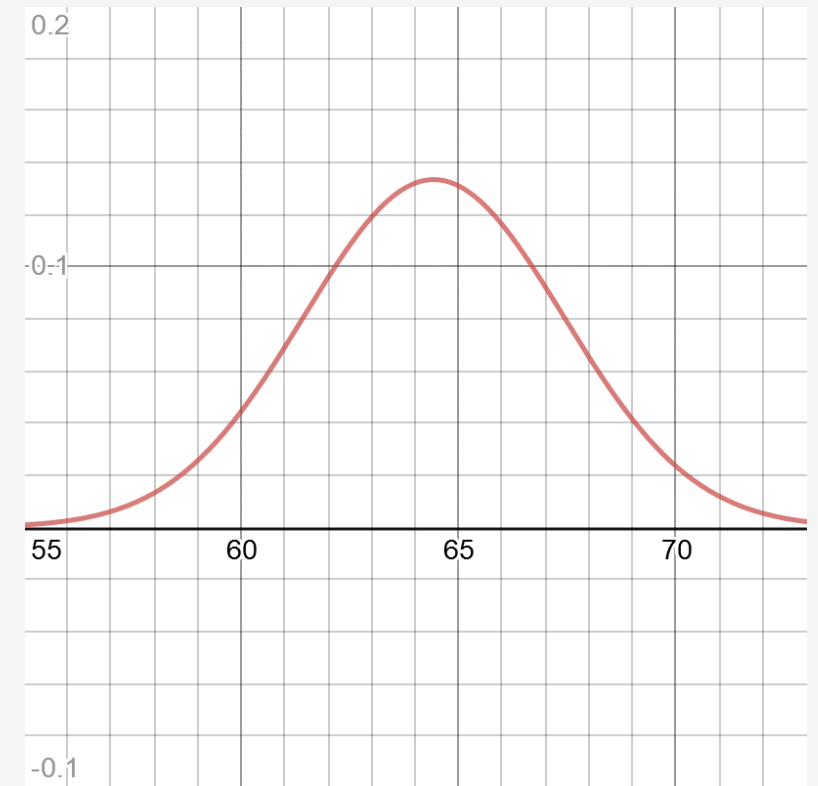
<https://www.desmos.com/calculator/yftpag0tse>

Discovering the Normal Distribution

The normal distribution has several important properties that make it useful:

- Symmetrical
- Most mass is at the center around the **mean**
- Has a spread (being narrow or wide) that is specified by **standard deviation**.
- The “tails” are the least likely outcomes, and approach zero infinitely but never touch zero.
- It resembles a lot of phenomena in nature and daily life, and even generalizes non-normal problems because of the central limit theorem.

We can expect any golden retriever to have a weight most likely around 64.43 (the mean), but highly unlikely around 55 or 73.



<https://www.desmos.com/calculator/yftpag0tse>

The Normal Distribution Formula

$\mu = \text{mean}$

$\sigma = \text{standard deviation}$

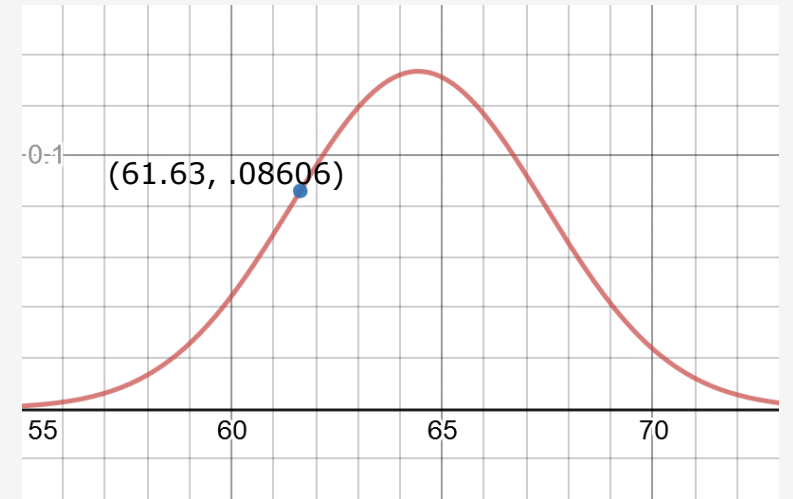
$x = \text{observed value}$

$f(x) = \text{probability density function}$

Python function:

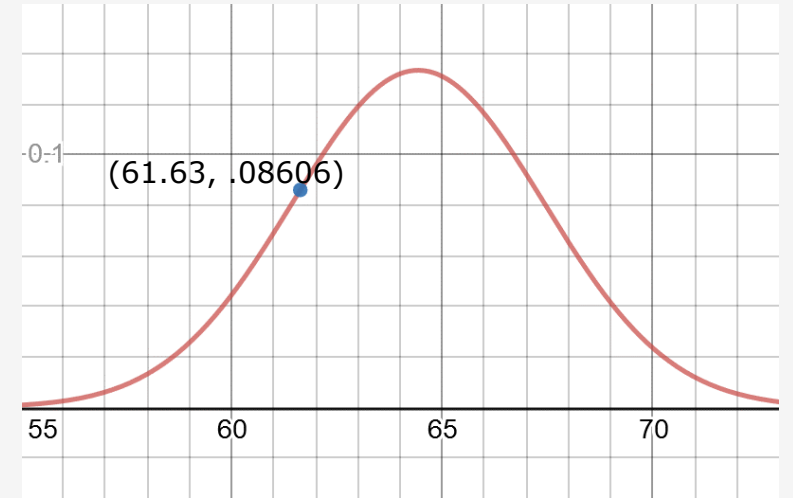
```
# normal distribution, returns likelihood
def normal_pdf(x: float, mean: float, std_dev: float) -> float:
    return (1.0 / (2.0 * math.pi * std_dev ** 2) ** 0.5) * math.exp(-1.0 * ((x - mean) ** 2 / (2.0 * std_dev ** 2)))
```

Discovering the Normal Distribution



Discovering the Normal Distribution

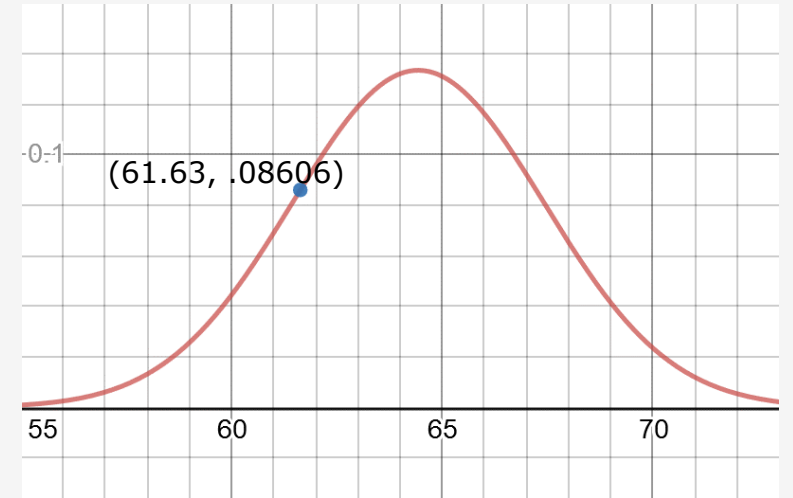
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Discovering the Normal Distribution

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The answer may surprise you: It is 0%!

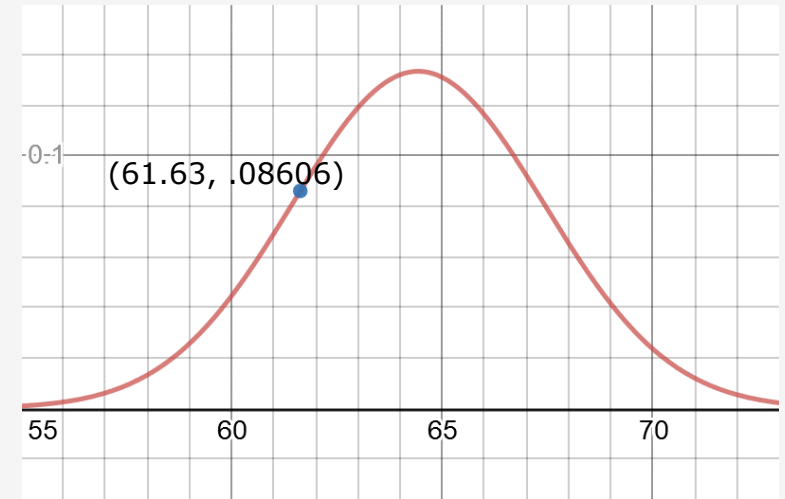


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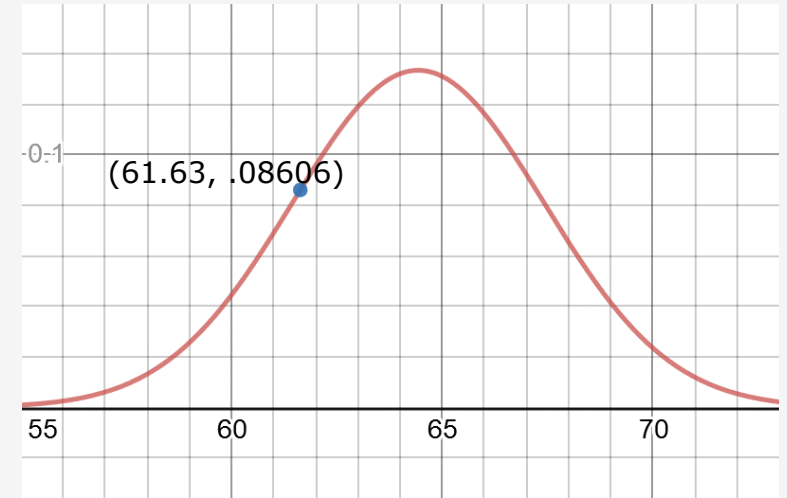
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The probability of getting a specific value is virtually impossible **unless** it is already observed.



Discovering the Normal Distribution

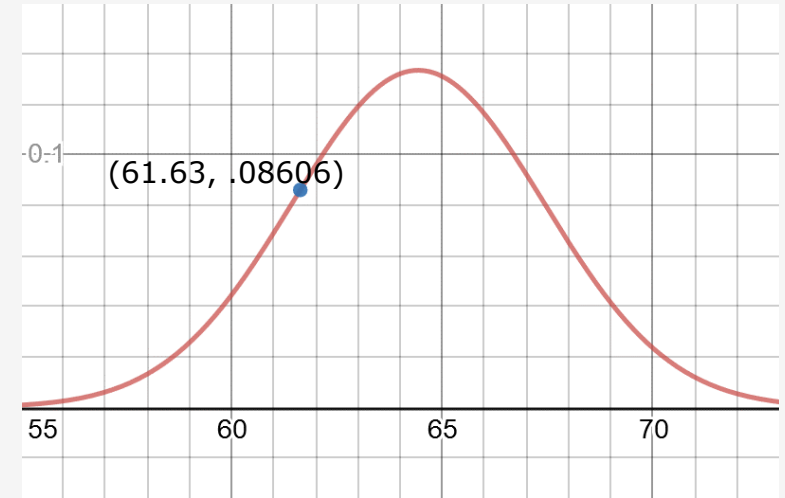
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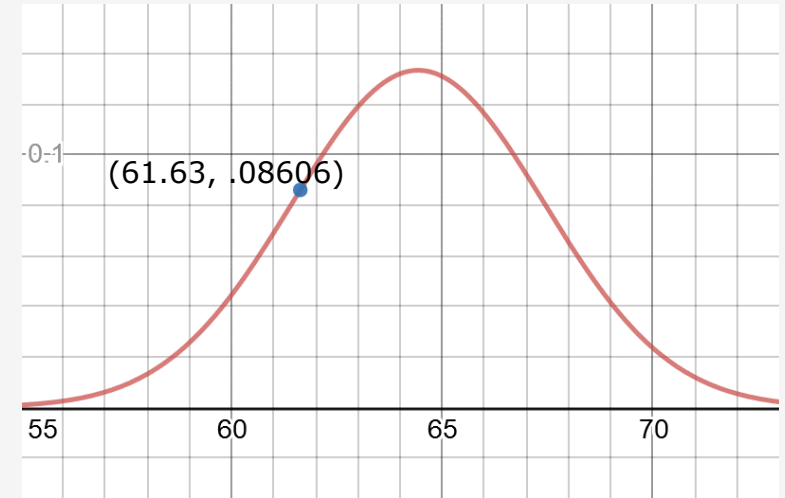
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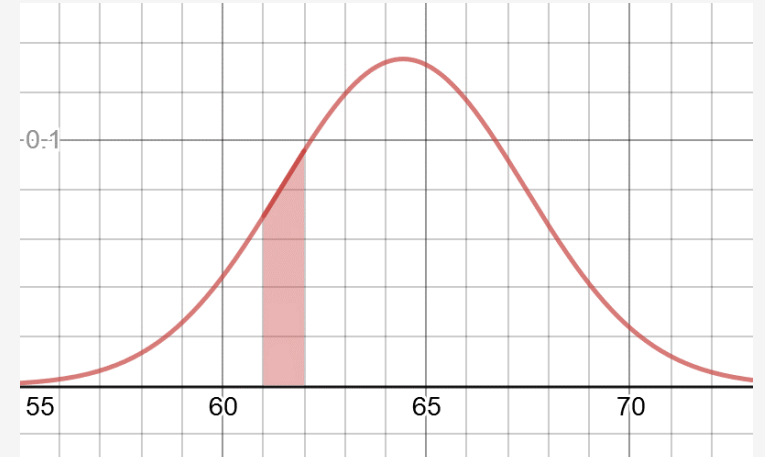
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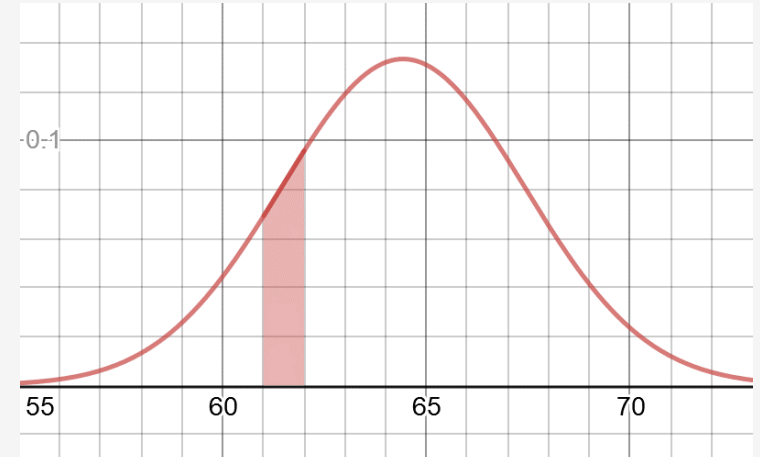


Calculating Probabilities



Calculating Probabilities

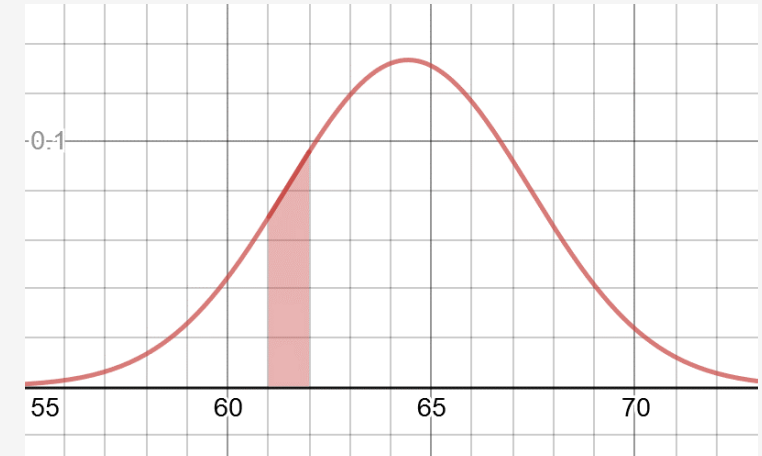
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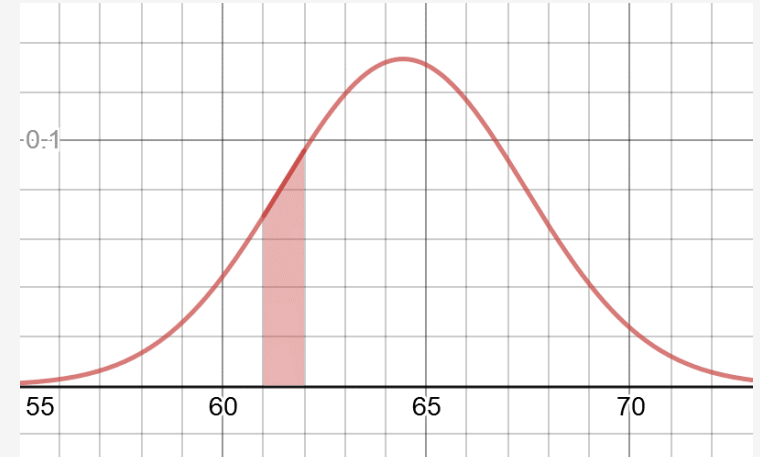


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Rather than ask:

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A more productive question would be:



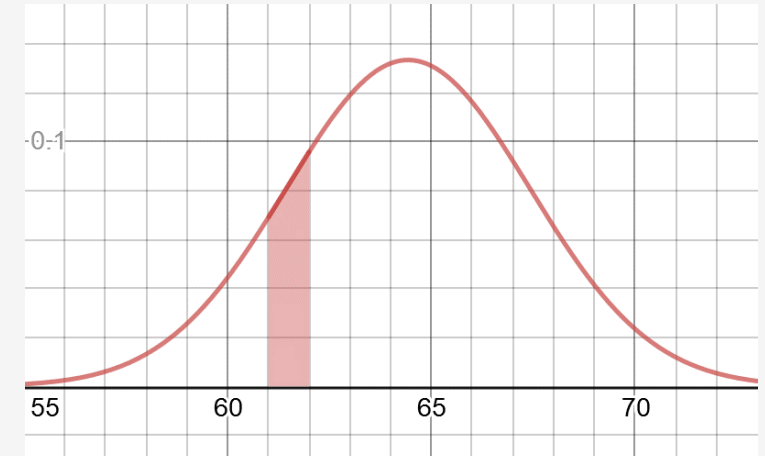
Calculating Probabilities

Rather than ask:

“What is the probability a golden retriever will be exactly 61.63 pounds?”

A more productive question would be:

“What is the probability a golden retriever will be **between** 61 and 62 pounds?”



Calculating Probabilities

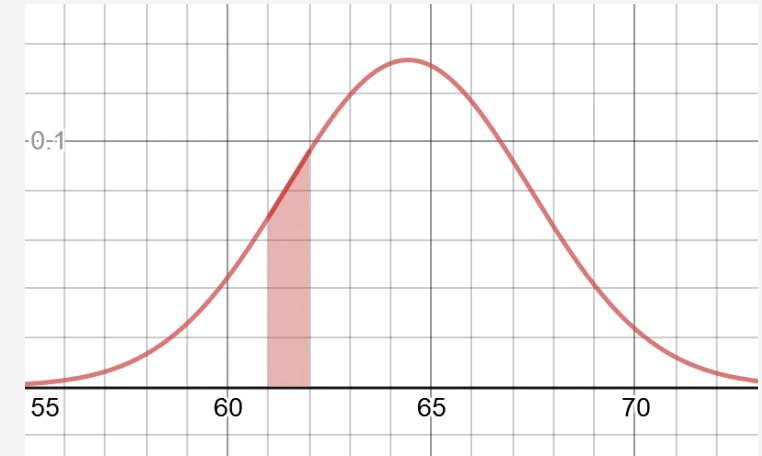
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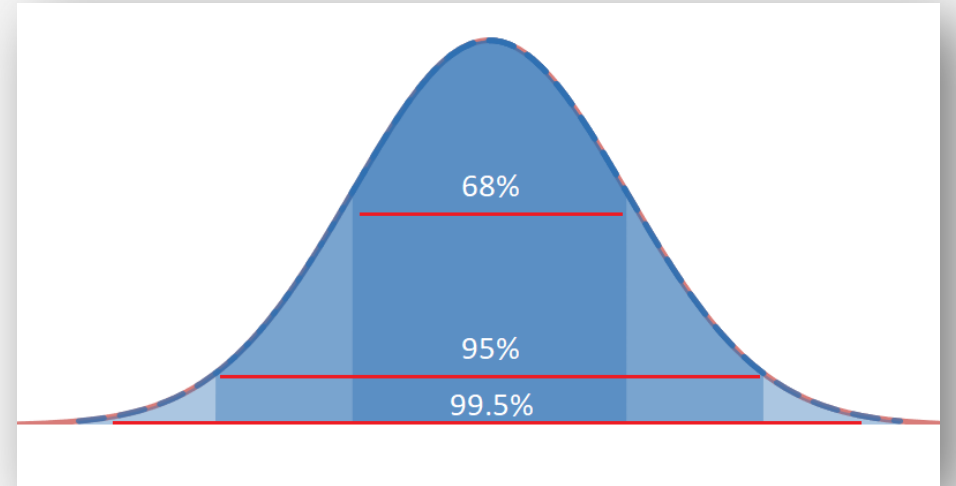
A more productive question would be:

“What is the probability a golden retriever will be **between** 61 and 62 pounds?”

The answer is 8.25%, and we will learn how to calculate this using integration just like we did with the beta distribution.

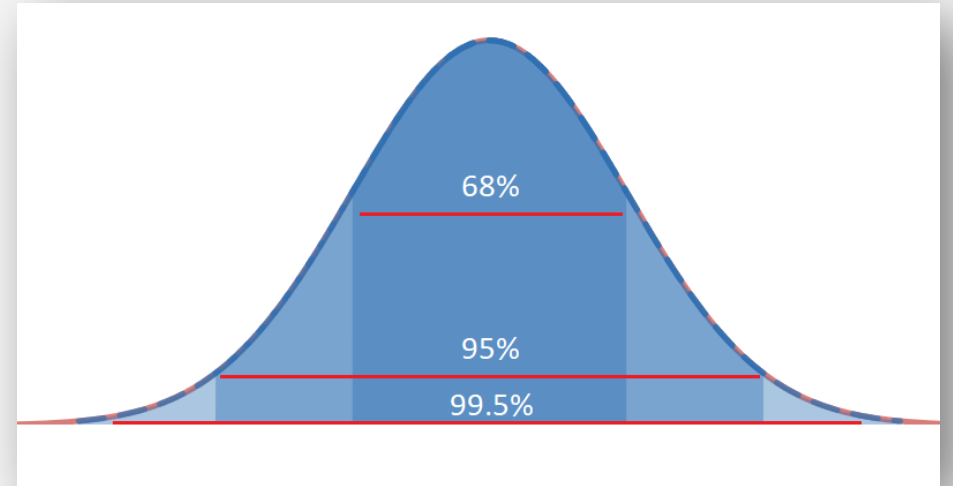


68-95-99.5 Rule



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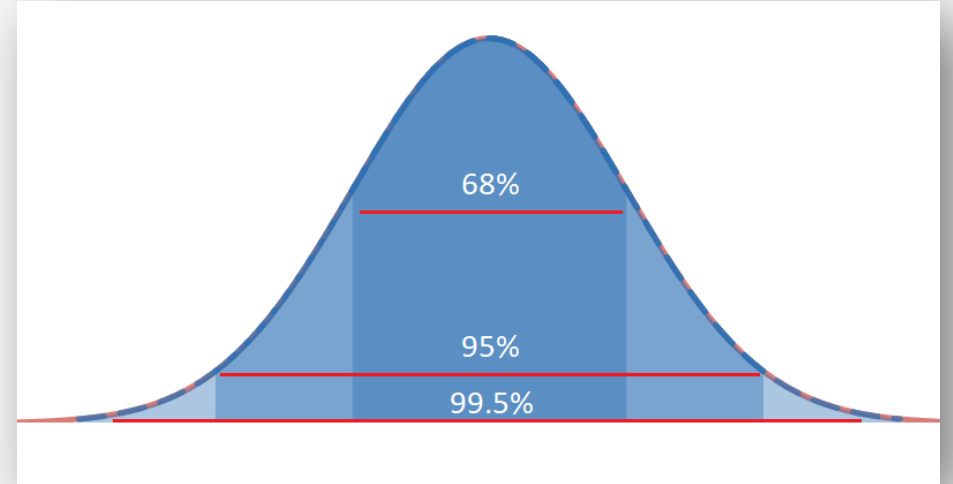
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This means that if I have a mean of 10 and standard deviation of 3, then...

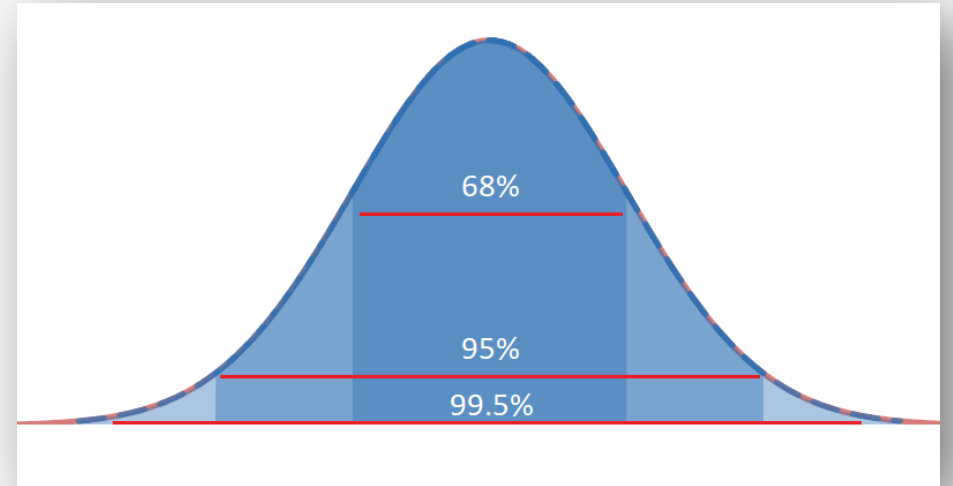


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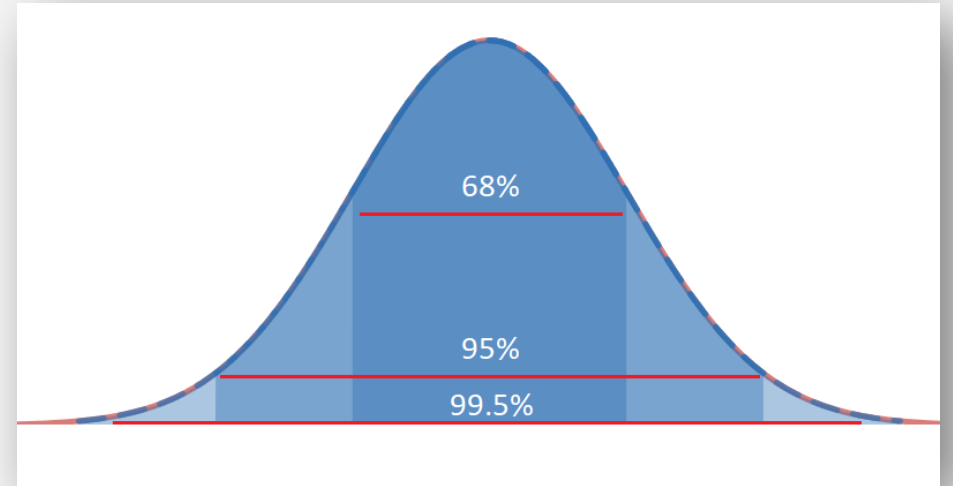
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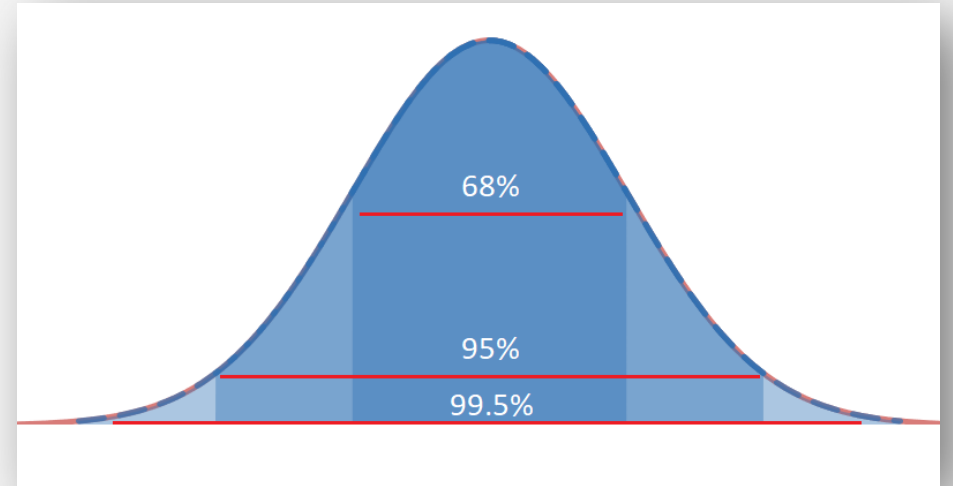
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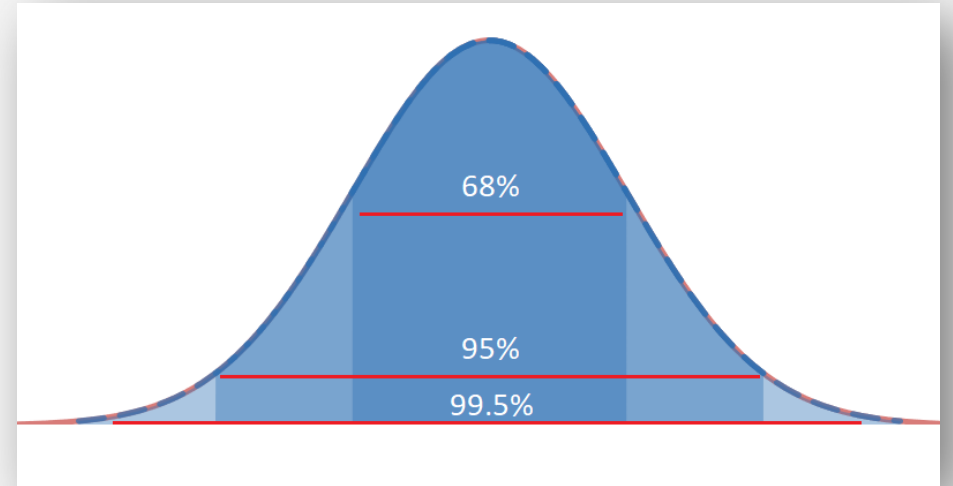
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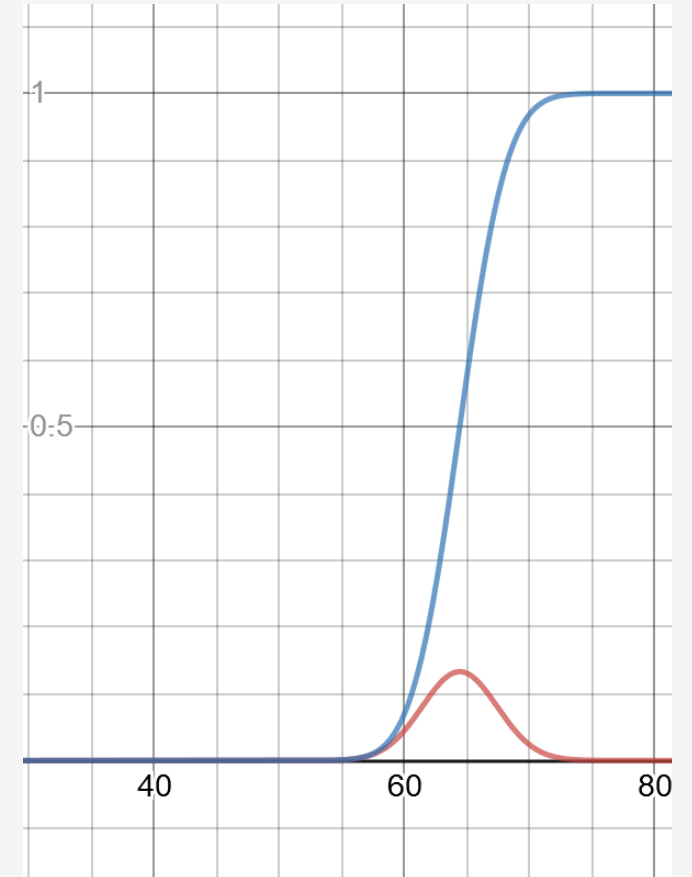
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You can use this rule to quickly calculate probabilities when using standard deviations.



Cumulative Density Function (CDF)

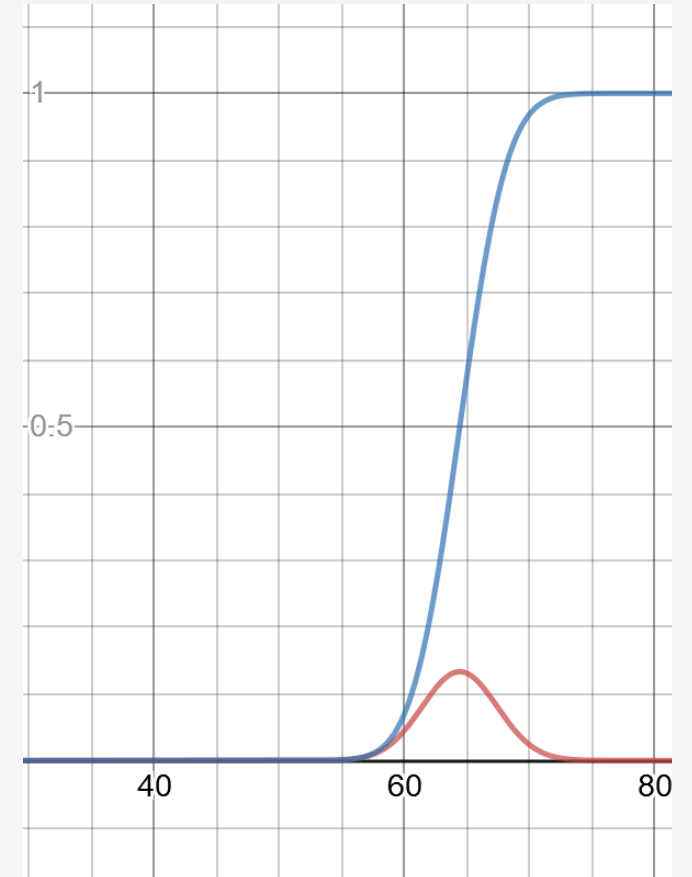
```
def
normal_cdf(x:float,mean:float=0.0,std_dev:float=1.0):
    return 0.5*(1+math.erf((x-mean)/
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```



Cumulative Density Function (CDF)

The way we used integration to find the probability between a range is perfectly valid and intuitive.

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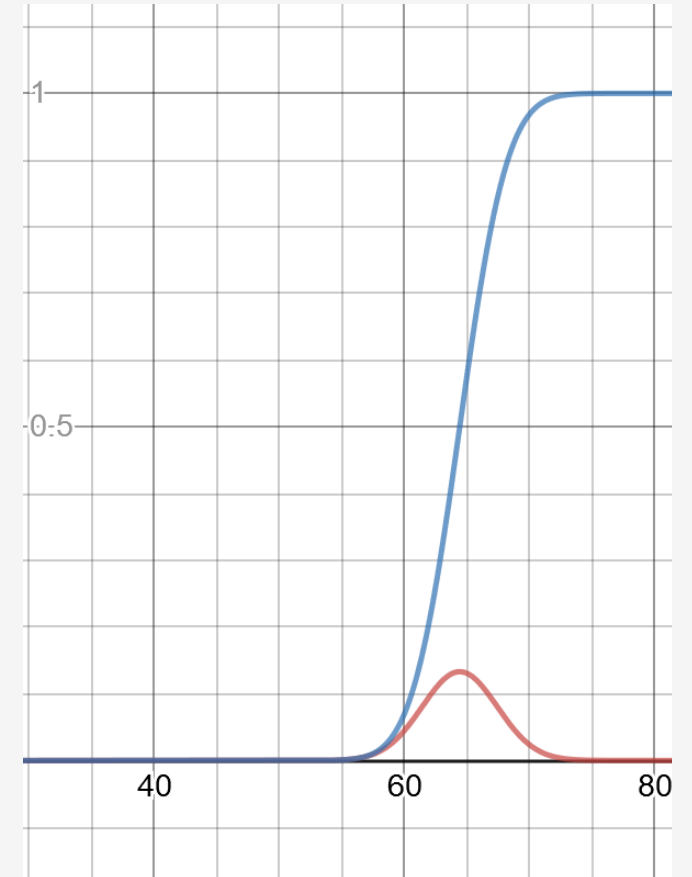


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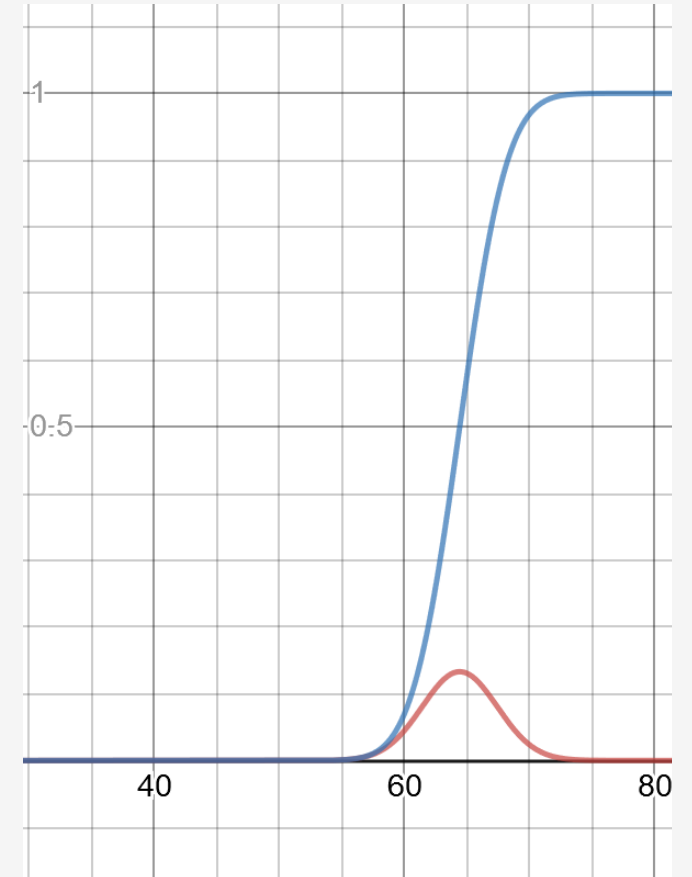
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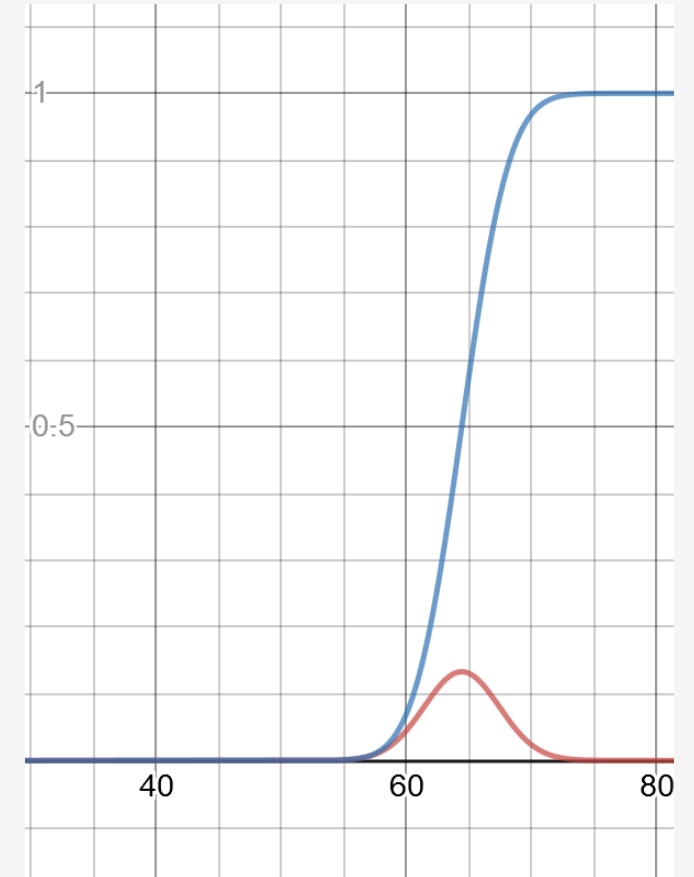
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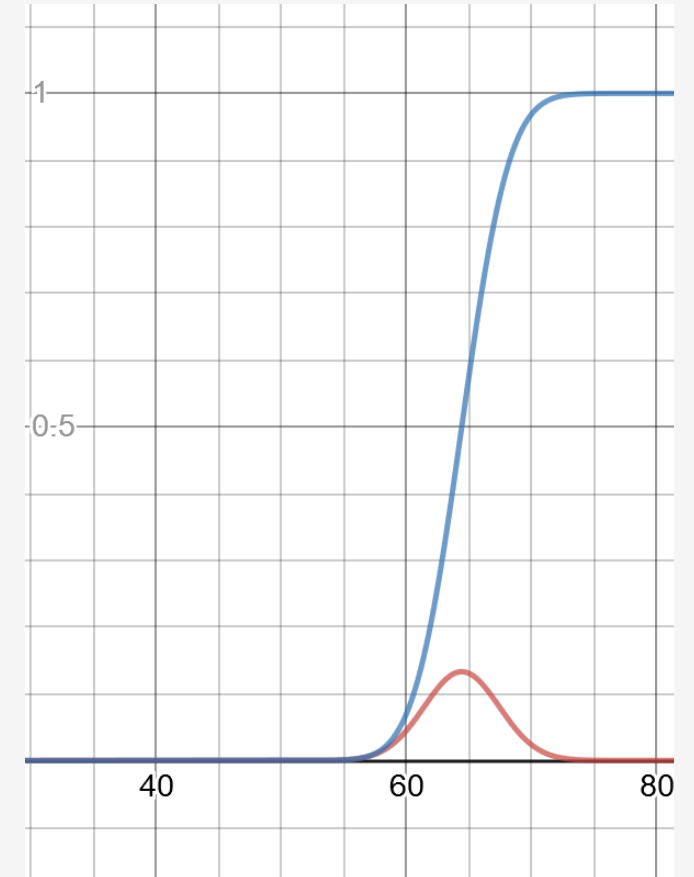
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```

We do cheat a bit here, but we'll leave it a black box as it is beyond the scope of this class.



Cumulative Density Function (CDF)

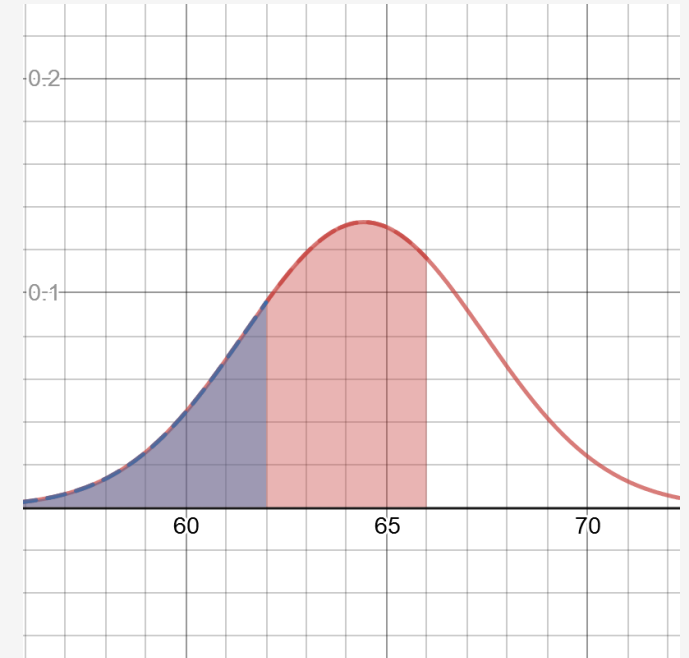
```
from scipy.stats import norm
```

```
mean = 64.43
```

```
std_dev = 2.99
```

```
x = norm.cdf(66, mean, std_dev) - norm.cdf(62, mean,  
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```

```
print(x)
```



Cumulative Density Function (CDF)

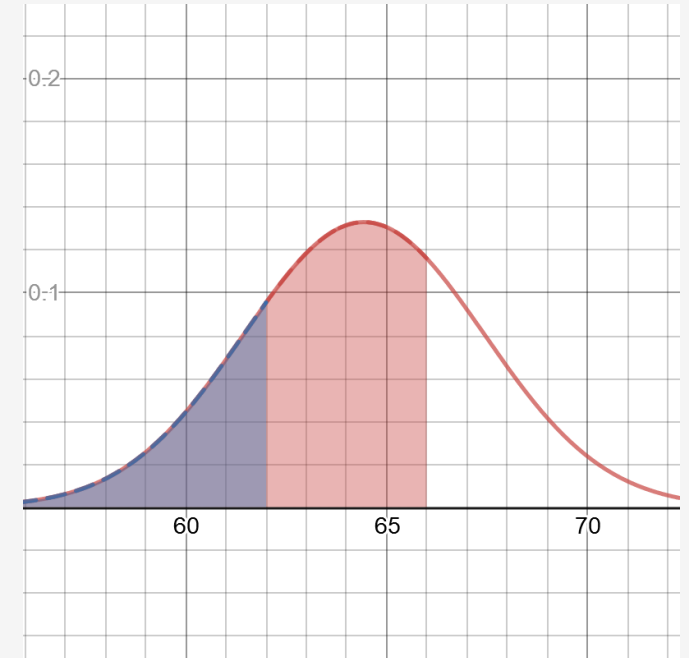
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Cumulative Density Function (CDF)

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I first find the area up to 66 (highlighted in red to the

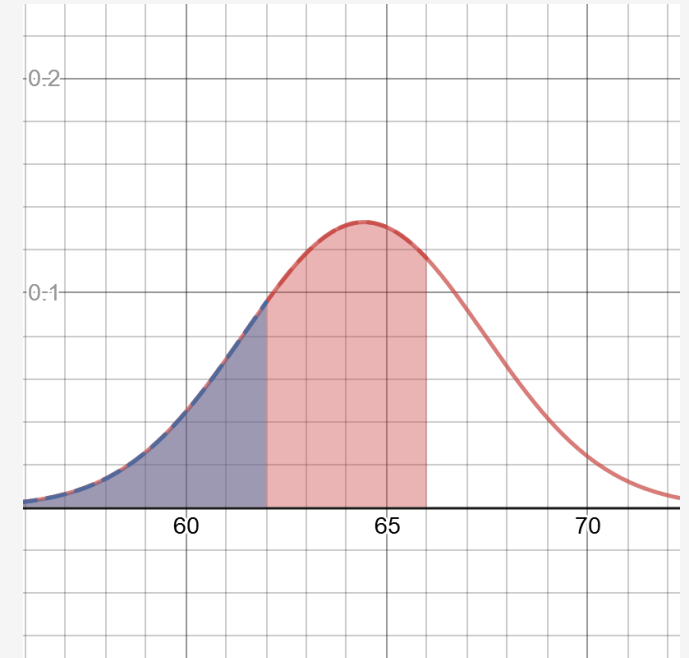
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```
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```

highlighted in



Katacoda Interactive Scenario – Normal Distribution



Discovering the Normal Distribution

Step 1 of 4 ▶

Normal Distribution I

You collected weight measurements of 50 adult golden retrievers. You can view the data by running the following code:

```
from
urllib.request import urlopen

# Retrieve golden retriever
weights
```

Copy to Editor

scratch.pynormal_distribution.py

```
1 from urllib.request import urlopen
2
3 # Retrieve golden retriever weights
4 weights = [float(w) for w in urlopen("https://bit.ly/3cXuuf
5     .read().decode('utf-8').split("\n") if w]
6
7 for w in weights:
8     print(w)
9
10
```

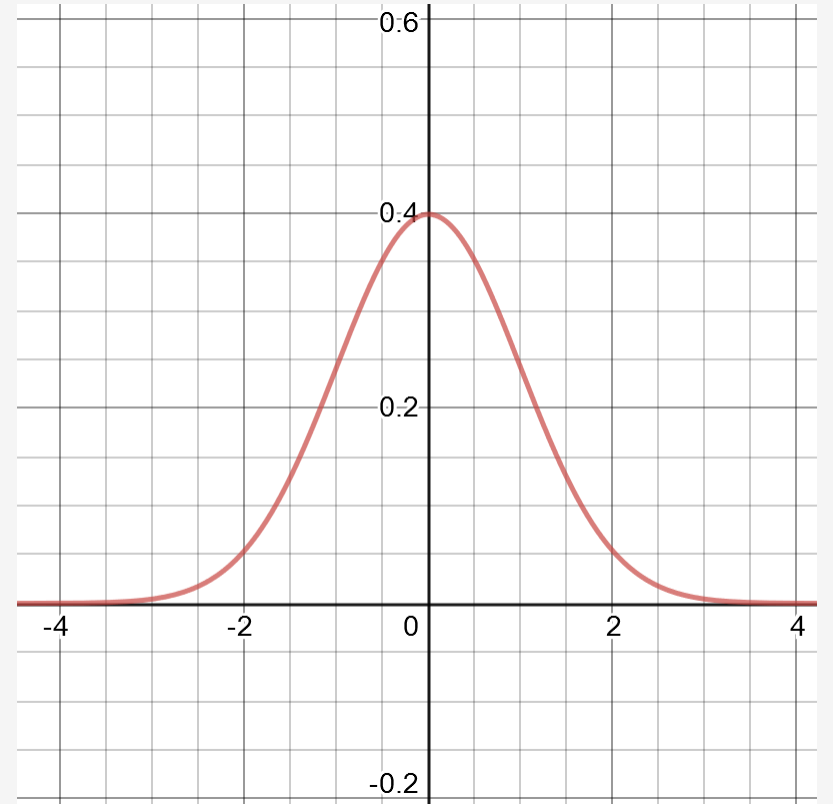
Terminal +

```
Introduction
$ cd /home/scrapbook/tutorial
$
```

Powered by  Katacoda

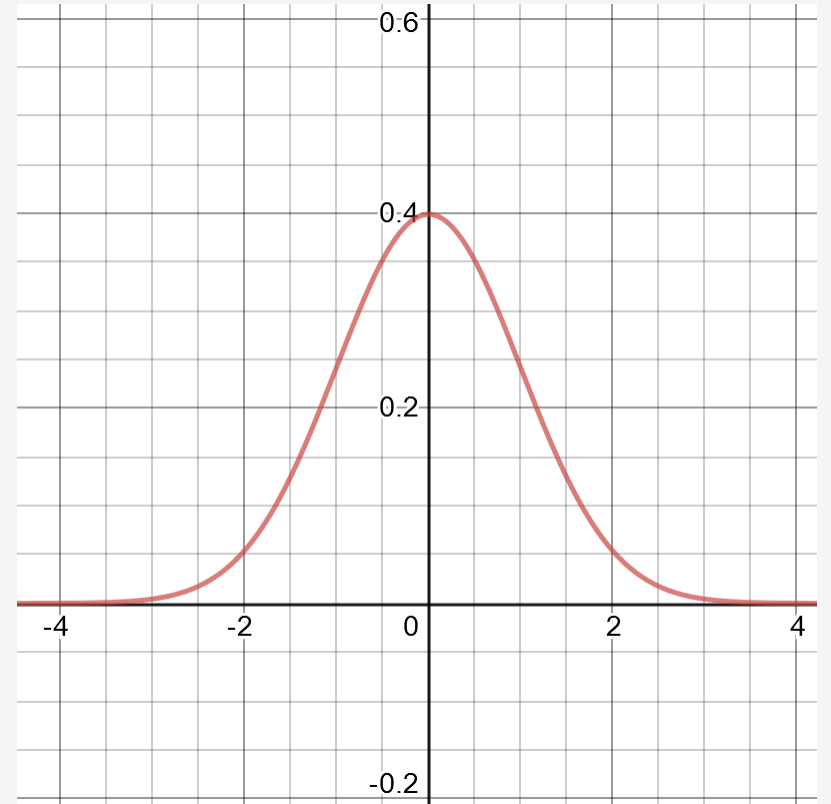
<https://learning.oreilly.com/scenarios/probability-from-scratch/9781492080572/>

Standard Normal Distribution



Standard Normal Distribution

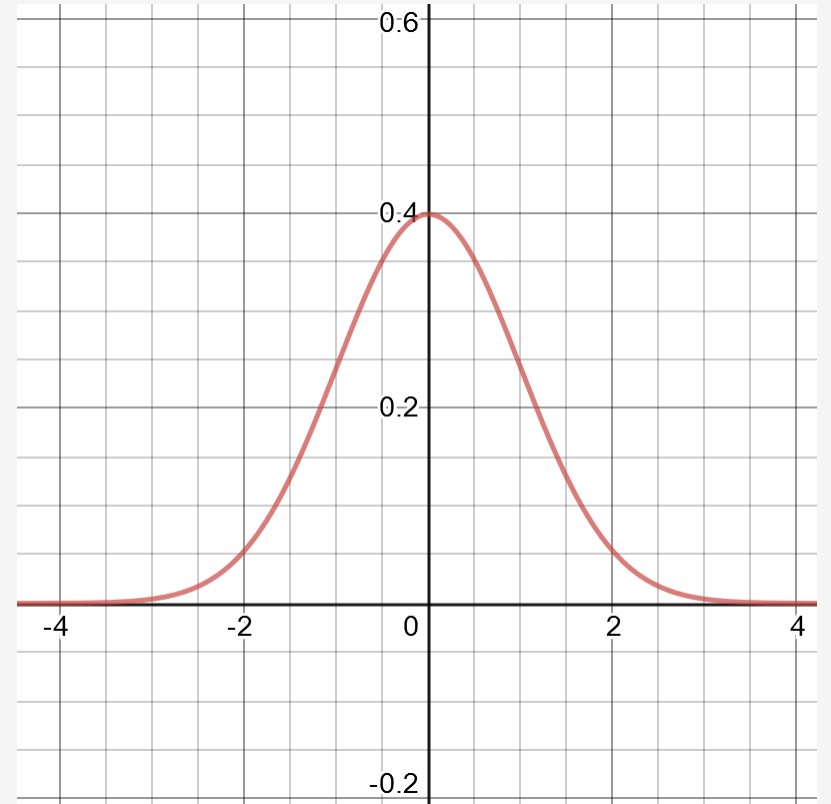
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Sometimes a normal distribution will be converted into a standard normal distribution.

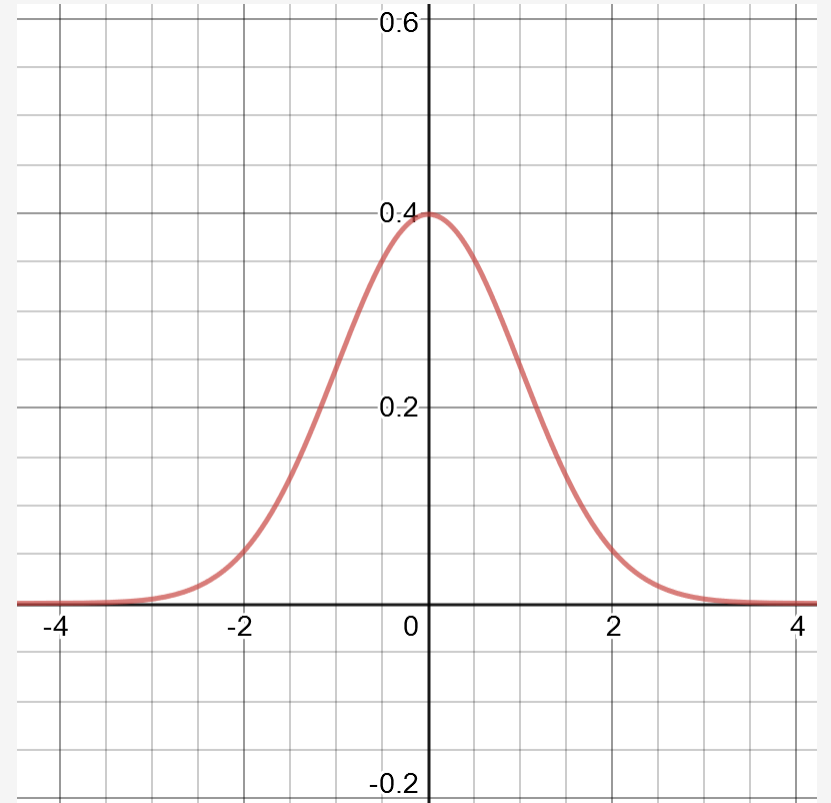


Standard Normal Distribution

A special type of normal distribution is the **standard normal distribution**, which has a mean of 0 and standard deviation of 1.

Sometimes a normal distribution will be converted into a standard normal distribution.

This creates a convenient way to express all values in terms of the standard deviation, known as **z scores**.



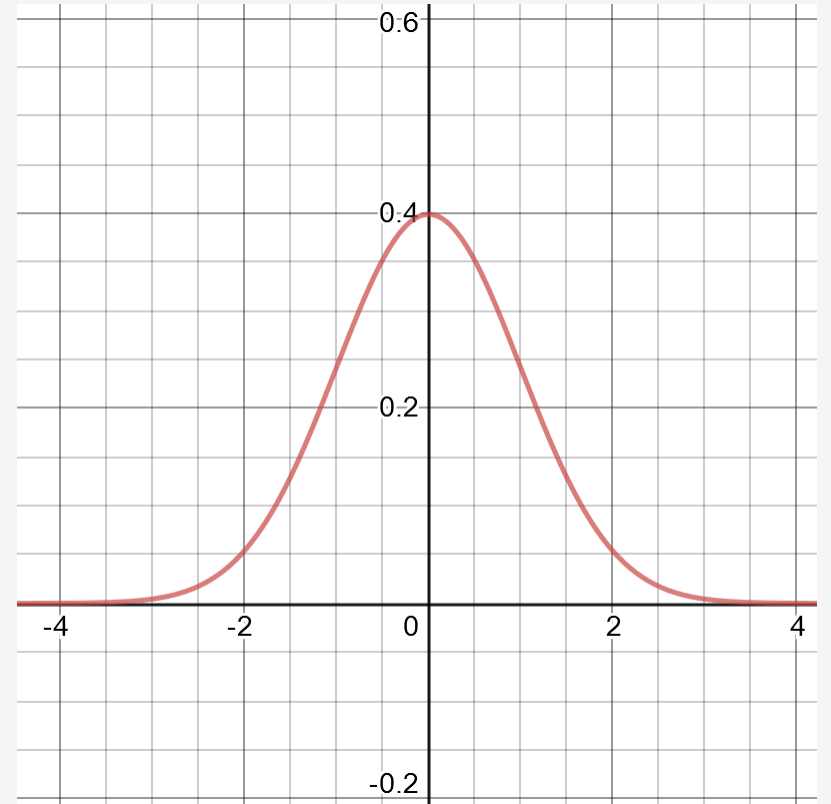
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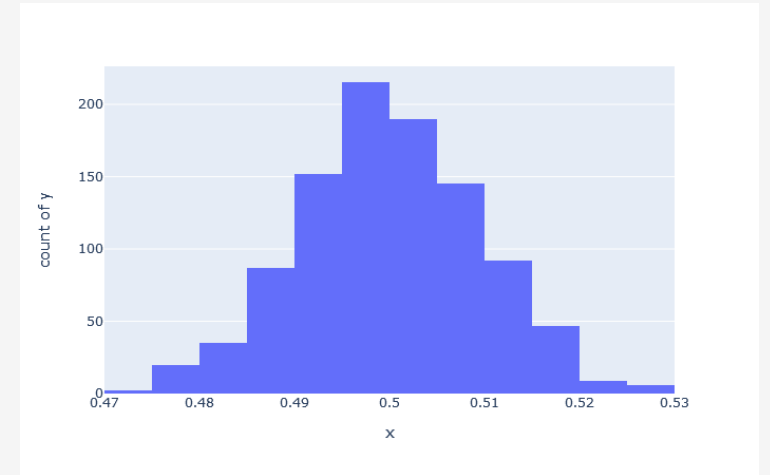
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This creates a convenient way to express all values in terms of the standard deviation, known as **z scores**.

Turning several normal distributions into standard normal distributions also makes them easier to compare, as comparisons can be made relative to their means rather than absolute values.



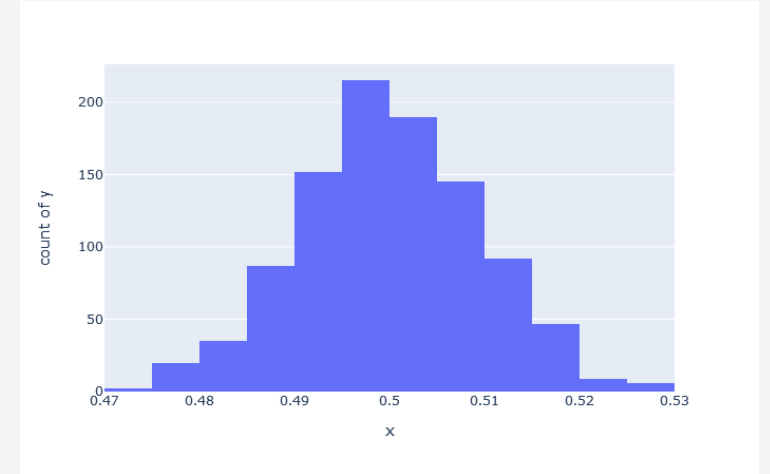
Central Limit Theorem



The above data came from a uniform distribution, and yet the mean of each sample of 1000 data points shows a normal distribution!

Central Limit Theorem

The normal distribution shows up in an important phenomena called the **central limit theorem**, which states that the means of different samples will form a normal distribution no matter what the underlying distribution is.

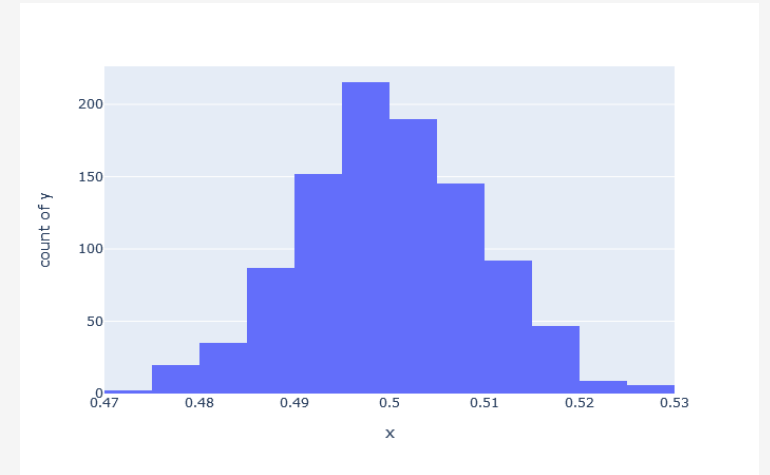


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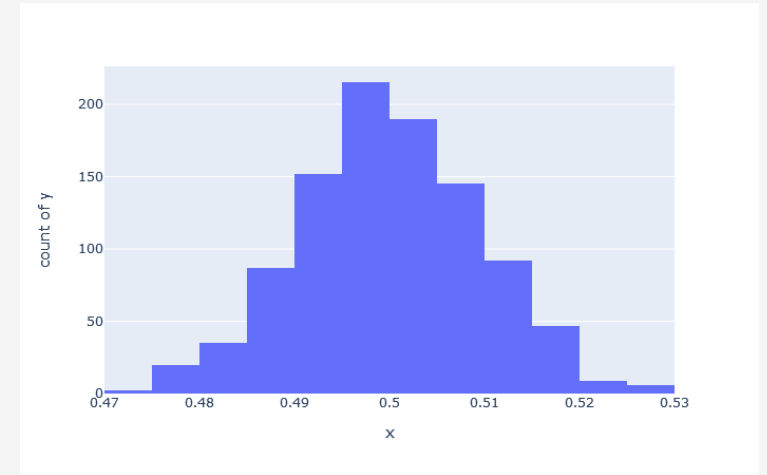
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Say I generated 1000 samples, each containing 1000 random numbers coming from a uniform distribution between 0.0 and 1.0.

If I take the average of each sample and plot them in a histogram, we will see they form a normal distribution!



The above data came from a uniform distribution, and yet the mean of each sample of 1000 data points shows a normal distribution!

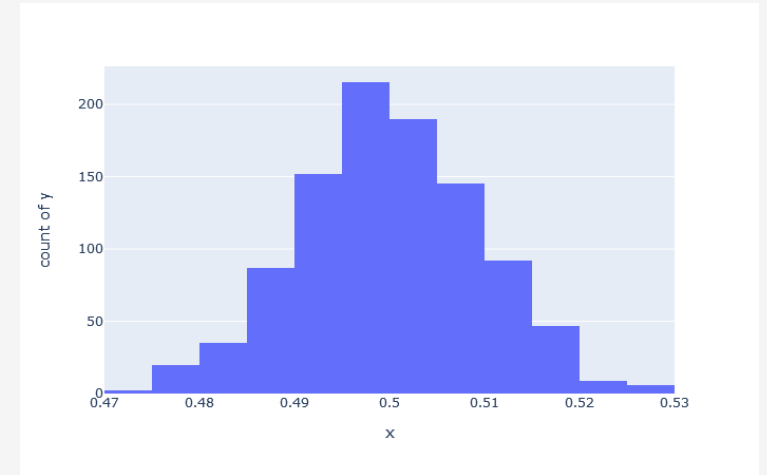
Central Limit Theorem

The normal distribution shows up in an important phenomena called the **central limit theorem**, which states that the means of different samples will form a normal distribution no matter what the underlying distribution is.

Say I generated 1000 samples, each containing 1000 random numbers coming from a uniform distribution between 0.0 and 1.0.

If I take the average of each sample and plot them in a histogram, we will see they form a normal distribution!

Even if the data is completely random or has a distribution different from a normal distribution, the averages of the samples will form a normal distribution.



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Central Limit Theorem

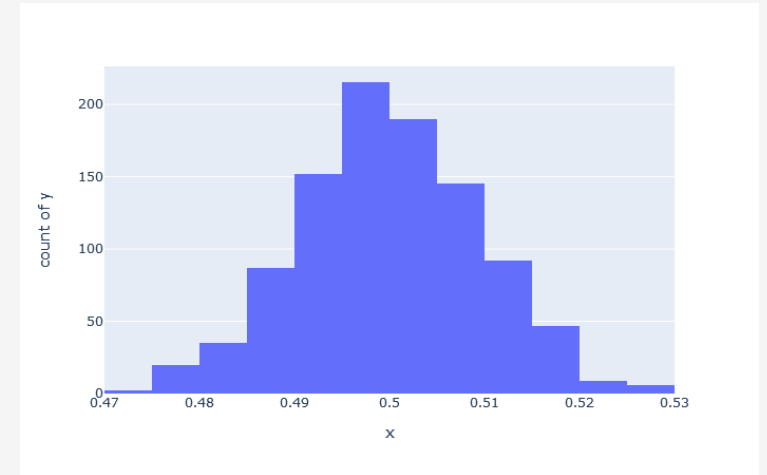
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This is another reason why the normal distribution is useful, as it shows up even in data that is not normally distributed.



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Central Limit Theorem

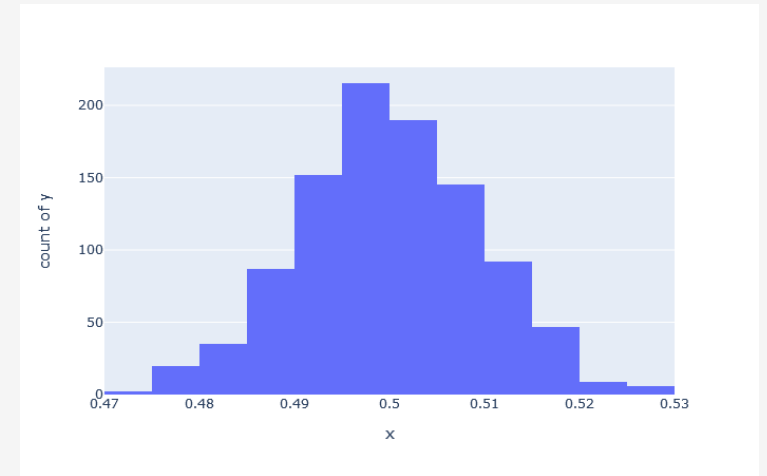
```
# Samples of the uniform distribution will average out to a normal distribution.

import random

import plotly.express as px

# Central limit theorem, 1000 samples each with 1000 random numbers between 0.0 and 1.0
x_values = [(sum([random.uniform(0.0, 1.0) for i in range(1000)]) / 1000.0) for _ in
range(1000)]
y_values = [1 for _ in range(1000)]

px.histogram(x=x_values, y = y_values, nbins=20).show()
```



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Other Distribution Types

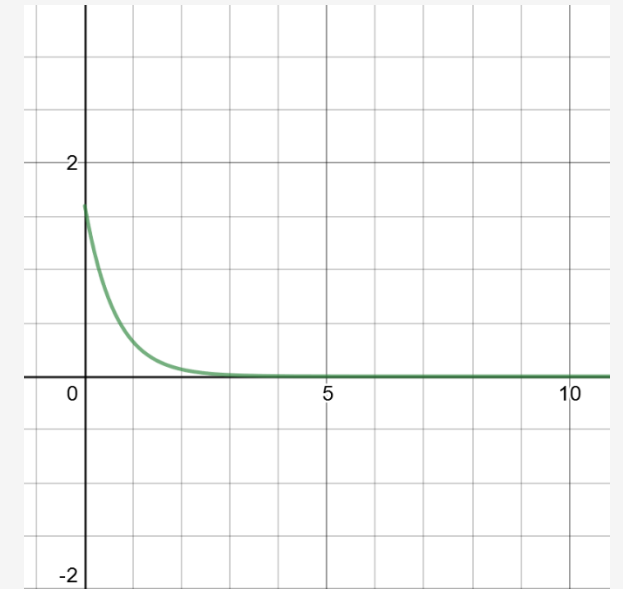
There are many types of probability distributions, some much more niche than others, but here are a few other notable ones:

Poisson – Discrete distribution of probabilities for **x** number of events in a fixed amount of time.

Exponential – Distribution of time between events in a Poisson process

Gamma – A generalization of a two-parameter distribution that includes exponential, Erlang, and chi-squared distributions

Weibull – Often used for survival analysis, failure analysis, and reliability engineering.

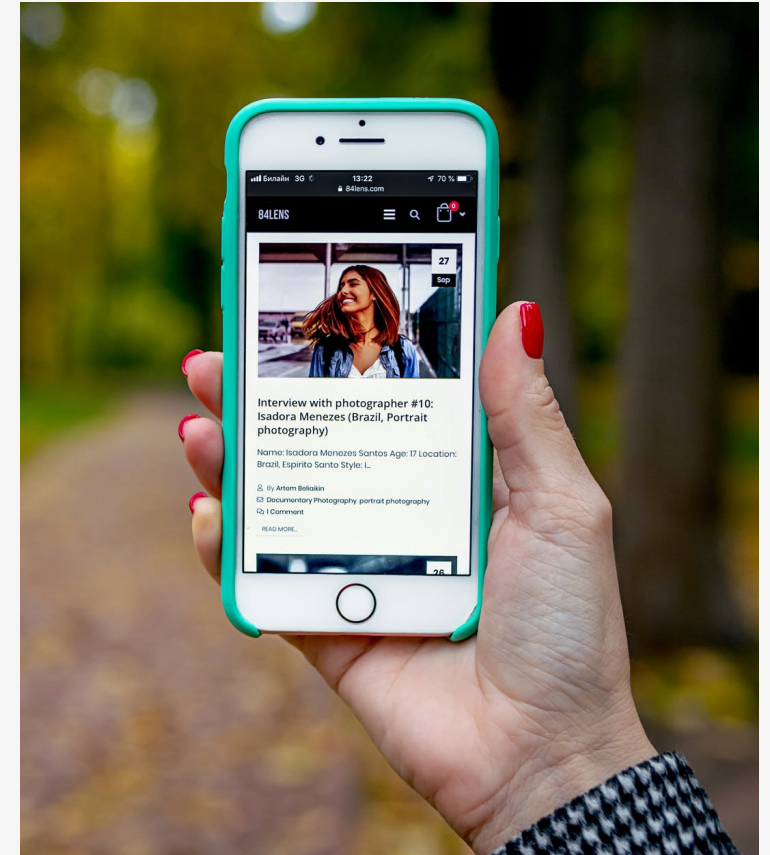


Exponential Distribution

Exercise

A market researcher estimates that the Z-Phone smart phone has a mean consumer life of 42 months with a standard deviation of 8 months.

Assuming a normal distribution, what is the probability a given random Z-Phone will last between 20 and 30 months?



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A market researcher estimates that the Z-Phone smart phone has a mean consumer life of 42 months with a standard deviation of 8 months.

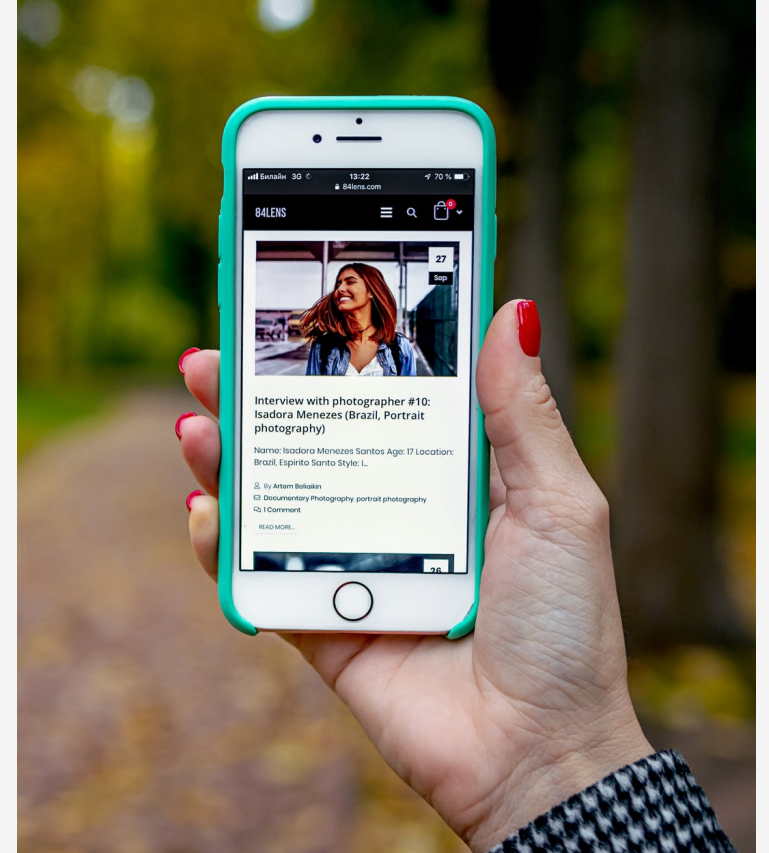
Assuming a normal distribution, what is the probability a given random Z-Phone will last between 20 and 30 months?

```
from scipy.stats import norm

mean = 42
std_dev = 8

x = norm.cdf(30, mean, std_dev) - norm.cdf(20, mean, std_dev)

print(x)
```



There is a 6.3827% probability the Z-Phone will last between 20 and 30 months.